Proyecto_practico

July 15, 2026

1 Actividad - Proyecto práctico

Asignatura: Aprendizaje por Refuerzo
Entorno: `SpaceInvaders-v0`
Algoritmo principal: Deep Q-Network (**DQN**)
Objetivo del grupo: alcanzar al menos 20 puntos con *reward clipping* durante 100 episodios consecutivos.

Componentes

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Repositorio

- GitHub: 08MIAR_Aprendizaje_por_Refuerzo
- Entorno ALE: Space Invaders

1.1 Índice

1.2 PARTE 1. Instalación y requisitos previos

Las prácticas se han preparado para poder ejecutarse en Google Colab. No obstante, la visualización de entornos `gym` puede presentar incompatibilidades en Colab, por lo que también se incluye una configuración para ejecución local.

El flujo recomendado es el siguiente:

1. **Local:** preparar el entorno siguiendo la sección 1.1.
2. **Colab o local:** configurar las rutas de trabajo en la sección 1.2.
3. **Colab:** montar Google Drive siguiendo la sección 1.3.
4. **Colab o local:** instalar las librerías necesarias indicadas en la sección 1.4.

1.2.1 1.1 Preparación del entorno local

En ejecución local, se recomienda crear un entorno virtual específico para evitar conflictos de versiones entre TensorFlow, Gym, Keras-RL y las dependencias de Atari.

1.2.2 1.2 Detección del entorno de ejecución

Esta celda identifica si el notebook se está ejecutando en Google Colab o en local y define las rutas base de trabajo.

[7]:
```
# ATENCIÓN!! Modificar ruta relativa a la práctica si es distinta (drive_root)
mount='/content/gdrive'
drive_root = mount + "/My Drive/Colab Notebooks/VIU IA/08_AR_MIAR/Proyecto"

try:
  from google.colab import drive
  IN_COLAB=True
```

```
except:
  IN_COLAB=False
```

1.2.3 1.3 Montaje de Google Drive en Colab

Esta sección solo debe ejecutarse en Google Colab. En ejecución local, no es necesario montar Google Drive.

[8]:
```
# Switch to the directory on the Google Drive that you want to use
import os
if IN_COLAB:
  print("We're running Colab")

  if IN_COLAB:
    # Mount the Google Drive at mount
    print("Colab: mounting Google drive on ", mount)

    drive.mount(mount)

    # Create drive_root if it doesn't exist
    create_drive_root = True
    if create_drive_root:
      print("\nColab: making sure ", drive_root, " exists.")
      os.makedirs(drive_root, exist_ok=True)

    # Change to the directory
    print("\nColab: Changing directory to ", drive_root)
    %cd $drive_root
# Verify we're in the correct working directory
%pwd
print("Archivos en el directorio: ")
print(os.listdir())
```

```
We're running Colab
Colab: mounting Google drive on  /content/gdrive
Mounted at /content/gdrive

Colab: making sure  /content/gdrive/My Drive/Colab Notebooks/VIU
IA/08_AR_MIAR/Proyecto  exists.

Colab: Changing directory to  /content/gdrive/My Drive/Colab Notebooks/VIU
IA/08_AR_MIAR/Proyecto
/content/gdrive/My Drive/Colab Notebooks/VIU IA/08_AR_MIAR/Proyecto
Archivos en el directorio:
['Logo_VIU.png', 'Proyecto_práctico_v8_Resultados.ipynb',
'Proyecto_práctico_v5_Propuesta2.ipynb',
```

```
'Proyecto_práctico_v9_Tabla_resumen.ipynb',
'Proyecto_practico_v10_graficas_indice_corregido.ipynb',
'Proyecto_práctico.ipynb', 'Proyecto_práctico_v3_Train_Base.ipynb',
'Proyecto_práctico_v7_Propuesta3.ipynb', 'README.md', 'Proyecto_práctico
Esqueleto + 3 mejoras sin ejecutar.ipynb',
'Proyecto_práctico_v4_Propuesta1.ipynb',
'Proyecto_práctico_v6_Refactorizacion.ipynb',
'dqn_SpaceInvaders-v0_Resultados', 'dqn_SpaceInvaders-v0_Train_Base',
'dqn_SpaceInvaders-v0_Train_Propuesta3',
'dqn_SpaceInvaders-v0_Train_Propuesta2',
'dqn_SpaceInvaders-v0_Train_Propuesta1',
'Proyecto_practico_v11_Resultado_Test.ipynb']
```

1.2.4 1.4 Instalación de librerías

Se instalan las dependencias necesarias para TensorFlow, Gym, Atari y Keras-RL. La instalación está condicionada para ejecutarse principalmente en Colab.

[ ]:
```
if IN_COLAB:
  %pip install tensorflow==2.15.0
  %pip install tf-keras==2.15.0
  %pip install gym==0.17.3
  %pip install git+https://github.com/Kojoley/atari-py.git
  %pip install keras-rl2==1.0.5
else:
  %pip install typing_extensions==3.7.4
  %pip install gym==0.17.3
  %pip install git+https://github.com/Kojoley/atari-py.git
  %pip install pyglet==1.5.0
  %pip install h5py==3.1.0
  %pip install Pillow==9.5.0
  %pip install keras-rl2==1.0.5
  %pip install Keras==2.2.4
  %pip install tensorflow==2.5.3
  %pip install torch==2.0.1
  %pip install agents==1.4.0
```

1.3 PARTE 2. Enunciado, objetivo y criterio de evaluación

Consideraciones a tener en cuenta:

- El entorno sobre el que trabajaremos será el indicado en el listado correspondiente de cada grupo y el algoritmo que usaremos será *DQN*.
- Para nuestro ejercicio, el requisito mínimo será alcanzado cuando el agente consiga una **media de recompensa por encima de los puntos indicados en el listado por grupos en**

modo test. Por ello, esta media de la recompensa se calculará a partir del código de test en la última celda del notebook.

Este proyecto práctico consta de tres partes:

1. Implementar la red neuronal que se usará en la solución
2. Implementar las distintas piezas de la solución DQN y probar al menos 3 propuestas diferentes de mejora.
3. Justificar la respuesta en relación a los resultados obtenidos e incluir al menos 3 gráficas relevantes comparando las 3 propuestas.

Rúbrica: Se valorará la originalidad en la solución aportada, así como la capacidad de discutir los resultados de forma detallada. El requisito mínimo servirá para aprobar la actividad, bajo premisa de que la discusión del resultado será apropiada.

IMPORTANTE:

- Si no se consigue una puntuación óptima, responder sobre la mejor puntuación obtenida.
- Para entrenamientos largos, recordad que podéis usar checkpoints de vuestros modelos para retomar los entrenamientos. En este caso, recordad cambiar los parámetros adecuadamente (sobre todo los relacionados con el proceso de exploración).
- Se deberá entregar unicamente el notebook y los pesos del mejor modelo en un fichero .zip, de forma organizada.
- Cada alumno deberá de subir la solución de forma individual.

1.4 PARTE 3. Desarrollo de la solución DQN

En esta parte se define el entorno de Atari, el preprocesamiento de observaciones, la red neuronal convolucional y los agentes DQN que se compararán posteriormente.

[ ]:
```
import os
import logging

# Mostrar solo los mensajes de error TensorFlow (0=Todo, 1=Info, 2=Warnings,␣
↪3=Errors)
os.environ['TF_CPP_MIN_LOG_LEVEL'] = '3'

import tensorflow as tf
tf.get_logger().setLevel(logging.ERROR)
```

1.4.1 3.1 Importación de librerías

[ ]:
```
#Comprobamos que se carga la versión esperada de TF y detecta la GPU
import os
import tensorflow.keras as tf
import tensorflow as tf1

os.environ['TF_USE_LEGACY_KERAS']="1"
```

```
from keras import __version__
tf.__version__ = __version__

print(tf1.__version__)
print(tf1.config.list_physical_devices('GPU'))
print(tf1.config.list_logical_devices('GPU'))
```

```
2.15.0
[PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU')]
[LogicalDevice(name='/device:GPU:0', device_type='GPU')]
```

[ ]:
```
from __future__ import division

from PIL import Image
import numpy as np
import gym

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten, Conv2D, InputLayer, Permute

if IN_COLAB:
  from tensorflow.keras.optimizers.legacy import Adam
else:
  from tensorflow.keras.optimizers.legacy import Adam

import tensorflow.keras.backend as K

from rl.agents.dqn import DQNAgent
from rl.policy import LinearAnnealedPolicy, BoltzmannQPolicy, EpsGreedyQPolicy
from rl.memory import SequentialMemory
from rl.core import Processor
from rl.callbacks import FileLogger, ModelIntervalCheckpoint
```

1.4.2 3.2 Configuración del entorno y preprocesamiento Atari

Se define el formato de entrada de los frames, se inicializa el entorno `SpaceInvaders-v0` y se implementa el procesador encargado de adaptar las observaciones y recompensas al formato esperado por DQN.

[ ]:
```
# Formato de las imágenes de entrada
INPUT_SHAPE = (84, 84)
WINDOW_LENGTH = 4

# Configuración del entorno y extracción del número de acciones
env_name = 'SpaceInvaders-v0'
env = gym.make(env_name)
```

```python
np.random.seed(123)
env.seed(123)
nb_actions = env.action_space.n
```

[ ]:
```python
print("Numero de acciones disponibles:" + str(nb_actions))

print("Formato de las observaciones:")
env.observation_space
```

```
Numero de acciones disponibles:6
Formato de las observaciones:
```

[ ]:
```
Box(0, 255, (210, 160, 3), uint8)
```

[ ]:
```python
class AtariProcessor(Processor):
    def process_observation(self, observation):
        assert observation.ndim == 3  # (height, width, channel)
        img = Image.fromarray(observation)
        img = img.resize(INPUT_SHAPE).convert('L')
        processed_observation = np.array(img)
        assert processed_observation.shape == INPUT_SHAPE
        return processed_observation.astype('uint8')

    def process_state_batch(self, batch):
        processed_batch = batch.astype('float32') / 255.
        return processed_batch

    def process_reward(self, reward):
        return np.clip(reward, -1., 1.)
```

1.4.3 3.3 Caso 0: DQN base (*baseline*)

Este primer agente se utiliza como referencia para comprobar que el flujo de trabajo funciona correctamente y para comparar las mejoras introducidas en las propuestas posteriores.

La arquitectura se inspira en la red convolucional clásica utilizada en los trabajos iniciales de DQN para Atari: varias capas convolucionales para extraer patrones espaciales de los frames y capas densas finales para estimar el valor Q de cada acción.

3.3.1 Implementación de la red neuronal

[ ]:
```python
# Formato de los datos de entrada
input_shape = (WINDOW_LENGTH,) + INPUT_SHAPE

# Definición del modelo
model = Sequential(name="DQN_SpaceInvaders")

# Datos de entrada
```

```python
model.add(InputLayer(input_shape=input_shape, name="input_frames"))

# Compatibilidad de la entrada
model.add(Permute((2, 3, 1), name="permute_channels"))

# Capas convolucionales
model.add(Conv2D(32, (8, 8), strides=(4, 4), activation="relu", name="conv1"))
model.add(Conv2D(64, (4, 4), strides=(2, 2), activation="relu", name="conv2"))
model.add(Conv2D(64, (3, 3), strides=(1, 1), activation="relu", name="conv3"))

# Capa Flatten
model.add(Flatten(name="flatten"))

# Capas Densas
model.add(Dense(512, activation="relu", name="dense_hidden"))
model.add(Dense(nb_actions, activation="linear", name="output_values"))

# Imprimir información del modelo
print(model.summary())
```

```
Model: "DQN_SpaceInvaders"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 permute_channels (Permute)  (None, 84, 84, 4)         0

 conv1 (Conv2D)              (None, 20, 20, 32)        8224

 conv2 (Conv2D)              (None, 9, 9, 64)          32832

 conv3 (Conv2D)              (None, 7, 7, 64)          36928

 flatten (Flatten)           (None, 3136)              0

 dense_hidden (Dense)        (None, 512)               1606144

 output_values (Dense)       (None, 6)                 3078

=================================================================
Total params: 1687206 (6.44 MB)
Trainable params: 1687206 (6.44 MB)
Non-trainable params: 0 (0.00 Byte)
_________________________________________________________________
None
```

3.3.2 Implementación del agente DQN

[ ]:
```python
# Configuración de la memoria del algoritmo:
#    limit:         número máximo de transacciones
#    window_length: fotogramas consecutivos utilizados para formar un estado
memory = SequentialMemory(limit=1000000, window_length=WINDOW_LENGTH)
# Procesado de imágenes: convertir imagenes a escala de grises y normalización␣
↪de los píxeles
processor = AtariProcessor()

# Estrategia de exploración:
#    EpsGreedyQPolicy: estrategia basada en epsilon
#    attr:             atributo epsilon
#    value_max:        valor inicial (máximo)
#    value_min:        valor mínimo (final) con el tiempo irá disminuyendo␣
↪hasta este valor
#    value_test:       valor eps para el modo de prueba
#    nb_steps:         número de pasos, la disminución de eps será gradual␣
↪durante este número de pasos
policy = LinearAnnealedPolicy(EpsGreedyQPolicy(), attr='eps',
                              value_max=1., value_min=.1, value_test=.05,
                              nb_steps=2000000)

# Configuración del agente DQN
#    model:               red neuronal
#    nb_actions:          número de acciones del juego
#    policy:              estrategia de exploración
#    memory:              memoria del algoritmo
#    processor:           procesado de imágenes
#    nb_steps_warmup:     número de pasos en los que el algoritmo no entrena,␣
↪solo juega para llenar la memoria con experiencias iniciales
#    gamma:               factor de actualización de las recompensas
#    target_model_update: número de pasos hasta actualizar los valores de la␣
↪red de predicción a la red de target
#    train_interval:      la red se entrena en intervalo de fotogramas para␣
↪optimización y alineamiento con los fotogramas del estado
dqn = DQNAgent(model=model, nb_actions=nb_actions, policy=policy,
               memory=memory, processor=processor,
               nb_steps_warmup=50000, gamma=.99,
               target_model_update=10000,
               train_interval=4)

# Compilación del agente
#    Adam:          optimizador Adam
#    learning_rate: tasa de aprendizaje
#    metrics:       métrica MAE (Error Absoluto Medio)
dqn.compile(Adam(learning_rate=.00025), metrics=['mae'])
```

```
2026-07-04 13:35:42.563164: W tensorflow/c/c_api.cc:305] Operation
```

```
'{name:'conv2_4/kernel/Assign' id:1090 op device:{requested: '', assigned: ''}
def:{{{node conv2_4/kernel/Assign}} =
AssignVariableOp[_has_manual_control_dependencies=true, dtype=DT_FLOAT,
validate_shape=false](conv2_4/kernel,
conv2_4/kernel/Initializer/stateless_random_uniform)}}' was changed by setting
attribute after it was run by a session. This mutation will have no effect, and
will trigger an error in the future. Either don't modify nodes after running
them or create a new session.
```

[ ]:
```python
# Training part
weights_filename = 'dqn_{}_weights.h5f'.format(env_name)
checkpoint_weights_filename = 'dqn_' + env_name + '_weights_{step}.h5f'
log_filename = 'dqn_{}_log.json'.format(env_name)
callbacks = [ModelIntervalCheckpoint(checkpoint_weights_filename,␣
 ↪interval=100000)]
callbacks += [FileLogger(log_filename, interval=100)]

# Comenzar el entrenamiento
dqn.fit(env, callbacks=callbacks, nb_steps=2000000, log_interval=10000,␣
 ↪visualize=False)
#dqn.fit(env, callbacks=callbacks, nb_steps=20000, log_interval=100,␣
 ↪visualize=False)

dqn.save_weights(weights_filename, overwrite=True)
```

```
Training for 2000000 steps …
Interval 1 (0 steps performed)
10000/10000 [==============================] - 41s 4ms/step - reward: 0.0123
13 episodes - episode_reward: 9.231 [2.000, 18.000] - ale.lives: 2.096

Interval 2 (10000 steps performed)
10000/10000 [==============================] - 44s 4ms/step - reward: 0.0133
16 episodes - episode_reward: 8.438 [3.000, 15.000] - ale.lives: 2.083

Interval 3 (20000 steps performed)
10000/10000 [==============================] - 45s 4ms/step - reward: 0.0129
16 episodes - episode_reward: 8.000 [4.000, 17.000] - ale.lives: 2.081

Interval 4 (30000 steps performed)
10000/10000 [==============================] - 46s 5ms/step - reward: 0.0148
15 episodes - episode_reward: 9.333 [4.000, 18.000] - ale.lives: 2.113

Interval 5 (40000 steps performed)
10000/10000 [==============================] - 45s 4ms/step - reward: 0.0128
12 episodes - episode_reward: 10.750 [3.000, 20.000] - ale.lives: 2.301

Interval 6 (50000 steps performed)
```

```
    1/10000 […] - ETA: 2:31 - reward: 0.0000e+00

2026-07-04 13:39:31.676361: W tensorflow/c/c_api.cc:305] Operation
'{name:'output_values_4/BiasAdd' id:1174 op device:{requested: '', assigned: ''}
def:{{{node output_values_4/BiasAdd}} = BiasAdd[T=DT_FLOAT,
_has_manual_control_dependencies=true,
data_format="NHWC"](output_values_4/MatMul,
output_values_4/BiasAdd/ReadVariableOp)}}' was changed by setting attribute
after it was run by a session. This mutation will have no effect, and will
trigger an error in the future. Either don't modify nodes after running them or
create a new session.
2026-07-04 13:39:31.858314: W tensorflow/c/c_api.cc:305] Operation
'{name:'loss_11/AddN' id:1407 op device:{requested: '', assigned: ''}
def:{{{node loss_11/AddN}} = AddN[N=2, T=DT_FLOAT,
_has_manual_control_dependencies=true](loss_11/mul, loss_11/mul_1)}}' was
changed by setting attribute after it was run by a session. This mutation will
have no effect, and will trigger an error in the future. Either don't modify
nodes after running them or create a new session.
2026-07-04 13:39:31.904606: W tensorflow/c/c_api.cc:305] Operation
'{name:'training_1/Adam/output_values_3/bias/v/Assign' id:1735 op
device:{requested: '', assigned: ''} def:{{{node
training_1/Adam/output_values_3/bias/v/Assign}} =
AssignVariableOp[_has_manual_control_dependencies=true, dtype=DT_FLOAT,
validate_shape=false](training_1/Adam/output_values_3/bias/v,
training_1/Adam/output_values_3/bias/v/Initializer/zeros)}}' was changed by
setting attribute after it was run by a session. This mutation will have no
effect, and will trigger an error in the future. Either don't modify nodes after
running them or create a new session.

10000/10000 [==============================] - 260s 26ms/step - reward: 0.0138
16 episodes - episode_reward: 9.062 [3.000, 16.000] - loss: 0.007 - mae: 0.044 -
mean_q: 0.059 - mean_eps: 0.975 - ale.lives: 2.067

Interval 7 (60000 steps performed)
10000/10000 [==============================] - 254s 25ms/step - reward: 0.0147
15 episodes - episode_reward: 9.933 [5.000, 22.000] - loss: 0.007 - mae: 0.062 -
mean_q: 0.080 - mean_eps: 0.971 - ale.lives: 2.063

Interval 8 (70000 steps performed)
10000/10000 [==============================] - 251s 25ms/step - reward: 0.0133
12 episodes - episode_reward: 10.750 [5.000, 35.000] - loss: 0.007 - mae: 0.080
- mean_q: 0.103 - mean_eps: 0.966 - ale.lives: 2.078

Interval 9 (80000 steps performed)
10000/10000 [==============================] - 245s 25ms/step - reward: 0.0126
13 episodes - episode_reward: 9.231 [5.000, 14.000] - loss: 0.007 - mae: 0.103 -
mean_q: 0.133 - mean_eps: 0.962 - ale.lives: 2.038

Interval 10 (90000 steps performed)
```

```
10000/10000 [==============================] - 244s 24ms/step - reward: 0.0120
14 episodes - episode_reward: 9.214 [2.000, 22.000] - loss: 0.006 - mae: 0.123 -
mean_q: 0.158 - mean_eps: 0.957 - ale.lives: 2.092

Interval 11 (100000 steps performed)
10000/10000 [==============================] - 248s 25ms/step - reward: 0.0139
18 episodes - episode_reward: 7.778 [4.000, 11.000] - loss: 0.006 - mae: 0.141 -
mean_q: 0.179 - mean_eps: 0.953 - ale.lives: 2.111

Interval 12 (110000 steps performed)
10000/10000 [==============================] - 247s 25ms/step - reward: 0.0146
14 episodes - episode_reward: 9.857 [5.000, 16.000] - loss: 0.006 - mae: 0.165 -
mean_q: 0.209 - mean_eps: 0.948 - ale.lives: 2.199

Interval 13 (120000 steps performed)
10000/10000 [==============================] - 257s 26ms/step - reward: 0.0139
13 episodes - episode_reward: 10.769 [3.000, 20.000] - loss: 0.006 - mae: 0.186
- mean_q: 0.235 - mean_eps: 0.944 - ale.lives: 2.083

Interval 14 (130000 steps performed)
10000/10000 [==============================] - 261s 26ms/step - reward: 0.0139
15 episodes - episode_reward: 9.467 [4.000, 23.000] - loss: 0.006 - mae: 0.211 -
mean_q: 0.265 - mean_eps: 0.939 - ale.lives: 2.096

Interval 15 (140000 steps performed)
10000/10000 [==============================] - 253s 25ms/step - reward: 0.0140
14 episodes - episode_reward: 9.786 [5.000, 18.000] - loss: 0.006 - mae: 0.235 -
mean_q: 0.295 - mean_eps: 0.935 - ale.lives: 2.176

Interval 16 (150000 steps performed)
10000/10000 [==============================] - 255s 26ms/step - reward: 0.0149
15 episodes - episode_reward: 10.000 [3.000, 25.000] - loss: 0.007 - mae: 0.267
- mean_q: 0.333 - mean_eps: 0.930 - ale.lives: 2.190

Interval 17 (160000 steps performed)
10000/10000 [==============================] - 254s 25ms/step - reward: 0.0143
13 episodes - episode_reward: 10.923 [6.000, 18.000] - loss: 0.008 - mae: 0.289
- mean_q: 0.359 - mean_eps: 0.926 - ale.lives: 2.189

Interval 18 (170000 steps performed)
10000/10000 [==============================] - 256s 26ms/step - reward: 0.0137
14 episodes - episode_reward: 10.286 [3.000, 17.000] - loss: 0.008 - mae: 0.318
- mean_q: 0.395 - mean_eps: 0.921 - ale.lives: 1.998

Interval 19 (180000 steps performed)
10000/10000 [==============================] - 260s 26ms/step - reward: 0.0153
15 episodes - episode_reward: 10.200 [5.000, 16.000] - loss: 0.008 - mae: 0.348
- mean_q: 0.432 - mean_eps: 0.917 - ale.lives: 2.234
```

```
Interval 20 (190000 steps performed)
10000/10000 [==============================] - 260s 26ms/step - reward: 0.0147
13 episodes - episode_reward: 11.308 [6.000, 17.000] - loss: 0.008 - mae: 0.371
- mean_q: 0.460 - mean_eps: 0.912 - ale.lives: 2.146

Interval 21 (200000 steps performed)
10000/10000 [==============================] - 270s 27ms/step - reward: 0.0126
16 episodes - episode_reward: 7.438 [2.000, 12.000] - loss: 0.008 - mae: 0.408 -
mean_q: 0.505 - mean_eps: 0.908 - ale.lives: 2.029

Interval 22 (210000 steps performed)
10000/10000 [==============================] - 266s 27ms/step - reward: 0.0131
15 episodes - episode_reward: 9.000 [3.000, 20.000] - loss: 0.008 - mae: 0.435 -
mean_q: 0.537 - mean_eps: 0.903 - ale.lives: 2.125

Interval 23 (220000 steps performed)
10000/10000 [==============================] - 268s 27ms/step - reward: 0.0137
12 episodes - episode_reward: 10.583 [4.000, 16.000] - loss: 0.008 - mae: 0.460
- mean_q: 0.568 - mean_eps: 0.899 - ale.lives: 2.083

Interval 24 (230000 steps performed)
10000/10000 [==============================] - 267s 27ms/step - reward: 0.0148
15 episodes - episode_reward: 10.333 [4.000, 21.000] - loss: 0.009 - mae: 0.497
- mean_q: 0.613 - mean_eps: 0.894 - ale.lives: 1.991

Interval 25 (240000 steps performed)
10000/10000 [==============================] - 269s 27ms/step - reward: 0.0151
13 episodes - episode_reward: 12.077 [6.000, 25.000] - loss: 0.009 - mae: 0.518
- mean_q: 0.641 - mean_eps: 0.890 - ale.lives: 2.212

Interval 26 (250000 steps performed)
10000/10000 [==============================] - 269s 27ms/step - reward: 0.0149
12 episodes - episode_reward: 12.250 [5.000, 27.000] - loss: 0.010 - mae: 0.556
- mean_q: 0.687 - mean_eps: 0.885 - ale.lives: 2.024

Interval 27 (260000 steps performed)
10000/10000 [==============================] - 272s 27ms/step - reward: 0.0146
19 episodes - episode_reward: 7.474 [2.000, 20.000] - loss: 0.010 - mae: 0.584 -
mean_q: 0.720 - mean_eps: 0.881 - ale.lives: 2.197

Interval 28 (270000 steps performed)
10000/10000 [==============================] - 273s 27ms/step - reward: 0.0136
13 episodes - episode_reward: 10.923 [5.000, 18.000] - loss: 0.011 - mae: 0.629
- mean_q: 0.776 - mean_eps: 0.876 - ale.lives: 2.042

Interval 29 (280000 steps performed)
10000/10000 [==============================] - 277s 28ms/step - reward: 0.0142
```

```
15 episodes - episode_reward: 9.467 [2.000, 17.000] - loss: 0.011 - mae: 0.666 -
mean_q: 0.822 - mean_eps: 0.872 - ale.lives: 2.130

Interval 30 (290000 steps performed)
10000/10000 [==============================] - 273s 27ms/step - reward: 0.0158
12 episodes - episode_reward: 12.750 [6.000, 29.000] - loss: 0.011 - mae: 0.712
- mean_q: 0.877 - mean_eps: 0.867 - ale.lives: 2.047

Interval 31 (300000 steps performed)
10000/10000 [==============================] - 284s 28ms/step - reward: 0.0141
14 episodes - episode_reward: 10.143 [3.000, 24.000] - loss: 0.011 - mae: 0.728
- mean_q: 0.898 - mean_eps: 0.863 - ale.lives: 1.986

Interval 32 (310000 steps performed)
10000/10000 [==============================] - 289s 29ms/step - reward: 0.0155
14 episodes - episode_reward: 10.286 [3.000, 26.000] - loss: 0.011 - mae: 0.746
- mean_q: 0.919 - mean_eps: 0.858 - ale.lives: 2.056

Interval 33 (320000 steps performed)
10000/10000 [==============================] - 283s 28ms/step - reward: 0.0154
14 episodes - episode_reward: 12.000 [4.000, 20.000] - loss: 0.012 - mae: 0.787
- mean_q: 0.968 - mean_eps: 0.854 - ale.lives: 2.148

Interval 34 (330000 steps performed)
10000/10000 [==============================] - 276s 28ms/step - reward: 0.0157
14 episodes - episode_reward: 11.071 [6.000, 19.000] - loss: 0.012 - mae: 0.822
- mean_q: 1.013 - mean_eps: 0.849 - ale.lives: 2.153

Interval 35 (340000 steps performed)
10000/10000 [==============================] - 270s 27ms/step - reward: 0.0171
15 episodes - episode_reward: 11.133 [5.000, 20.000] - loss: 0.012 - mae: 0.863
- mean_q: 1.063 - mean_eps: 0.845 - ale.lives: 2.115

Interval 36 (350000 steps performed)
10000/10000 [==============================] - 271s 27ms/step - reward: 0.0169
15 episodes - episode_reward: 11.067 [5.000, 19.000] - loss: 0.012 - mae: 0.895
- mean_q: 1.101 - mean_eps: 0.840 - ale.lives: 2.150

Interval 37 (360000 steps performed)
10000/10000 [==============================] - 273s 27ms/step - reward: 0.0148
17 episodes - episode_reward: 8.588 [5.000, 19.000] - loss: 0.012 - mae: 0.943 -
mean_q: 1.159 - mean_eps: 0.836 - ale.lives: 2.185

Interval 38 (370000 steps performed)
10000/10000 [==============================] - 280s 28ms/step - reward: 0.0165
16 episodes - episode_reward: 10.812 [7.000, 17.000] - loss: 0.012 - mae: 0.991
- mean_q: 1.218 - mean_eps: 0.831 - ale.lives: 2.100
```

```
Interval 39 (380000 steps performed)
10000/10000 [==============================] - 280s 28ms/step - reward: 0.0166
16 episodes - episode_reward: 10.438 [4.000, 19.000] - loss: 0.012 - mae: 1.017
- mean_q: 1.250 - mean_eps: 0.827 - ale.lives: 2.137

Interval 40 (390000 steps performed)
10000/10000 [==============================] - 295s 29ms/step - reward: 0.0161
16 episodes - episode_reward: 10.062 [3.000, 20.000] - loss: 0.012 - mae: 1.035
- mean_q: 1.271 - mean_eps: 0.822 - ale.lives: 2.125

Interval 41 (400000 steps performed)
10000/10000 [==============================] - 285s 28ms/step - reward: 0.0152
15 episodes - episode_reward: 10.133 [3.000, 20.000] - loss: 0.013 - mae: 1.074
- mean_q: 1.317 - mean_eps: 0.818 - ale.lives: 2.055

Interval 42 (410000 steps performed)
10000/10000 [==============================] - 282s 28ms/step - reward: 0.0171
14 episodes - episode_reward: 11.857 [7.000, 19.000] - loss: 0.013 - mae: 1.127
- mean_q: 1.382 - mean_eps: 0.813 - ale.lives: 2.052

Interval 43 (420000 steps performed)
10000/10000 [==============================] - 286s 29ms/step - reward: 0.0163
15 episodes - episode_reward: 11.400 [6.000, 20.000] - loss: 0.014 - mae: 1.157
- mean_q: 1.418 - mean_eps: 0.809 - ale.lives: 2.125

Interval 44 (430000 steps performed)
10000/10000 [==============================] - 284s 28ms/step - reward: 0.0168
14 episodes - episode_reward: 11.714 [7.000, 23.000] - loss: 0.013 - mae: 1.179
- mean_q: 1.443 - mean_eps: 0.804 - ale.lives: 2.215

Interval 45 (440000 steps performed)
10000/10000 [==============================] - 290s 29ms/step - reward: 0.0157
13 episodes - episode_reward: 12.000 [6.000, 20.000] - loss: 0.013 - mae: 1.207
- mean_q: 1.477 - mean_eps: 0.800 - ale.lives: 2.149

Interval 46 (450000 steps performed)
10000/10000 [==============================] - 286s 29ms/step - reward: 0.0161
17 episodes - episode_reward: 9.353 [3.000, 14.000] - loss: 0.013 - mae: 1.240 -
mean_q: 1.516 - mean_eps: 0.795 - ale.lives: 2.218

Interval 47 (460000 steps performed)
10000/10000 [==============================] - 287s 29ms/step - reward: 0.0165
16 episodes - episode_reward: 10.688 [4.000, 22.000] - loss: 0.014 - mae: 1.292
- mean_q: 1.580 - mean_eps: 0.791 - ale.lives: 2.072

Interval 48 (470000 steps performed)
10000/10000 [==============================] - 288s 29ms/step - reward: 0.0181
12 episodes - episode_reward: 14.167 [7.000, 24.000] - loss: 0.015 - mae: 1.340
```

```
- mean_q: 1.636 - mean_eps: 0.786 - ale.lives: 2.259

Interval 49 (480000 steps performed)
10000/10000 [==============================] - 317s 32ms/step - reward: 0.0171
15 episodes - episode_reward: 12.200 [7.000, 21.000] - loss: 0.014 - mae: 1.365
- mean_q: 1.667 - mean_eps: 0.782 - ale.lives: 2.175

Interval 50 (490000 steps performed)
10000/10000 [==============================] - 308s 31ms/step - reward: 0.0168
16 episodes - episode_reward: 10.000 [2.000, 17.000] - loss: 0.015 - mae: 1.402
- mean_q: 1.712 - mean_eps: 0.777 - ale.lives: 2.101

Interval 51 (500000 steps performed)
10000/10000 [==============================] - 311s 31ms/step - reward: 0.0181
13 episodes - episode_reward: 14.077 [3.000, 28.000] - loss: 0.014 - mae: 1.425
- mean_q: 1.740 - mean_eps: 0.773 - ale.lives: 2.068

Interval 52 (510000 steps performed)
10000/10000 [==============================] - 311s 31ms/step - reward: 0.0162
16 episodes - episode_reward: 10.000 [4.000, 16.000] - loss: 0.014 - mae: 1.456
- mean_q: 1.775 - mean_eps: 0.768 - ale.lives: 2.191

Interval 53 (520000 steps performed)
10000/10000 [==============================] - 327s 33ms/step - reward: 0.0168
13 episodes - episode_reward: 12.615 [7.000, 19.000] - loss: 0.015 - mae: 1.480
- mean_q: 1.805 - mean_eps: 0.764 - ale.lives: 2.220

Interval 54 (530000 steps performed)
10000/10000 [==============================] - 331s 33ms/step - reward: 0.0187
15 episodes - episode_reward: 12.733 [7.000, 22.000] - loss: 0.015 - mae: 1.502
- mean_q: 1.831 - mean_eps: 0.759 - ale.lives: 2.126

Interval 55 (540000 steps performed)
10000/10000 [==============================] - 321s 32ms/step - reward: 0.0186
15 episodes - episode_reward: 12.267 [3.000, 21.000] - loss: 0.015 - mae: 1.537
- mean_q: 1.872 - mean_eps: 0.755 - ale.lives: 2.217

Interval 56 (550000 steps performed)
10000/10000 [==============================] - 322s 32ms/step - reward: 0.0178
14 episodes - episode_reward: 12.786 [5.000, 22.000] - loss: 0.015 - mae: 1.579
- mean_q: 1.922 - mean_eps: 0.750 - ale.lives: 2.183

Interval 57 (560000 steps performed)
10000/10000 [==============================] - 322s 32ms/step - reward: 0.0184
13 episodes - episode_reward: 14.231 [5.000, 31.000] - loss: 0.015 - mae: 1.604
- mean_q: 1.953 - mean_eps: 0.746 - ale.lives: 2.099

Interval 58 (570000 steps performed)
```

```
10000/10000 [==============================] - 328s 33ms/step - reward: 0.0190
13 episodes - episode_reward: 14.538 [8.000, 23.000] - loss: 0.016 - mae: 1.651
- mean_q: 2.008 - mean_eps: 0.741 - ale.lives: 2.061

Interval 59 (580000 steps performed)
10000/10000 [==============================] - 318s 32ms/step - reward: 0.0175
17 episodes - episode_reward: 10.529 [6.000, 19.000] - loss: 0.016 - mae: 1.691
- mean_q: 2.056 - mean_eps: 0.737 - ale.lives: 2.188

Interval 60 (590000 steps performed)
10000/10000 [==============================] - 333s 33ms/step - reward: 0.0172
15 episodes - episode_reward: 11.733 [3.000, 23.000] - loss: 0.016 - mae: 1.722
- mean_q: 2.092 - mean_eps: 0.732 - ale.lives: 2.056

Interval 61 (600000 steps performed)
10000/10000 [==============================] - 335s 33ms/step - reward: 0.0175
16 episodes - episode_reward: 10.375 [6.000, 16.000] - loss: 0.016 - mae: 1.731
- mean_q: 2.104 - mean_eps: 0.728 - ale.lives: 2.144

Interval 62 (610000 steps performed)
10000/10000 [==============================] - 331s 33ms/step - reward: 0.0177
15 episodes - episode_reward: 12.200 [5.000, 20.000] - loss: 0.017 - mae: 1.779
- mean_q: 2.163 - mean_eps: 0.723 - ale.lives: 2.111

Interval 63 (620000 steps performed)
10000/10000 [==============================] - 347s 35ms/step - reward: 0.0184
13 episodes - episode_reward: 13.846 [5.000, 23.000] - loss: 0.016 - mae: 1.796
- mean_q: 2.182 - mean_eps: 0.719 - ale.lives: 2.277

Interval 64 (630000 steps performed)
10000/10000 [==============================] - 325s 32ms/step - reward: 0.0191
15 episodes - episode_reward: 12.733 [7.000, 21.000] - loss: 0.016 - mae: 1.828
- mean_q: 2.222 - mean_eps: 0.714 - ale.lives: 2.129

Interval 65 (640000 steps performed)
10000/10000 [==============================] - 339s 34ms/step - reward: 0.0204
14 episodes - episode_reward: 14.857 [4.000, 22.000] - loss: 0.017 - mae: 1.858
- mean_q: 2.256 - mean_eps: 0.710 - ale.lives: 2.176

Interval 66 (650000 steps performed)
10000/10000 [==============================] - 346s 35ms/step - reward: 0.0186
14 episodes - episode_reward: 12.429 [5.000, 21.000] - loss: 0.017 - mae: 1.892
- mean_q: 2.298 - mean_eps: 0.705 - ale.lives: 2.160

Interval 67 (660000 steps performed)
10000/10000 [==============================] - 343s 34ms/step - reward: 0.0171
16 episodes - episode_reward: 11.000 [4.000, 19.000] - loss: 0.017 - mae: 1.915
- mean_q: 2.325 - mean_eps: 0.701 - ale.lives: 2.124
```

```
Interval 68 (670000 steps performed)
10000/10000 [==============================] - 333s 33ms/step - reward: 0.0162
17 episodes - episode_reward: 9.765 [4.000, 19.000] - loss: 0.017 - mae: 1.940 -
mean_q: 2.355 - mean_eps: 0.696 - ale.lives: 2.210

Interval 69 (680000 steps performed)
10000/10000 [==============================] - 332s 33ms/step - reward: 0.0188
14 episodes - episode_reward: 13.500 [4.000, 26.000] - loss: 0.018 - mae: 1.988
- mean_q: 2.412 - mean_eps: 0.692 - ale.lives: 2.205

Interval 70 (690000 steps performed)
10000/10000 [==============================] - 332s 33ms/step - reward: 0.0207
13 episodes - episode_reward: 15.769 [9.000, 27.000] - loss: 0.017 - mae: 2.007
- mean_q: 2.436 - mean_eps: 0.687 - ale.lives: 2.135

Interval 71 (700000 steps performed)
10000/10000 [==============================] - 342s 34ms/step - reward: 0.0181
14 episodes - episode_reward: 13.357 [4.000, 21.000] - loss: 0.018 - mae: 2.074
- mean_q: 2.516 - mean_eps: 0.683 - ale.lives: 2.078

Interval 72 (710000 steps performed)
10000/10000 [==============================] - 349s 35ms/step - reward: 0.0177
15 episodes - episode_reward: 11.800 [4.000, 20.000] - loss: 0.022 - mae: 2.112
- mean_q: 2.562 - mean_eps: 0.678 - ale.lives: 2.272

Interval 73 (720000 steps performed)
10000/10000 [==============================] - 327s 33ms/step - reward: 0.0159
16 episodes - episode_reward: 9.562 [4.000, 21.000] - loss: 0.019 - mae: 2.130 -
mean_q: 2.584 - mean_eps: 0.674 - ale.lives: 2.110

Interval 74 (730000 steps performed)
10000/10000 [==============================] - 334s 33ms/step - reward: 0.0172
15 episodes - episode_reward: 11.067 [5.000, 19.000] - loss: 0.019 - mae: 2.152
- mean_q: 2.609 - mean_eps: 0.669 - ale.lives: 2.164

Interval 75 (740000 steps performed)
10000/10000 [==============================] - 326s 33ms/step - reward: 0.0181
15 episodes - episode_reward: 12.733 [5.000, 19.000] - loss: 0.020 - mae: 2.201
- mean_q: 2.667 - mean_eps: 0.665 - ale.lives: 2.144

Interval 76 (750000 steps performed)
10000/10000 [==============================] - 350s 35ms/step - reward: 0.0175
14 episodes - episode_reward: 12.429 [4.000, 29.000] - loss: 0.023 - mae: 2.254
- mean_q: 2.732 - mean_eps: 0.660 - ale.lives: 2.070

Interval 77 (760000 steps performed)
10000/10000 [==============================] - 358s 36ms/step - reward: 0.0181
```

```
12 episodes - episode_reward: 13.417 [7.000, 21.000] - loss: 0.021 - mae: 2.293
- mean_q: 2.777 - mean_eps: 0.656 - ale.lives: 2.105

Interval 78 (770000 steps performed)
10000/10000 [==============================] - 363s 36ms/step - reward: 0.0160
16 episodes - episode_reward: 11.500 [2.000, 25.000] - loss: 0.021 - mae: 2.318
- mean_q: 2.805 - mean_eps: 0.651 - ale.lives: 2.143

Interval 79 (780000 steps performed)
10000/10000 [==============================] - 350s 35ms/step - reward: 0.0174
13 episodes - episode_reward: 12.615 [4.000, 22.000] - loss: 0.020 - mae: 2.296
- mean_q: 2.778 - mean_eps: 0.647 - ale.lives: 2.110

Interval 80 (790000 steps performed)
10000/10000 [==============================] - 352s 35ms/step - reward: 0.0171
15 episodes - episode_reward: 12.133 [5.000, 21.000] - loss: 0.019 - mae: 2.304
- mean_q: 2.788 - mean_eps: 0.642 - ale.lives: 2.168

Interval 81 (800000 steps performed)
10000/10000 [==============================] - 364s 36ms/step - reward: 0.0172
13 episodes - episode_reward: 13.154 [6.000, 26.000] - loss: 0.021 - mae: 2.335
- mean_q: 2.825 - mean_eps: 0.638 - ale.lives: 2.288

Interval 82 (810000 steps performed)
10000/10000 [==============================] - 359s 36ms/step - reward: 0.0170
12 episodes - episode_reward: 13.000 [6.000, 21.000] - loss: 0.020 - mae: 2.338
- mean_q: 2.828 - mean_eps: 0.633 - ale.lives: 2.237

Interval 83 (820000 steps performed)
10000/10000 [==============================] - 370s 37ms/step - reward: 0.0186
12 episodes - episode_reward: 15.750 [7.000, 27.000] - loss: 0.022 - mae: 2.382
- mean_q: 2.881 - mean_eps: 0.629 - ale.lives: 2.008

Interval 84 (830000 steps performed)
10000/10000 [==============================] - 367s 37ms/step - reward: 0.0169
14 episodes - episode_reward: 12.357 [7.000, 22.000] - loss: 0.021 - mae: 2.399
- mean_q: 2.899 - mean_eps: 0.624 - ale.lives: 1.980

Interval 85 (840000 steps performed)
10000/10000 [==============================] - 396s 40ms/step - reward: 0.0165
13 episodes - episode_reward: 12.000 [6.000, 23.000] - loss: 0.021 - mae: 2.428
- mean_q: 2.933 - mean_eps: 0.620 - ale.lives: 2.060

Interval 86 (850000 steps performed)
10000/10000 [==============================] - 391s 39ms/step - reward: 0.0180
14 episodes - episode_reward: 13.929 [8.000, 21.000] - loss: 0.021 - mae: 2.432
- mean_q: 2.939 - mean_eps: 0.615 - ale.lives: 2.176
```

```
Interval 87 (860000 steps performed)
10000/10000 [==============================] - 386s 39ms/step - reward: 0.0172
14 episodes - episode_reward: 12.357 [7.000, 20.000] - loss: 0.022 - mae: 2.458
- mean_q: 2.971 - mean_eps: 0.611 - ale.lives: 2.139

Interval 88 (870000 steps performed)
10000/10000 [==============================] - 394s 39ms/step - reward: 0.0178
14 episodes - episode_reward: 12.571 [5.000, 23.000] - loss: 0.019 - mae: 2.468
- mean_q: 2.984 - mean_eps: 0.606 - ale.lives: 2.094

Interval 89 (880000 steps performed)
10000/10000 [==============================] - 397s 40ms/step - reward: 0.0189
14 episodes - episode_reward: 13.357 [7.000, 24.000] - loss: 0.021 - mae: 2.517
- mean_q: 3.045 - mean_eps: 0.602 - ale.lives: 2.052

Interval 90 (890000 steps performed)
10000/10000 [==============================] - 391s 39ms/step - reward: 0.0191
13 episodes - episode_reward: 14.538 [5.000, 21.000] - loss: 0.021 - mae: 2.544
- mean_q: 3.075 - mean_eps: 0.597 - ale.lives: 2.124

Interval 91 (900000 steps performed)
10000/10000 [==============================] - 385s 38ms/step - reward: 0.0193
13 episodes - episode_reward: 14.923 [8.000, 25.000] - loss: 0.021 - mae: 2.545
- mean_q: 3.074 - mean_eps: 0.593 - ale.lives: 2.248

Interval 92 (910000 steps performed)
10000/10000 [==============================] - 395s 39ms/step - reward: 0.0201
13 episodes - episode_reward: 15.538 [4.000, 25.000] - loss: 0.020 - mae: 2.550
- mean_q: 3.080 - mean_eps: 0.588 - ale.lives: 2.161

Interval 93 (920000 steps performed)
10000/10000 [==============================] - 372s 37ms/step - reward: 0.0200
12 episodes - episode_reward: 16.333 [9.000, 26.000] - loss: 0.020 - mae: 2.562
- mean_q: 3.095 - mean_eps: 0.584 - ale.lives: 2.067

Interval 94 (930000 steps performed)
10000/10000 [==============================] - 402s 40ms/step - reward: 0.0172
13 episodes - episode_reward: 12.769 [6.000, 20.000] - loss: 0.021 - mae: 2.585
- mean_q: 3.123 - mean_eps: 0.579 - ale.lives: 2.113

Interval 95 (940000 steps performed)
10000/10000 [==============================] - 393s 39ms/step - reward: 0.0190
14 episodes - episode_reward: 14.286 [7.000, 30.000] - loss: 0.024 - mae: 2.615
- mean_q: 3.163 - mean_eps: 0.575 - ale.lives: 2.084

Interval 96 (950000 steps performed)
10000/10000 [==============================] - 397s 40ms/step - reward: 0.0193
12 episodes - episode_reward: 15.833 [4.000, 27.000] - loss: 0.030 - mae: 2.617
```

```
- mean_q: 3.162 - mean_eps: 0.570 - ale.lives: 2.181

Interval 97 (960000 steps performed)
10000/10000 [==============================] - 469s 47ms/step - reward: 0.0198
12 episodes - episode_reward: 15.417 [8.000, 27.000] - loss: 0.022 - mae: 2.633
- mean_q: 3.179 - mean_eps: 0.566 - ale.lives: 2.099

Interval 98 (970000 steps performed)
10000/10000 [==============================] - 481s 48ms/step - reward: 0.0191
12 episodes - episode_reward: 17.333 [10.000, 26.000] - loss: 0.023 - mae: 2.675
- mean_q: 3.230 - mean_eps: 0.561 - ale.lives: 2.180

Interval 99 (980000 steps performed)
10000/10000 [==============================] - 461s 46ms/step - reward: 0.0195
12 episodes - episode_reward: 15.750 [5.000, 23.000] - loss: 0.022 - mae: 2.693
- mean_q: 3.250 - mean_eps: 0.557 - ale.lives: 2.060

Interval 100 (990000 steps performed)
10000/10000 [==============================] - 404s 40ms/step - reward: 0.0191
13 episodes - episode_reward: 14.923 [9.000, 24.000] - loss: 0.023 - mae: 2.684
- mean_q: 3.241 - mean_eps: 0.552 - ale.lives: 2.205

Interval 101 (1000000 steps performed)
10000/10000 [==============================] - 408s 41ms/step - reward: 0.0175
12 episodes - episode_reward: 15.167 [6.000, 21.000] - loss: 0.024 - mae: 2.722
- mean_q: 3.286 - mean_eps: 0.548 - ale.lives: 2.053

Interval 102 (1010000 steps performed)
10000/10000 [==============================] - 402s 40ms/step - reward: 0.0171
12 episodes - episode_reward: 13.500 [8.000, 24.000] - loss: 0.024 - mae: 2.789
- mean_q: 3.367 - mean_eps: 0.543 - ale.lives: 2.099

Interval 103 (1020000 steps performed)
10000/10000 [==============================] - 391s 39ms/step - reward: 0.0182
12 episodes - episode_reward: 15.333 [9.000, 22.000] - loss: 0.025 - mae: 2.795
- mean_q: 3.374 - mean_eps: 0.539 - ale.lives: 2.283

Interval 104 (1030000 steps performed)
10000/10000 [==============================] - 395s 39ms/step - reward: 0.0189
10 episodes - episode_reward: 18.800 [6.000, 30.000] - loss: 0.023 - mae: 2.793
- mean_q: 3.372 - mean_eps: 0.534 - ale.lives: 2.167

Interval 105 (1040000 steps performed)
10000/10000 [==============================] - 396s 40ms/step - reward: 0.0175
12 episodes - episode_reward: 14.750 [6.000, 31.000] - loss: 0.023 - mae: 2.791
- mean_q: 3.368 - mean_eps: 0.530 - ale.lives: 2.189

Interval 106 (1050000 steps performed)
```

```
10000/10000 [==============================] - 399s 40ms/step - reward: 0.0177
13 episodes - episode_reward: 14.077 [7.000, 27.000] - loss: 0.023 - mae: 2.806
- mean_q: 3.387 - mean_eps: 0.525 - ale.lives: 2.031

Interval 107 (1060000 steps performed)
10000/10000 [==============================] - 404s 40ms/step - reward: 0.0193
12 episodes - episode_reward: 15.083 [8.000, 26.000] - loss: 0.022 - mae: 2.813
- mean_q: 3.396 - mean_eps: 0.521 - ale.lives: 2.134

Interval 108 (1070000 steps performed)
10000/10000 [==============================] - 406s 41ms/step - reward: 0.0194
10 episodes - episode_reward: 18.900 [7.000, 28.000] - loss: 0.023 - mae: 2.845
- mean_q: 3.436 - mean_eps: 0.516 - ale.lives: 2.022

Interval 109 (1080000 steps performed)
10000/10000 [==============================] - 397s 40ms/step - reward: 0.0186
12 episodes - episode_reward: 16.250 [8.000, 21.000] - loss: 0.024 - mae: 2.839
- mean_q: 3.427 - mean_eps: 0.512 - ale.lives: 2.251

Interval 110 (1090000 steps performed)
10000/10000 [==============================] - 394s 39ms/step - reward: 0.0193
13 episodes - episode_reward: 14.615 [4.000, 27.000] - loss: 0.023 - mae: 2.843
- mean_q: 3.432 - mean_eps: 0.507 - ale.lives: 2.079

Interval 111 (1100000 steps performed)
10000/10000 [==============================] - 411s 41ms/step - reward: 0.0184
12 episodes - episode_reward: 16.000 [9.000, 27.000] - loss: 0.022 - mae: 2.868
- mean_q: 3.462 - mean_eps: 0.503 - ale.lives: 2.099

Interval 112 (1110000 steps performed)
10000/10000 [==============================] - 393s 39ms/step - reward: 0.0188
13 episodes - episode_reward: 13.615 [5.000, 21.000] - loss: 0.023 - mae: 2.909
- mean_q: 3.512 - mean_eps: 0.498 - ale.lives: 2.131

Interval 113 (1120000 steps performed)
10000/10000 [==============================] - 399s 40ms/step - reward: 0.0197
12 episodes - episode_reward: 16.917 [6.000, 32.000] - loss: 0.023 - mae: 2.912
- mean_q: 3.513 - mean_eps: 0.494 - ale.lives: 2.050

Interval 114 (1130000 steps performed)
10000/10000 [==============================] - 398s 40ms/step - reward: 0.0194
12 episodes - episode_reward: 15.167 [7.000, 24.000] - loss: 0.023 - mae: 2.915
- mean_q: 3.518 - mean_eps: 0.489 - ale.lives: 2.046

Interval 115 (1140000 steps performed)
10000/10000 [==============================] - 395s 40ms/step - reward: 0.0169
14 episodes - episode_reward: 12.071 [4.000, 27.000] - loss: 0.024 - mae: 2.931
- mean_q: 3.537 - mean_eps: 0.485 - ale.lives: 2.167
```

```
Interval 116 (1150000 steps performed)
10000/10000 [==============================] - 396s 40ms/step - reward: 0.0193
12 episodes - episode_reward: 16.167 [5.000, 24.000] - loss: 0.023 - mae: 2.927
- mean_q: 3.532 - mean_eps: 0.480 - ale.lives: 2.044

Interval 117 (1160000 steps performed)
10000/10000 [==============================] - 415s 41ms/step - reward: 0.0194
13 episodes - episode_reward: 15.846 [4.000, 26.000] - loss: 0.024 - mae: 2.964
- mean_q: 3.577 - mean_eps: 0.476 - ale.lives: 2.155

Interval 118 (1170000 steps performed)
10000/10000 [==============================] - 409s 41ms/step - reward: 0.0192
12 episodes - episode_reward: 15.750 [8.000, 28.000] - loss: 0.024 - mae: 2.998
- mean_q: 3.619 - mean_eps: 0.471 - ale.lives: 2.059

Interval 119 (1180000 steps performed)
10000/10000 [==============================] - 398s 40ms/step - reward: 0.0163
13 episodes - episode_reward: 12.769 [6.000, 18.000] - loss: 0.024 - mae: 3.032
- mean_q: 3.659 - mean_eps: 0.467 - ale.lives: 2.037

Interval 120 (1190000 steps performed)
10000/10000 [==============================] - 397s 40ms/step - reward: 0.0208
12 episodes - episode_reward: 17.500 [5.000, 28.000] - loss: 0.024 - mae: 3.060
- mean_q: 3.693 - mean_eps: 0.462 - ale.lives: 2.022

Interval 121 (1200000 steps performed)
10000/10000 [==============================] - 397s 40ms/step - reward: 0.0184
15 episodes - episode_reward: 12.467 [3.000, 26.000] - loss: 0.024 - mae: 3.089
- mean_q: 3.727 - mean_eps: 0.458 - ale.lives: 2.034

Interval 122 (1210000 steps performed)
10000/10000 [==============================] - 394s 39ms/step - reward: 0.0202
12 episodes - episode_reward: 16.500 [7.000, 30.000] - loss: 0.024 - mae: 3.138
- mean_q: 3.787 - mean_eps: 0.453 - ale.lives: 1.976

Interval 123 (1220000 steps performed)
10000/10000 [==============================] - 395s 39ms/step - reward: 0.0200
12 episodes - episode_reward: 16.917 [7.000, 29.000] - loss: 0.024 - mae: 3.154
- mean_q: 3.805 - mean_eps: 0.449 - ale.lives: 2.189

Interval 124 (1230000 steps performed)
10000/10000 [==============================] - 394s 39ms/step - reward: 0.0189
13 episodes - episode_reward: 14.769 [4.000, 29.000] - loss: 0.025 - mae: 3.186
- mean_q: 3.843 - mean_eps: 0.444 - ale.lives: 2.134

Interval 125 (1240000 steps performed)
10000/10000 [==============================] - 393s 39ms/step - reward: 0.0206
```

```
10 episodes - episode_reward: 19.600 [8.000, 33.000] - loss: 0.027 - mae: 3.203
- mean_q: 3.863 - mean_eps: 0.440 - ale.lives: 1.988

Interval 126 (1250000 steps performed)
10000/10000 [==============================] - 397s 40ms/step - reward: 0.0198
15 episodes - episode_reward: 13.800 [4.000, 25.000] - loss: 0.024 - mae: 3.180
- mean_q: 3.835 - mean_eps: 0.435 - ale.lives: 2.072

Interval 127 (1260000 steps performed)
10000/10000 [==============================] - 398s 40ms/step - reward: 0.0235
11 episodes - episode_reward: 20.636 [12.000, 30.000] - loss: 0.025 - mae: 3.220
- mean_q: 3.886 - mean_eps: 0.431 - ale.lives: 2.191

Interval 128 (1270000 steps performed)
10000/10000 [==============================] - 398s 40ms/step - reward: 0.0218
12 episodes - episode_reward: 17.833 [13.000, 28.000] - loss: 0.026 - mae: 3.235
- mean_q: 3.904 - mean_eps: 0.426 - ale.lives: 2.272

Interval 129 (1280000 steps performed)
10000/10000 [==============================] - 400s 40ms/step - reward: 0.0210
12 episodes - episode_reward: 17.917 [7.000, 34.000] - loss: 0.024 - mae: 3.268
- mean_q: 3.944 - mean_eps: 0.422 - ale.lives: 2.051

Interval 130 (1290000 steps performed)
10000/10000 [==============================] - 393s 39ms/step - reward: 0.0219
12 episodes - episode_reward: 17.500 [4.000, 31.000] - loss: 0.027 - mae: 3.269
- mean_q: 3.944 - mean_eps: 0.417 - ale.lives: 1.984

Interval 131 (1300000 steps performed)
10000/10000 [==============================] - 390s 39ms/step - reward: 0.0206
13 episodes - episode_reward: 16.538 [6.000, 25.000] - loss: 0.027 - mae: 3.289
- mean_q: 3.969 - mean_eps: 0.413 - ale.lives: 2.184

Interval 132 (1310000 steps performed)
10000/10000 [==============================] - 391s 39ms/step - reward: 0.0199
13 episodes - episode_reward: 15.615 [7.000, 27.000] - loss: 0.025 - mae: 3.283
- mean_q: 3.963 - mean_eps: 0.408 - ale.lives: 2.049

Interval 133 (1320000 steps performed)
10000/10000 [==============================] - 390s 39ms/step - reward: 0.0230
12 episodes - episode_reward: 19.083 [6.000, 34.000] - loss: 0.025 - mae: 3.260
- mean_q: 3.933 - mean_eps: 0.404 - ale.lives: 2.074

Interval 134 (1330000 steps performed)
10000/10000 [==============================] - 393s 39ms/step - reward: 0.0198
14 episodes - episode_reward: 13.929 [3.000, 32.000] - loss: 0.025 - mae: 3.284
- mean_q: 3.964 - mean_eps: 0.399 - ale.lives: 2.117
```

```
Interval 135 (1340000 steps performed)
10000/10000 [==============================] - 396s 40ms/step - reward: 0.0211
12 episodes - episode_reward: 16.417 [10.000, 30.000] - loss: 0.025 - mae: 3.284
- mean_q: 3.963 - mean_eps: 0.395 - ale.lives: 1.934

Interval 136 (1350000 steps performed)
10000/10000 [==============================] - 392s 39ms/step - reward: 0.0233
12 episodes - episode_reward: 19.333 [6.000, 31.000] - loss: 0.025 - mae: 3.310
- mean_q: 3.995 - mean_eps: 0.390 - ale.lives: 1.967

Interval 137 (1360000 steps performed)
10000/10000 [==============================] - 393s 39ms/step - reward: 0.0212
12 episodes - episode_reward: 19.250 [8.000, 32.000] - loss: 0.025 - mae: 3.360
- mean_q: 4.055 - mean_eps: 0.386 - ale.lives: 2.067

Interval 138 (1370000 steps performed)
10000/10000 [==============================] - 395s 39ms/step - reward: 0.0184
12 episodes - episode_reward: 15.667 [5.000, 24.000] - loss: 0.025 - mae: 3.392
- mean_q: 4.092 - mean_eps: 0.381 - ale.lives: 2.181

Interval 139 (1380000 steps performed)
10000/10000 [==============================] - 396s 40ms/step - reward: 0.0209
13 episodes - episode_reward: 15.846 [9.000, 25.000] - loss: 0.025 - mae: 3.402
- mean_q: 4.103 - mean_eps: 0.377 - ale.lives: 2.096

Interval 140 (1390000 steps performed)
10000/10000 [==============================] - 399s 40ms/step - reward: 0.0228
12 episodes - episode_reward: 19.250 [9.000, 28.000] - loss: 0.027 - mae: 3.448
- mean_q: 4.159 - mean_eps: 0.372 - ale.lives: 2.190

Interval 141 (1400000 steps performed)
10000/10000 [==============================] - 402s 40ms/step - reward: 0.0213
11 episodes - episode_reward: 18.455 [10.000, 31.000] - loss: 0.026 - mae: 3.450
- mean_q: 4.163 - mean_eps: 0.368 - ale.lives: 2.108

Interval 142 (1410000 steps performed)
10000/10000 [==============================] - 396s 40ms/step - reward: 0.0207
11 episodes - episode_reward: 18.364 [10.000, 34.000] - loss: 0.027 - mae: 3.470
- mean_q: 4.186 - mean_eps: 0.363 - ale.lives: 2.144

Interval 143 (1420000 steps performed)
10000/10000 [==============================] - 392s 39ms/step - reward: 0.0219
13 episodes - episode_reward: 17.308 [3.000, 31.000] - loss: 0.025 - mae: 3.485
- mean_q: 4.205 - mean_eps: 0.359 - ale.lives: 2.048

Interval 144 (1430000 steps performed)
10000/10000 [==============================] - 395s 40ms/step - reward: 0.0211
11 episodes - episode_reward: 19.091 [11.000, 30.000] - loss: 0.027 - mae: 3.494
```

```
- mean_q: 4.216 - mean_eps: 0.354 - ale.lives: 2.004

Interval 145 (1440000 steps performed)
10000/10000 [==============================] - 395s 40ms/step - reward: 0.0203
13 episodes - episode_reward: 15.154 [5.000, 31.000] - loss: 0.027 - mae: 3.537
- mean_q: 4.267 - mean_eps: 0.350 - ale.lives: 2.021

Interval 146 (1450000 steps performed)
10000/10000 [==============================] - 396s 40ms/step - reward: 0.0219
12 episodes - episode_reward: 18.917 [9.000, 28.000] - loss: 0.027 - mae: 3.539
- mean_q: 4.270 - mean_eps: 0.345 - ale.lives: 2.136

Interval 147 (1460000 steps performed)
10000/10000 [==============================] - 396s 40ms/step - reward: 0.0201
13 episodes - episode_reward: 15.538 [6.000, 30.000] - loss: 0.029 - mae: 3.578
- mean_q: 4.317 - mean_eps: 0.341 - ale.lives: 2.113

Interval 148 (1470000 steps performed)
10000/10000 [==============================] - 396s 40ms/step - reward: 0.0216
12 episodes - episode_reward: 17.333 [6.000, 24.000] - loss: 0.028 - mae: 3.603
- mean_q: 4.347 - mean_eps: 0.336 - ale.lives: 2.158

Interval 149 (1480000 steps performed)
10000/10000 [==============================] - 398s 40ms/step - reward: 0.0211
11 episodes - episode_reward: 19.636 [8.000, 35.000] - loss: 0.026 - mae: 3.576
- mean_q: 4.314 - mean_eps: 0.332 - ale.lives: 1.934

Interval 150 (1490000 steps performed)
10000/10000 [==============================] - 399s 40ms/step - reward: 0.0233
13 episodes - episode_reward: 17.692 [9.000, 29.000] - loss: 0.026 - mae: 3.589
- mean_q: 4.332 - mean_eps: 0.327 - ale.lives: 2.157

Interval 151 (1500000 steps performed)
10000/10000 [==============================] - 396s 40ms/step - reward: 0.0198
12 episodes - episode_reward: 16.833 [9.000, 27.000] - loss: 0.027 - mae: 3.617
- mean_q: 4.363 - mean_eps: 0.323 - ale.lives: 2.117

Interval 152 (1510000 steps performed)
10000/10000 [==============================] - 394s 39ms/step - reward: 0.0189
14 episodes - episode_reward: 14.071 [7.000, 22.000] - loss: 0.027 - mae: 3.637
- mean_q: 4.386 - mean_eps: 0.318 - ale.lives: 2.064

Interval 153 (1520000 steps performed)
10000/10000 [==============================] - 397s 40ms/step - reward: 0.0198
11 episodes - episode_reward: 16.273 [8.000, 28.000] - loss: 0.026 - mae: 3.661
- mean_q: 4.415 - mean_eps: 0.314 - ale.lives: 2.290

Interval 154 (1530000 steps performed)
```

```
10000/10000 [==============================] - 395s 40ms/step - reward: 0.0221
12 episodes - episode_reward: 18.583 [8.000, 31.000] - loss: 0.028 - mae: 3.683
- mean_q: 4.441 - mean_eps: 0.309 - ale.lives: 2.043

Interval 155 (1540000 steps performed)
10000/10000 [==============================] - 391s 39ms/step - reward: 0.0207
13 episodes - episode_reward: 17.000 [8.000, 27.000] - loss: 0.029 - mae: 3.698
- mean_q: 4.461 - mean_eps: 0.305 - ale.lives: 2.033

Interval 156 (1550000 steps performed)
10000/10000 [==============================] - 390s 39ms/step - reward: 0.0227
12 episodes - episode_reward: 17.333 [10.000, 35.000] - loss: 0.028 - mae: 3.693
- mean_q: 4.455 - mean_eps: 0.300 - ale.lives: 2.047

Interval 157 (1560000 steps performed)
10000/10000 [==============================] - 392s 39ms/step - reward: 0.0209
12 episodes - episode_reward: 19.000 [10.000, 30.000] - loss: 0.029 - mae: 3.746
- mean_q: 4.519 - mean_eps: 0.296 - ale.lives: 2.089

Interval 158 (1570000 steps performed)
10000/10000 [==============================] - 389s 39ms/step - reward: 0.0215
11 episodes - episode_reward: 18.636 [11.000, 27.000] - loss: 0.028 - mae: 3.746
- mean_q: 4.518 - mean_eps: 0.291 - ale.lives: 2.220

Interval 159 (1580000 steps performed)
10000/10000 [==============================] - 389s 39ms/step - reward: 0.0197
12 episodes - episode_reward: 16.250 [8.000, 31.000] - loss: 0.028 - mae: 3.781
- mean_q: 4.559 - mean_eps: 0.287 - ale.lives: 2.276

Interval 160 (1590000 steps performed)
10000/10000 [==============================] - 389s 39ms/step - reward: 0.0189
13 episodes - episode_reward: 15.692 [8.000, 33.000] - loss: 0.028 - mae: 3.784
- mean_q: 4.564 - mean_eps: 0.282 - ale.lives: 2.108

Interval 161 (1600000 steps performed)
10000/10000 [==============================] - 389s 39ms/step - reward: 0.0209
12 episodes - episode_reward: 15.667 [7.000, 27.000] - loss: 0.028 - mae: 3.787
- mean_q: 4.566 - mean_eps: 0.278 - ale.lives: 2.090

Interval 162 (1610000 steps performed)
10000/10000 [==============================] - 389s 39ms/step - reward: 0.0209
13 episodes - episode_reward: 17.769 [5.000, 27.000] - loss: 0.026 - mae: 3.814
- mean_q: 4.598 - mean_eps: 0.273 - ale.lives: 2.137

Interval 163 (1620000 steps performed)
10000/10000 [==============================] - 388s 39ms/step - reward: 0.0211
11 episodes - episode_reward: 19.091 [11.000, 32.000] - loss: 0.028 - mae: 3.850
- mean_q: 4.643 - mean_eps: 0.269 - ale.lives: 2.158
```

```
Interval 164 (1630000 steps performed)
10000/10000 [==============================] - 389s 39ms/step - reward: 0.0199
13 episodes - episode_reward: 15.308 [5.000, 30.000] - loss: 0.029 - mae: 3.876
- mean_q: 4.674 - mean_eps: 0.264 - ale.lives: 2.116

Interval 165 (1640000 steps performed)
10000/10000 [==============================] - 390s 39ms/step - reward: 0.0184
12 episodes - episode_reward: 14.500 [6.000, 22.000] - loss: 0.027 - mae: 3.891
- mean_q: 4.692 - mean_eps: 0.260 - ale.lives: 2.118

Interval 166 (1650000 steps performed)
10000/10000 [==============================] - 388s 39ms/step - reward: 0.0197
12 episodes - episode_reward: 17.250 [11.000, 24.000] - loss: 0.026 - mae: 3.893
- mean_q: 4.692 - mean_eps: 0.255 - ale.lives: 2.176

Interval 167 (1660000 steps performed)
10000/10000 [==============================] - 391s 39ms/step - reward: 0.0207
10 episodes - episode_reward: 19.300 [7.000, 25.000] - loss: 0.028 - mae: 3.931
- mean_q: 4.738 - mean_eps: 0.251 - ale.lives: 2.108

Interval 168 (1670000 steps performed)
10000/10000 [==============================] - 391s 39ms/step - reward: 0.0181
14 episodes - episode_reward: 13.714 [4.000, 25.000] - loss: 0.028 - mae: 3.944
- mean_q: 4.754 - mean_eps: 0.246 - ale.lives: 2.076

Interval 169 (1680000 steps performed)
10000/10000 [==============================] - 389s 39ms/step - reward: 0.0189
13 episodes - episode_reward: 13.538 [6.000, 25.000] - loss: 0.027 - mae: 3.959
- mean_q: 4.772 - mean_eps: 0.242 - ale.lives: 2.191

Interval 170 (1690000 steps performed)
10000/10000 [==============================] - 389s 39ms/step - reward: 0.0206
11 episodes - episode_reward: 19.727 [15.000, 27.000] - loss: 0.029 - mae: 3.947
- mean_q: 4.758 - mean_eps: 0.237 - ale.lives: 1.966

Interval 171 (1700000 steps performed)
10000/10000 [==============================] - 388s 39ms/step - reward: 0.0207
10 episodes - episode_reward: 20.600 [11.000, 30.000] - loss: 0.028 - mae: 3.943
- mean_q: 4.754 - mean_eps: 0.233 - ale.lives: 2.281

Interval 172 (1710000 steps performed)
10000/10000 [==============================] - 390s 39ms/step - reward: 0.0207
11 episodes - episode_reward: 18.727 [9.000, 25.000] - loss: 0.027 - mae: 3.949
- mean_q: 4.761 - mean_eps: 0.228 - ale.lives: 2.170

Interval 173 (1720000 steps performed)
10000/10000 [==============================] - 389s 39ms/step - reward: 0.0190
```

```
12 episodes - episode_reward: 15.000 [8.000, 21.000] - loss: 0.026 - mae: 3.949
- mean_q: 4.759 - mean_eps: 0.224 - ale.lives: 1.961

Interval 174 (1730000 steps performed)
10000/10000 [==============================] - 390s 39ms/step - reward: 0.0191
14 episodes - episode_reward: 14.571 [7.000, 27.000] - loss: 0.026 - mae: 3.959
- mean_q: 4.773 - mean_eps: 0.219 - ale.lives: 2.168

Interval 175 (1740000 steps performed)
10000/10000 [==============================] - 394s 39ms/step - reward: 0.0211
11 episodes - episode_reward: 19.636 [12.000, 24.000] - loss: 0.028 - mae: 3.951
- mean_q: 4.762 - mean_eps: 0.215 - ale.lives: 2.053

Interval 176 (1750000 steps performed)
10000/10000 [==============================] - 394s 39ms/step - reward: 0.0231
11 episodes - episode_reward: 18.636 [11.000, 31.000] - loss: 0.025 - mae: 3.986
- mean_q: 4.803 - mean_eps: 0.210 - ale.lives: 2.012

Interval 177 (1760000 steps performed)
10000/10000 [==============================] - 393s 39ms/step - reward: 0.0199
12 episodes - episode_reward: 16.750 [4.000, 26.000] - loss: 0.026 - mae: 3.957
- mean_q: 4.769 - mean_eps: 0.206 - ale.lives: 2.101

Interval 178 (1770000 steps performed)
10000/10000 [==============================] - 397s 40ms/step - reward: 0.0203
11 episodes - episode_reward: 19.909 [8.000, 30.000] - loss: 0.027 - mae: 3.947
- mean_q: 4.757 - mean_eps: 0.201 - ale.lives: 2.038

Interval 179 (1780000 steps performed)
10000/10000 [==============================] - 393s 39ms/step - reward: 0.0209
12 episodes - episode_reward: 17.833 [9.000, 32.000] - loss: 0.027 - mae: 3.981
- mean_q: 4.798 - mean_eps: 0.197 - ale.lives: 2.149

Interval 180 (1790000 steps performed)
10000/10000 [==============================] - 394s 39ms/step - reward: 0.0185
11 episodes - episode_reward: 15.000 [4.000, 27.000] - loss: 0.027 - mae: 4.001
- mean_q: 4.822 - mean_eps: 0.192 - ale.lives: 2.057

Interval 181 (1800000 steps performed)
10000/10000 [==============================] - 394s 39ms/step - reward: 0.0197
12 episodes - episode_reward: 17.667 [4.000, 36.000] - loss: 0.026 - mae: 4.000
- mean_q: 4.822 - mean_eps: 0.188 - ale.lives: 2.103

Interval 182 (1810000 steps performed)
10000/10000 [==============================] - 391s 39ms/step - reward: 0.0223
13 episodes - episode_reward: 16.385 [8.000, 25.000] - loss: 0.027 - mae: 4.025
- mean_q: 4.853 - mean_eps: 0.183 - ale.lives: 2.105
```

```
Interval 183 (1820000 steps performed)
10000/10000 [==============================] - 394s 39ms/step - reward: 0.0209
12 episodes - episode_reward: 18.917 [8.000, 30.000] - loss: 0.026 - mae: 4.069
- mean_q: 4.904 - mean_eps: 0.179 - ale.lives: 2.098

Interval 184 (1830000 steps performed)
10000/10000 [==============================] - 391s 39ms/step - reward: 0.0206
12 episodes - episode_reward: 15.667 [6.000, 26.000] - loss: 0.026 - mae: 4.064
- mean_q: 4.898 - mean_eps: 0.174 - ale.lives: 1.943

Interval 185 (1840000 steps performed)
10000/10000 [==============================] - 392s 39ms/step - reward: 0.0195
12 episodes - episode_reward: 16.500 [9.000, 32.000] - loss: 0.027 - mae: 4.060
- mean_q: 4.894 - mean_eps: 0.170 - ale.lives: 2.074

Interval 186 (1850000 steps performed)
10000/10000 [==============================] - 396s 40ms/step - reward: 0.0212
12 episodes - episode_reward: 18.167 [9.000, 28.000] - loss: 0.026 - mae: 4.075
- mean_q: 4.912 - mean_eps: 0.165 - ale.lives: 2.133

Interval 187 (1860000 steps performed)
10000/10000 [==============================] - 393s 39ms/step - reward: 0.0194
12 episodes - episode_reward: 16.667 [5.000, 26.000] - loss: 0.028 - mae: 4.091
- mean_q: 4.930 - mean_eps: 0.161 - ale.lives: 2.095

Interval 188 (1870000 steps performed)
10000/10000 [==============================] - 393s 39ms/step - reward: 0.0209
13 episodes - episode_reward: 15.923 [5.000, 27.000] - loss: 0.026 - mae: 4.073
- mean_q: 4.910 - mean_eps: 0.156 - ale.lives: 2.099

Interval 189 (1880000 steps performed)
10000/10000 [==============================] - 392s 39ms/step - reward: 0.0244
10 episodes - episode_reward: 23.000 [11.000, 32.000] - loss: 0.027 - mae: 4.087
- mean_q: 4.927 - mean_eps: 0.152 - ale.lives: 2.112

Interval 190 (1890000 steps performed)
10000/10000 [==============================] - 393s 39ms/step - reward: 0.0208
12 episodes - episode_reward: 18.333 [12.000, 30.000] - loss: 0.027 - mae: 4.089
- mean_q: 4.928 - mean_eps: 0.147 - ale.lives: 2.171

Interval 191 (1900000 steps performed)
10000/10000 [==============================] - 390s 39ms/step - reward: 0.0194
14 episodes - episode_reward: 14.214 [2.000, 25.000] - loss: 0.026 - mae: 4.092
- mean_q: 4.932 - mean_eps: 0.143 - ale.lives: 2.010

Interval 192 (1910000 steps performed)
10000/10000 [==============================] - 392s 39ms/step - reward: 0.0173
13 episodes - episode_reward: 12.923 [3.000, 21.000] - loss: 0.025 - mae: 4.063
```

```
- mean_q: 4.897 - mean_eps: 0.138 - ale.lives: 2.116

Interval 193 (1920000 steps performed)
10000/10000 [==============================] - 393s 39ms/step - reward: 0.0223
11 episodes - episode_reward: 19.364 [8.000, 34.000] - loss: 0.026 - mae: 4.074
- mean_q: 4.910 - mean_eps: 0.134 - ale.lives: 2.180

Interval 194 (1930000 steps performed)
10000/10000 [==============================] - 393s 39ms/step - reward: 0.0199
12 episodes - episode_reward: 16.000 [6.000, 26.000] - loss: 0.025 - mae: 4.023
- mean_q: 4.849 - mean_eps: 0.129 - ale.lives: 2.046

Interval 195 (1940000 steps performed)
10000/10000 [==============================] - 418s 42ms/step - reward: 0.0215
13 episodes - episode_reward: 17.385 [6.000, 28.000] - loss: 0.025 - mae: 4.047
- mean_q: 4.878 - mean_eps: 0.125 - ale.lives: 1.936

Interval 196 (1950000 steps performed)
10000/10000 [==============================] - 444s 44ms/step - reward: 0.0211
11 episodes - episode_reward: 19.000 [9.000, 27.000] - loss: 0.027 - mae: 4.065
- mean_q: 4.902 - mean_eps: 0.120 - ale.lives: 2.154

Interval 197 (1960000 steps performed)
10000/10000 [==============================] - 497s 50ms/step - reward: 0.0201
14 episodes - episode_reward: 14.714 [4.000, 26.000] - loss: 0.027 - mae: 4.064
- mean_q: 4.899 - mean_eps: 0.116 - ale.lives: 2.168

Interval 198 (1970000 steps performed)
10000/10000 [==============================] - 501s 50ms/step - reward: 0.0230
13 episodes - episode_reward: 17.615 [7.000, 29.000] - loss: 0.025 - mae: 4.048
- mean_q: 4.879 - mean_eps: 0.111 - ale.lives: 2.262

Interval 199 (1980000 steps performed)
10000/10000 [==============================] - 501s 50ms/step - reward: 0.0201
9 episodes - episode_reward: 21.333 [11.000, 30.000] - loss: 0.025 - mae: 4.051
- mean_q: 4.881 - mean_eps: 0.107 - ale.lives: 2.207

Interval 200 (1990000 steps performed)
10000/10000 [==============================] - 475s 47ms/step - reward: 0.0210
done, took 70391.034 seconds
```

[ ]:

```
# Testing part to calculate the mean reward
weights_filename = 'dqn_{}_weights.h5f'.format(env_name)
dqn.load_weights(weights_filename)
dqn.test(env, nb_episodes=10, visualize=False)
```

```
Testing for 10 episodes …
Episode 1: reward: 24.000, steps: 1175
```

```
Episode 2: reward: 13.000, steps: 863
Episode 3: reward: 18.000, steps: 993
Episode 4: reward: 25.000, steps: 1249
Episode 5: reward: 13.000, steps: 848
Episode 6: reward: 8.000, steps: 603
Episode 7: reward: 6.000, steps: 485
Episode 8: reward: 14.000, steps: 942
Episode 9: reward: 5.000, steps: 466
Episode 10: reward: 6.000, steps: 389
```

[ ]: <keras.src.callbacks.History at 0x7c36c417f310>

3.3.3 Generación de vídeo del agente base Esta sección permite grabar una ejecución del agente.

[ ]:
```
import tensorflow as tf
tf.config.list_physical_devices('GPU')
```

[ ]: [PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU')]

[ ]:
```
# Testing part
from gym import wrappers

if IN_COLAB:
    env = wrappers.Monitor(env, "/content/drive/", force=True)
else:
    env = wrappers.Monitor(env, "./videos_grabados/", force=True)
env.reset()

weights_filename = 'dqn_{}_weights.h5f'.format(env_name)
dqn.load_weights(weights_filename)
dqn.test(env, nb_episodes=10, visualize=False)
```

```
Testing for 10 episodes …
Episode 1: reward: 15.000, steps: 934
Episode 2: reward: 34.000, steps: 1342
Episode 3: reward: 3.000, steps: 363
Episode 4: reward: 11.000, steps: 804
Episode 5: reward: 9.000, steps: 601
Episode 6: reward: 23.000, steps: 1133
Episode 7: reward: 9.000, steps: 723
Episode 8: reward: 7.000, steps: 634
Episode 9: reward: 12.000, steps: 768
Episode 10: reward: 18.000, steps: 1166
```

[ ]: <keras.src.callbacks.History at 0x7c3845658650>

```
[ ]: import io
import base64
from IPython.display import HTML

if IN_COLAB:
    video = io.open('/content/drive/openaigym.video.%s.video000008.mp4' % env.
 ↪file_infix, 'r+b').read()
else:
    video = io.open('./videos_grabados/openaigym.video.%s.video000008.mp4' %␣
 ↪env.file_infix, 'r+b').read()

encoded = base64.b64encode(video)
HTML(data='''
    <video width="360" height="auto" alt="test" controls><source src="data:
 ↪video/mp4;base64,{0}" type="video/mp4" /></video>'''
.format(encoded.decode('ascii')))
```

```
[ ]: <IPython.core.display.HTML object>
```

1.4.4 3.4 Propuesta 1: Double DQN (DDQN)

Como primera propuesta de mejora utilizaremos Double DQN (DDQN). El fundamento de esta evolución es corregir una de las debilidades clásicas de DQN: **la sobreestimación de los valores Q**.

En DQN estándar, la misma red se usa indirectamente para seleccionar la mejor acción y para estimar su valor, lo que puede propiciar que el agente favorezca acciones cuyo valor ha sido estimado de forma demasiado optimista por factores como ruido o inestabilidad durante el entrenamiento.

En Double DQN se trata de corregir este efecto separando ambos pasos: la red principal selecciona la acción y la red objetivo estima su valor. Esto puede conducir a un entrenamiento más estable, reduciendo las oscilaciones y con mayor capacidad de generalización de la política aprendida, lo cual es especialmente aplicable a entornos de tipo Atari, donde las recompensas son ruidosas, escasas y dependientes de una larga secuencia de decisiones previas.

Implementación: se instancia el agente activando el parámetro nativo `enable_double_dqn=True`.

3.4.1 Implementación de la red neuronal Se reutiliza la arquitectura del caso base para aislar el efecto de Double DQN.

3.4.2 Implementación del agente DQN

```
[ ]: # Configuración de la memoria del algoritmo:
#    limit:          número máximo de transacciones
#    window_length: fotogramas consecutivos utilizados para formar un estado
memory_propuesta_1 = SequentialMemory(limit=1000000,␣
 ↪window_length=WINDOW_LENGTH)
# Procesado de imágenes: convertir imagenes a escala de grises y normalización␣
 ↪de los píxeles
```

```python
processor_propuesta_1 = AtariProcessor()

# Estrategia de exploración:
#    EpsGreedyQPolicy: estrategia basada en epsilon
#    attr:             atributo epsilon
#    value_max:        valor inicial (máximo)
#    value_min:        valor mínimo (final) con el tiempo irá disminuyendo␣
↪hasta este valor
#    value_test:       valor eps para el modo de prueba
#    nb_steps:         número de pasos, la disminución de eps será gradual␣
↪durante este número de pasos
policy_propuesta_1 = LinearAnnealedPolicy(EpsGreedyQPolicy(), attr='eps',
                              value_max=1., value_min=.1, value_test=.05,
                              nb_steps=2000000)

# Configuración del agente DQN
#    model:               red neuronal
#    nb_actions:          número de acciones del juego
#    policy:              estrategia de exploración
#    memory:              memoria del algoritmo
#    processor:           procesado de imágenes
#    nb_steps_warmup:     número de pasos en los que el algoritmo no entrena,␣
↪solo juega para llenar la memoria con experiencias iniciales
#    gamma:               factor de actualización de las recompensas
#    target_model_update: número de pasos hasta actualizar los valores de la␣
↪red de predicción a la red de target
#    train_interval:      la red se entrena en intervalo de fotogramas para␣
↪optimización y alineamiento con los fotogramas del estado
#    enable_double_dqn:   DDQN
dqn_propuesta_1 = DQNAgent(model=model, nb_actions=nb_actions,␣
↪policy=policy_propuesta_1,
               memory=memory_propuesta_1, processor=processor_propuesta_1,
               nb_steps_warmup=50000, gamma=.99,
               target_model_update=10000,
               train_interval=4, enable_double_dqn=True)

# Compilación del agente
#    Adam:          optimizador Adam
#    learning_rate: tasa de aprendizaje
#    metrics:       métrica MAE (Error Absoluto Medio)
dqn_propuesta_1.compile(Adam(learning_rate=.00025), metrics=['mae'])
```

[ ]:
```python
# Training part
weights_filename = 'dqn_{}_weights.h5f'.format(env_name)
checkpoint_weights_filename = 'dqn_' + env_name + '_weights_{step}.h5f'
log_filename = 'dqn_{}_log.json'.format(env_name)
```

```python
callbacks = [ModelIntervalCheckpoint(checkpoint_weights_filename,␣
 ↪interval=100000)]
callbacks += [FileLogger(log_filename, interval=100)]

# Comenzar el entrenamiento
dqn_propuesta_1.fit(env, callbacks=callbacks, nb_steps=1000000,␣
 ↪log_interval=10000, visualize=False)
#dqn_propuesta_1.fit(env, callbacks=callbacks, nb_steps=20000,␣
 ↪log_interval=100, visualize=False)

dqn_propuesta_1.save_weights(weights_filename, overwrite=True)
```

```
Training for 1000000 steps …
Interval 1 (0 steps performed)
```

```
/home/mrlyon/miniconda3/envs/miar_rl/lib/python3.11/site-
packages/keras/src/engine/training_v1.py:2359: UserWarning:
`Model.state_updates` will be removed in a future version. This property should
not be used in TensorFlow 2.0, as `updates` are applied automatically.
  updates=self.state_updates,
```

```
10000/10000 [==============================] - 41s 4ms/step - reward: 0.0126
16 episodes - episode_reward: 7.500 [4.000, 12.000] - ale.lives: 2.247

Interval 2 (10000 steps performed)
10000/10000 [==============================] - 40s 4ms/step - reward: 0.0133
15 episodes - episode_reward: 8.800 [3.000, 16.000] - ale.lives: 1.988

Interval 3 (20000 steps performed)
10000/10000 [==============================] - 41s 4ms/step - reward: 0.0138
14 episodes - episode_reward: 9.214 [3.000, 22.000] - ale.lives: 2.231

Interval 4 (30000 steps performed)
10000/10000 [==============================] - 35s 3ms/step - reward: 0.0136
15 episodes - episode_reward: 9.800 [5.000, 18.000] - ale.lives: 2.177

Interval 5 (40000 steps performed)
10000/10000 [==============================] - 33s 3ms/step - reward: 0.0140
16 episodes - episode_reward: 8.750 [4.000, 18.000] - ale.lives: 2.098

Interval 6 (50000 steps performed)
10000/10000 [==============================] - 253s 25ms/step - reward: 0.0133
15 episodes - episode_reward: 8.867 [2.000, 26.000] - loss: 0.006 - mae: 0.016 -
mean_q: 0.021 - mean_eps: 0.975 - ale.lives: 2.274

Interval 7 (60000 steps performed)
10000/10000 [==============================] - 254s 25ms/step - reward: 0.0122
15 episodes - episode_reward: 8.467 [3.000, 16.000] - loss: 0.006 - mae: 0.030 -
```

```
mean_q: 0.039 - mean_eps: 0.971 - ale.lives: 2.056

Interval 8 (70000 steps performed)
10000/10000 [==============================] - 255s 25ms/step - reward: 0.0150
15 episodes - episode_reward: 9.600 [4.000, 16.000] - loss: 0.007 - mae: 0.042 -
mean_q: 0.054 - mean_eps: 0.966 - ale.lives: 2.106

Interval 9 (80000 steps performed)
10000/10000 [==============================] - 254s 25ms/step - reward: 0.0104
16 episodes - episode_reward: 6.812 [3.000, 18.000] - loss: 0.007 - mae: 0.058 -
mean_q: 0.075 - mean_eps: 0.962 - ale.lives: 2.092

Interval 10 (90000 steps performed)
10000/10000 [==============================] - 254s 25ms/step - reward: 0.0131
13 episodes - episode_reward: 10.154 [0.000, 21.000] - loss: 0.006 - mae: 0.070
- mean_q: 0.090 - mean_eps: 0.957 - ale.lives: 2.047

Interval 11 (100000 steps performed)
10000/10000 [==============================] - 254s 25ms/step - reward: 0.0129
15 episodes - episode_reward: 8.400 [2.000, 19.000] - loss: 0.006 - mae: 0.093 -
mean_q: 0.120 - mean_eps: 0.953 - ale.lives: 2.121

Interval 12 (110000 steps performed)
10000/10000 [==============================] - 256s 26ms/step - reward: 0.0128
16 episodes - episode_reward: 8.188 [3.000, 12.000] - loss: 0.007 - mae: 0.110 -
mean_q: 0.140 - mean_eps: 0.948 - ale.lives: 2.015

Interval 13 (120000 steps performed)
10000/10000 [==============================] - 257s 26ms/step - reward: 0.0145
13 episodes - episode_reward: 10.385 [5.000, 21.000] - loss: 0.006 - mae: 0.124
- mean_q: 0.157 - mean_eps: 0.944 - ale.lives: 2.171

Interval 14 (130000 steps performed)
10000/10000 [==============================] - 256s 26ms/step - reward: 0.0123
15 episodes - episode_reward: 8.667 [5.000, 14.000] - loss: 0.006 - mae: 0.143 -
mean_q: 0.180 - mean_eps: 0.939 - ale.lives: 2.120

Interval 15 (140000 steps performed)
10000/10000 [==============================] - 259s 26ms/step - reward: 0.0136
16 episodes - episode_reward: 8.625 [5.000, 21.000] - loss: 0.006 - mae: 0.152 -
mean_q: 0.192 - mean_eps: 0.935 - ale.lives: 2.085

Interval 16 (150000 steps performed)
10000/10000 [==============================] - 258s 26ms/step - reward: 0.0122
14 episodes - episode_reward: 8.357 [3.000, 15.000] - loss: 0.006 - mae: 0.168 -
mean_q: 0.211 - mean_eps: 0.930 - ale.lives: 2.094

Interval 17 (160000 steps performed)
```

```
10000/10000 [==============================] - 261s 26ms/step - reward: 0.0124
12 episodes - episode_reward: 9.583 [2.000, 23.000] - loss: 0.006 - mae: 0.190 -
mean_q: 0.238 - mean_eps: 0.926 - ale.lives: 2.194

Interval 18 (170000 steps performed)
10000/10000 [==============================] - 262s 26ms/step - reward: 0.0129
15 episodes - episode_reward: 9.200 [4.000, 15.000] - loss: 0.006 - mae: 0.195 -
mean_q: 0.244 - mean_eps: 0.921 - ale.lives: 2.062

Interval 19 (180000 steps performed)
10000/10000 [==============================] - 264s 26ms/step - reward: 0.0139
15 episodes - episode_reward: 9.533 [5.000, 16.000] - loss: 0.006 - mae: 0.207 -
mean_q: 0.259 - mean_eps: 0.917 - ale.lives: 2.032

Interval 20 (190000 steps performed)
10000/10000 [==============================] - 267s 27ms/step - reward: 0.0129
15 episodes - episode_reward: 8.067 [5.000, 13.000] - loss: 0.006 - mae: 0.228 -
mean_q: 0.283 - mean_eps: 0.912 - ale.lives: 2.110

Interval 21 (200000 steps performed)
10000/10000 [==============================] - 267s 27ms/step - reward: 0.0130
13 episodes - episode_reward: 9.615 [2.000, 22.000] - loss: 0.007 - mae: 0.242 -
mean_q: 0.301 - mean_eps: 0.908 - ale.lives: 2.007

Interval 22 (210000 steps performed)
10000/10000 [==============================] - 268s 27ms/step - reward: 0.0146
13 episodes - episode_reward: 11.308 [4.000, 23.000] - loss: 0.007 - mae: 0.261
- mean_q: 0.324 - mean_eps: 0.903 - ale.lives: 1.944

Interval 23 (220000 steps performed)
10000/10000 [==============================] - 269s 27ms/step - reward: 0.0147
16 episodes - episode_reward: 9.500 [2.000, 18.000] - loss: 0.008 - mae: 0.286 -
mean_q: 0.355 - mean_eps: 0.899 - ale.lives: 2.092

Interval 24 (230000 steps performed)
10000/10000 [==============================] - 272s 27ms/step - reward: 0.0134
14 episodes - episode_reward: 9.286 [3.000, 27.000] - loss: 0.008 - mae: 0.310 -
mean_q: 0.386 - mean_eps: 0.894 - ale.lives: 2.061

Interval 25 (240000 steps performed)
10000/10000 [==============================] - 272s 27ms/step - reward: 0.0159
14 episodes - episode_reward: 11.643 [8.000, 22.000] - loss: 0.008 - mae: 0.335
- mean_q: 0.417 - mean_eps: 0.890 - ale.lives: 2.045

Interval 26 (250000 steps performed)
10000/10000 [==============================] - 274s 27ms/step - reward: 0.0150
14 episodes - episode_reward: 11.071 [3.000, 23.000] - loss: 0.008 - mae: 0.348
- mean_q: 0.433 - mean_eps: 0.885 - ale.lives: 2.166
```

```
Interval 27 (260000 steps performed)
10000/10000 [==============================] - 276s 28ms/step - reward: 0.0143
14 episodes - episode_reward: 10.071 [3.000, 18.000] - loss: 0.008 - mae: 0.359
- mean_q: 0.447 - mean_eps: 0.881 - ale.lives: 1.994

Interval 28 (270000 steps performed)
10000/10000 [==============================] - 278s 28ms/step - reward: 0.0131
13 episodes - episode_reward: 10.385 [6.000, 17.000] - loss: 0.008 - mae: 0.377
- mean_q: 0.469 - mean_eps: 0.876 - ale.lives: 2.140

Interval 29 (280000 steps performed)
10000/10000 [==============================] - 276s 28ms/step - reward: 0.0161
15 episodes - episode_reward: 10.333 [3.000, 19.000] - loss: 0.008 - mae: 0.396
- mean_q: 0.493 - mean_eps: 0.872 - ale.lives: 2.118

Interval 30 (290000 steps performed)
10000/10000 [==============================] - 279s 28ms/step - reward: 0.0154
17 episodes - episode_reward: 8.765 [5.000, 14.000] - loss: 0.008 - mae: 0.418 -
mean_q: 0.521 - mean_eps: 0.867 - ale.lives: 2.225

Interval 31 (300000 steps performed)
10000/10000 [==============================] - 280s 28ms/step - reward: 0.0149
15 episodes - episode_reward: 10.533 [6.000, 19.000] - loss: 0.008 - mae: 0.444
- mean_q: 0.552 - mean_eps: 0.863 - ale.lives: 2.032

Interval 32 (310000 steps performed)
10000/10000 [==============================] - 281s 28ms/step - reward: 0.0179
12 episodes - episode_reward: 14.417 [8.000, 28.000] - loss: 0.009 - mae: 0.471
- mean_q: 0.587 - mean_eps: 0.858 - ale.lives: 2.096

Interval 33 (320000 steps performed)
10000/10000 [==============================] - 283s 28ms/step - reward: 0.0154
16 episodes - episode_reward: 10.188 [5.000, 16.000] - loss: 0.009 - mae: 0.503
- mean_q: 0.625 - mean_eps: 0.854 - ale.lives: 2.063

Interval 34 (330000 steps performed)
10000/10000 [==============================] - 285s 28ms/step - reward: 0.0139
15 episodes - episode_reward: 8.667 [4.000, 18.000] - loss: 0.009 - mae: 0.530 -
mean_q: 0.658 - mean_eps: 0.849 - ale.lives: 2.178

Interval 35 (340000 steps performed)
10000/10000 [==============================] - 284s 28ms/step - reward: 0.0178
14 episodes - episode_reward: 12.857 [7.000, 22.000] - loss: 0.010 - mae: 0.566
- mean_q: 0.703 - mean_eps: 0.845 - ale.lives: 2.092

Interval 36 (350000 steps performed)
10000/10000 [==============================] - 286s 29ms/step - reward: 0.0175
```

```
16 episodes - episode_reward: 11.438 [6.000, 21.000] - loss: 0.009 - mae: 0.596
- mean_q: 0.739 - mean_eps: 0.840 - ale.lives: 2.094

Interval 37 (360000 steps performed)
10000/10000 [==============================] - 288s 29ms/step - reward: 0.0161
16 episodes - episode_reward: 9.688 [3.000, 19.000] - loss: 0.010 - mae: 0.634 -
mean_q: 0.785 - mean_eps: 0.836 - ale.lives: 2.180

Interval 38 (370000 steps performed)
10000/10000 [==============================] - 289s 29ms/step - reward: 0.0152
15 episodes - episode_reward: 10.533 [4.000, 22.000] - loss: 0.010 - mae: 0.674
- mean_q: 0.834 - mean_eps: 0.831 - ale.lives: 2.218

Interval 39 (380000 steps performed)
10000/10000 [==============================] - 293s 29ms/step - reward: 0.0181
14 episodes - episode_reward: 12.643 [3.000, 23.000] - loss: 0.010 - mae: 0.698
- mean_q: 0.862 - mean_eps: 0.827 - ale.lives: 2.117

Interval 40 (390000 steps performed)
10000/10000 [==============================] - 294s 29ms/step - reward: 0.0153
15 episodes - episode_reward: 10.267 [5.000, 21.000] - loss: 0.011 - mae: 0.723
- mean_q: 0.892 - mean_eps: 0.822 - ale.lives: 2.195

Interval 41 (400000 steps performed)
10000/10000 [==============================] - 294s 29ms/step - reward: 0.0176
14 episodes - episode_reward: 12.500 [5.000, 25.000] - loss: 0.011 - mae: 0.743
- mean_q: 0.917 - mean_eps: 0.818 - ale.lives: 2.027

Interval 42 (410000 steps performed)
10000/10000 [==============================] - 295s 30ms/step - reward: 0.0166
15 episodes - episode_reward: 11.333 [5.000, 20.000] - loss: 0.011 - mae: 0.781
- mean_q: 0.962 - mean_eps: 0.813 - ale.lives: 2.219

Interval 43 (420000 steps performed)
10000/10000 [==============================] - 298s 30ms/step - reward: 0.0148
13 episodes - episode_reward: 11.000 [5.000, 18.000] - loss: 0.011 - mae: 0.791
- mean_q: 0.975 - mean_eps: 0.809 - ale.lives: 2.090

Interval 44 (430000 steps performed)
10000/10000 [==============================] - 296s 30ms/step - reward: 0.0172
14 episodes - episode_reward: 12.000 [5.000, 21.000] - loss: 0.011 - mae: 0.823
- mean_q: 1.015 - mean_eps: 0.804 - ale.lives: 2.047

Interval 45 (440000 steps performed)
10000/10000 [==============================] - 298s 30ms/step - reward: 0.0176
16 episodes - episode_reward: 11.562 [5.000, 18.000] - loss: 0.012 - mae: 0.867
- mean_q: 1.067 - mean_eps: 0.800 - ale.lives: 2.087
```

```
Interval 46 (450000 steps performed)
10000/10000 [==============================] - 301s 30ms/step - reward: 0.0164
12 episodes - episode_reward: 12.000 [4.000, 25.000] - loss: 0.012 - mae: 0.890
- mean_q: 1.094 - mean_eps: 0.795 - ale.lives: 2.093

Interval 47 (460000 steps performed)
10000/10000 [==============================] - 302s 30ms/step - reward: 0.0176
15 episodes - episode_reward: 12.733 [7.000, 21.000] - loss: 0.012 - mae: 0.916
- mean_q: 1.126 - mean_eps: 0.791 - ale.lives: 2.190

Interval 48 (470000 steps performed)
10000/10000 [==============================] - 301s 30ms/step - reward: 0.0165
15 episodes - episode_reward: 10.533 [5.000, 19.000] - loss: 0.012 - mae: 0.949
- mean_q: 1.165 - mean_eps: 0.786 - ale.lives: 2.244

Interval 49 (480000 steps performed)
10000/10000 [==============================] - 306s 31ms/step - reward: 0.0175
15 episodes - episode_reward: 12.333 [6.000, 25.000] - loss: 0.013 - mae: 1.006
- mean_q: 1.233 - mean_eps: 0.782 - ale.lives: 2.162

Interval 50 (490000 steps performed)
10000/10000 [==============================] - 307s 31ms/step - reward: 0.0191
13 episodes - episode_reward: 14.231 [6.000, 22.000] - loss: 0.013 - mae: 1.047
- mean_q: 1.281 - mean_eps: 0.777 - ale.lives: 2.117

Interval 51 (500000 steps performed)
10000/10000 [==============================] - 309s 31ms/step - reward: 0.0190
14 episodes - episode_reward: 14.000 [4.000, 28.000] - loss: 0.013 - mae: 1.072
- mean_q: 1.312 - mean_eps: 0.773 - ale.lives: 2.134

Interval 52 (510000 steps performed)
10000/10000 [==============================] - 311s 31ms/step - reward: 0.0173
16 episodes - episode_reward: 10.562 [5.000, 19.000] - loss: 0.013 - mae: 1.109
- mean_q: 1.355 - mean_eps: 0.768 - ale.lives: 2.205

Interval 53 (520000 steps performed)
10000/10000 [==============================] - 311s 31ms/step - reward: 0.0165
15 episodes - episode_reward: 11.067 [4.000, 17.000] - loss: 0.013 - mae: 1.124
- mean_q: 1.373 - mean_eps: 0.764 - ale.lives: 2.049

Interval 54 (530000 steps performed)
10000/10000 [==============================] - 310s 31ms/step - reward: 0.0165
16 episodes - episode_reward: 10.125 [4.000, 22.000] - loss: 0.013 - mae: 1.161
- mean_q: 1.418 - mean_eps: 0.759 - ale.lives: 2.184

Interval 55 (540000 steps performed)
10000/10000 [==============================] - 313s 31ms/step - reward: 0.0178
15 episodes - episode_reward: 11.667 [8.000, 16.000] - loss: 0.013 - mae: 1.157
```

```
- mean_q: 1.413 - mean_eps: 0.755 - ale.lives: 2.050

Interval 56 (550000 steps performed)
10000/10000 [==============================] - 317s 32ms/step - reward: 0.0180
15 episodes - episode_reward: 12.267 [5.000, 21.000] - loss: 0.013 - mae: 1.176
- mean_q: 1.435 - mean_eps: 0.750 - ale.lives: 2.090

Interval 57 (560000 steps performed)
10000/10000 [==============================] - 318s 32ms/step - reward: 0.0187
13 episodes - episode_reward: 14.308 [5.000, 22.000] - loss: 0.013 - mae: 1.192
- mean_q: 1.454 - mean_eps: 0.746 - ale.lives: 2.175

Interval 58 (570000 steps performed)
10000/10000 [==============================] - 321s 32ms/step - reward: 0.0182
16 episodes - episode_reward: 11.875 [4.000, 22.000] - loss: 0.013 - mae: 1.211
- mean_q: 1.476 - mean_eps: 0.741 - ale.lives: 2.174

Interval 59 (580000 steps performed)
10000/10000 [==============================] - 322s 32ms/step - reward: 0.0183
15 episodes - episode_reward: 11.467 [7.000, 17.000] - loss: 0.013 - mae: 1.234
- mean_q: 1.504 - mean_eps: 0.737 - ale.lives: 2.189

Interval 60 (590000 steps performed)
10000/10000 [==============================] - 320s 32ms/step - reward: 0.0179
15 episodes - episode_reward: 12.333 [7.000, 20.000] - loss: 0.013 - mae: 1.247
- mean_q: 1.519 - mean_eps: 0.732 - ale.lives: 2.095

Interval 61 (600000 steps performed)
10000/10000 [==============================] - 324s 32ms/step - reward: 0.0178
17 episodes - episode_reward: 10.588 [4.000, 20.000] - loss: 0.014 - mae: 1.258
- mean_q: 1.533 - mean_eps: 0.728 - ale.lives: 2.176

Interval 62 (610000 steps performed)
10000/10000 [==============================] - 325s 32ms/step - reward: 0.0172
14 episodes - episode_reward: 12.071 [7.000, 21.000] - loss: 0.014 - mae: 1.284
- mean_q: 1.565 - mean_eps: 0.723 - ale.lives: 2.189

Interval 63 (620000 steps performed)
10000/10000 [==============================] - 326s 33ms/step - reward: 0.0203
12 episodes - episode_reward: 16.500 [7.000, 27.000] - loss: 0.014 - mae: 1.322
- mean_q: 1.610 - mean_eps: 0.719 - ale.lives: 2.123

Interval 64 (630000 steps performed)
10000/10000 [==============================] - 328s 33ms/step - reward: 0.0168
14 episodes - episode_reward: 12.357 [5.000, 22.000] - loss: 0.014 - mae: 1.350
- mean_q: 1.643 - mean_eps: 0.714 - ale.lives: 2.163

Interval 65 (640000 steps performed)
```

```
10000/10000 [==============================] - 329s 33ms/step - reward: 0.0185
15 episodes - episode_reward: 12.400 [4.000, 19.000] - loss: 0.014 - mae: 1.361
- mean_q: 1.656 - mean_eps: 0.710 - ale.lives: 2.155

Interval 66 (650000 steps performed)
10000/10000 [==============================] - 334s 33ms/step - reward: 0.0183
13 episodes - episode_reward: 13.308 [7.000, 24.000] - loss: 0.014 - mae: 1.357
- mean_q: 1.650 - mean_eps: 0.705 - ale.lives: 2.203

Interval 67 (660000 steps performed)
10000/10000 [==============================] - 334s 33ms/step - reward: 0.0190
14 episodes - episode_reward: 13.929 [6.000, 23.000] - loss: 0.014 - mae: 1.378
- mean_q: 1.676 - mean_eps: 0.701 - ale.lives: 2.182

Interval 68 (670000 steps performed)
10000/10000 [==============================] - 336s 34ms/step - reward: 0.0174
16 episodes - episode_reward: 11.500 [5.000, 28.000] - loss: 0.014 - mae: 1.404
- mean_q: 1.706 - mean_eps: 0.696 - ale.lives: 2.215

Interval 69 (680000 steps performed)
10000/10000 [==============================] - 337s 34ms/step - reward: 0.0194
13 episodes - episode_reward: 14.615 [6.000, 24.000] - loss: 0.015 - mae: 1.437
- mean_q: 1.746 - mean_eps: 0.692 - ale.lives: 2.130

Interval 70 (690000 steps performed)
10000/10000 [==============================] - 340s 34ms/step - reward: 0.0179
13 episodes - episode_reward: 13.538 [6.000, 19.000] - loss: 0.014 - mae: 1.465
- mean_q: 1.779 - mean_eps: 0.687 - ale.lives: 2.039

Interval 71 (700000 steps performed)
10000/10000 [==============================] - 339s 34ms/step - reward: 0.0188
16 episodes - episode_reward: 11.625 [4.000, 21.000] - loss: 0.015 - mae: 1.449
- mean_q: 1.759 - mean_eps: 0.683 - ale.lives: 2.181

Interval 72 (710000 steps performed)
10000/10000 [==============================] - 342s 34ms/step - reward: 0.0179
15 episodes - episode_reward: 12.533 [7.000, 21.000] - loss: 0.015 - mae: 1.474
- mean_q: 1.791 - mean_eps: 0.678 - ale.lives: 2.151

Interval 73 (720000 steps performed)
10000/10000 [==============================] - 344s 34ms/step - reward: 0.0175
13 episodes - episode_reward: 12.385 [2.000, 27.000] - loss: 0.015 - mae: 1.496
- mean_q: 1.815 - mean_eps: 0.674 - ale.lives: 2.335

Interval 74 (730000 steps performed)
10000/10000 [==============================] - 344s 34ms/step - reward: 0.0204
13 episodes - episode_reward: 16.308 [5.000, 28.000] - loss: 0.014 - mae: 1.517
- mean_q: 1.840 - mean_eps: 0.669 - ale.lives: 2.109
```

```
Interval 75 (740000 steps performed)
10000/10000 [==============================] - 345s 35ms/step - reward: 0.0189
14 episodes - episode_reward: 12.857 [6.000, 21.000] - loss: 0.015 - mae: 1.536
- mean_q: 1.863 - mean_eps: 0.665 - ale.lives: 2.227

Interval 76 (750000 steps performed)
10000/10000 [==============================] - 350s 35ms/step - reward: 0.0197
12 episodes - episode_reward: 17.250 [10.000, 27.000] - loss: 0.015 - mae: 1.567
- mean_q: 1.899 - mean_eps: 0.660 - ale.lives: 2.174

Interval 77 (760000 steps performed)
10000/10000 [==============================] - 351s 35ms/step - reward: 0.0192
14 episodes - episode_reward: 14.071 [5.000, 29.000] - loss: 0.015 - mae: 1.586
- mean_q: 1.924 - mean_eps: 0.656 - ale.lives: 2.064

Interval 78 (770000 steps performed)
10000/10000 [==============================] - 353s 35ms/step - reward: 0.0200
15 episodes - episode_reward: 12.933 [3.000, 25.000] - loss: 0.016 - mae: 1.632
- mean_q: 1.977 - mean_eps: 0.651 - ale.lives: 2.196

Interval 79 (780000 steps performed)
10000/10000 [==============================] - 356s 36ms/step - reward: 0.0201
15 episodes - episode_reward: 12.933 [6.000, 20.000] - loss: 0.016 - mae: 1.644
- mean_q: 1.994 - mean_eps: 0.647 - ale.lives: 2.063

Interval 80 (790000 steps performed)
10000/10000 [==============================] - 357s 36ms/step - reward: 0.0174
16 episodes - episode_reward: 11.312 [2.000, 22.000] - loss: 0.015 - mae: 1.666
- mean_q: 2.019 - mean_eps: 0.642 - ale.lives: 2.043

Interval 81 (800000 steps performed)
10000/10000 [==============================] - 361s 36ms/step - reward: 0.0189
15 episodes - episode_reward: 12.400 [5.000, 30.000] - loss: 0.016 - mae: 1.692
- mean_q: 2.049 - mean_eps: 0.638 - ale.lives: 2.109

Interval 82 (810000 steps performed)
10000/10000 [==============================] - 359s 36ms/step - reward: 0.0185
13 episodes - episode_reward: 14.000 [3.000, 23.000] - loss: 0.016 - mae: 1.713
- mean_q: 2.074 - mean_eps: 0.633 - ale.lives: 2.029

Interval 83 (820000 steps performed)
10000/10000 [==============================] - 365s 36ms/step - reward: 0.0179
17 episodes - episode_reward: 11.176 [3.000, 21.000] - loss: 0.016 - mae: 1.718
- mean_q: 2.079 - mean_eps: 0.629 - ale.lives: 2.080

Interval 84 (830000 steps performed)
10000/10000 [==============================] - 364s 36ms/step - reward: 0.0193
```

```
12 episodes - episode_reward: 14.917 [8.000, 19.000] - loss: 0.016 - mae: 1.726
- mean_q: 2.089 - mean_eps: 0.624 - ale.lives: 2.241

Interval 85 (840000 steps performed)
10000/10000 [==============================] - 366s 37ms/step - reward: 0.0185
15 episodes - episode_reward: 13.333 [4.000, 23.000] - loss: 0.016 - mae: 1.727
- mean_q: 2.089 - mean_eps: 0.620 - ale.lives: 2.244

Interval 86 (850000 steps performed)
10000/10000 [==============================] - 370s 37ms/step - reward: 0.0203
13 episodes - episode_reward: 15.231 [8.000, 28.000] - loss: 0.015 - mae: 1.735
- mean_q: 2.099 - mean_eps: 0.615 - ale.lives: 2.058

Interval 87 (860000 steps performed)
10000/10000 [==============================] - 370s 37ms/step - reward: 0.0201
14 episodes - episode_reward: 14.714 [5.000, 30.000] - loss: 0.016 - mae: 1.737
- mean_q: 2.102 - mean_eps: 0.611 - ale.lives: 2.133

Interval 88 (870000 steps performed)
10000/10000 [==============================] - 375s 37ms/step - reward: 0.0211
13 episodes - episode_reward: 15.231 [6.000, 31.000] - loss: 0.017 - mae: 1.743
- mean_q: 2.109 - mean_eps: 0.606 - ale.lives: 2.112

Interval 89 (880000 steps performed)
10000/10000 [==============================] - 376s 38ms/step - reward: 0.0197
15 episodes - episode_reward: 14.000 [4.000, 23.000] - loss: 0.017 - mae: 1.755
- mean_q: 2.126 - mean_eps: 0.602 - ale.lives: 2.009

Interval 90 (890000 steps performed)
10000/10000 [==============================] - 379s 38ms/step - reward: 0.0188
16 episodes - episode_reward: 11.188 [6.000, 24.000] - loss: 0.017 - mae: 1.783
- mean_q: 2.159 - mean_eps: 0.597 - ale.lives: 2.097

Interval 91 (900000 steps performed)
10000/10000 [==============================] - 384s 38ms/step - reward: 0.0189
14 episodes - episode_reward: 13.929 [4.000, 24.000] - loss: 0.016 - mae: 1.823
- mean_q: 2.205 - mean_eps: 0.593 - ale.lives: 2.273

Interval 92 (910000 steps performed)
10000/10000 [==============================] - 384s 38ms/step - reward: 0.0180
13 episodes - episode_reward: 13.000 [6.000, 21.000] - loss: 0.017 - mae: 1.830
- mean_q: 2.213 - mean_eps: 0.588 - ale.lives: 2.131

Interval 93 (920000 steps performed)
10000/10000 [==============================] - 386s 39ms/step - reward: 0.0212
12 episodes - episode_reward: 17.583 [7.000, 27.000] - loss: 0.017 - mae: 1.809
- mean_q: 2.189 - mean_eps: 0.584 - ale.lives: 1.985
```

```
Interval 94 (930000 steps performed)
10000/10000 [==============================] - 388s 39ms/step - reward: 0.0205
11 episodes - episode_reward: 19.455 [8.000, 34.000] - loss: 0.016 - mae: 1.838
- mean_q: 2.224 - mean_eps: 0.579 - ale.lives: 1.993

Interval 95 (940000 steps performed)
10000/10000 [==============================] - 390s 39ms/step - reward: 0.0211
13 episodes - episode_reward: 16.077 [7.000, 26.000] - loss: 0.016 - mae: 1.855
- mean_q: 2.243 - mean_eps: 0.575 - ale.lives: 2.009

Interval 96 (950000 steps performed)
10000/10000 [==============================] - 391s 39ms/step - reward: 0.0203
12 episodes - episode_reward: 17.167 [7.000, 32.000] - loss: 0.017 - mae: 1.880
- mean_q: 2.273 - mean_eps: 0.570 - ale.lives: 2.136

Interval 97 (960000 steps performed)
10000/10000 [==============================] - 394s 39ms/step - reward: 0.0196
12 episodes - episode_reward: 16.083 [6.000, 36.000] - loss: 0.017 - mae: 1.885
- mean_q: 2.279 - mean_eps: 0.566 - ale.lives: 2.091

Interval 98 (970000 steps performed)
10000/10000 [==============================] - 398s 40ms/step - reward: 0.0195
13 episodes - episode_reward: 14.538 [6.000, 23.000] - loss: 0.018 - mae: 1.878
- mean_q: 2.270 - mean_eps: 0.561 - ale.lives: 2.135

Interval 99 (980000 steps performed)
10000/10000 [==============================] - 398s 40ms/step - reward: 0.0190
13 episodes - episode_reward: 15.615 [7.000, 28.000] - loss: 0.017 - mae: 1.910
- mean_q: 2.308 - mean_eps: 0.557 - ale.lives: 2.142

Interval 100 (990000 steps performed)
10000/10000 [==============================] - 400s 40ms/step - reward: 0.0182
done, took 30199.150 seconds
```

[ ]:
```python
# Testing part to calculate the mean reward
weights_filename = 'dqn_{}_weights.h5f'.format(env_name)
dqn_propuesta_1.load_weights(weights_filename)
dqn_propuesta_1.test(env, nb_episodes=10, visualize=False)
```

```
Testing for 10 episodes …
Episode 1: reward: 17.000, steps: 1260
Episode 2: reward: 16.000, steps: 667
Episode 3: reward: 16.000, steps: 645
Episode 4: reward: 27.000, steps: 1433
Episode 5: reward: 17.000, steps: 909
Episode 6: reward: 19.000, steps: 788
Episode 7: reward: 11.000, steps: 554
Episode 8: reward: 22.000, steps: 1046
```

```
Episode 9: reward: 22.000, steps: 1006
Episode 10: reward: 12.000, steps: 868
```

[ ]: <keras.src.callbacks.History at 0x7c83e5bb8d50>

3.4.3 Generación de vídeo

[ ]:
```python
import tensorflow as tf
tf.config.list_physical_devices('GPU')
```

[ ]: [PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU')]

[ ]:
```python
# Testing part
from gym import wrappers

if IN_COLAB:
    env = wrappers.Monitor(env, "/content/drive/", force=True)
else:
    env = wrappers.Monitor(env, "./videos_grabados/", force=True)
env.reset()

weights_filename = 'dqn_{}_weights.h5f'.format(env_name)
dqn_propuesta_1.load_weights(weights_filename)
dqn_propuesta_1.test(env, nb_episodes=10, visualize=False)
```

```
Testing for 10 episodes …
Episode 1: reward: 22.000, steps: 1077
Episode 2: reward: 16.000, steps: 850
Episode 3: reward: 22.000, steps: 1239
Episode 4: reward: 9.000, steps: 508
Episode 5: reward: 15.000, steps: 1026
Episode 6: reward: 15.000, steps: 1101
Episode 7: reward: 11.000, steps: 678
Episode 8: reward: 15.000, steps: 917
Episode 9: reward: 16.000, steps: 1149
Episode 10: reward: 13.000, steps: 1059
```

[ ]: <keras.src.callbacks.History at 0x7c83e5b78510>

[ ]:
```python
import io
import base64
from IPython.display import HTML

if IN_COLAB:
    video = io.open('/content/drive/openaigym.video.%s.video000008.mp4' % env.
↪file_infix, 'r+b').read()
else:
```

```
    video = io.open('./videos_grabados/openaigym.video.%s.video000008.mp4' %␣
 ↪env.file_infix, 'r+b').read()

encoded = base64.b64encode(video)
HTML(data='''
    <video width="360" height="auto" alt="test" controls><source src="data:
 ↪video/mp4;base64,{0}" type="video/mp4" /></video>'''
.format(encoded.decode('ascii')))
```

[ ]: <IPython.core.display.HTML object>

1.4.5 3.5 Propuesta 2: actualización suave de la red objetivo

Como propuesta número 2, proponemos un cambio en la actualización de los pesos del modelo. Este cambio tiene como propósito favorecer una transición de conocimiento más gradual entre la red principal y la red target.

En DQN estándar, la actualización de los pesos de la red target se lleva a cabo de forma abrupta en un momento concreto, lo que puede provocar cierta inestabilidad en el entrenamiento y producir oscilaciones fuertes.

Con el fin de tratar de mitigar este efecto, se propone una actualización más suave, donde la red objetivo se va moviendo más progresivamente hacia la red principal. Este efecto se consigue llevando a cabo una progresiva combinación de los pesos de una y otra, con el objetivo de aportar estabilidad al entrenamiento y reducir las oscilaciones.

Implementación: se modifica `target_model_update` para realizar una actualización suave de la red objetivo.

3.5.1 Implementación de la red neuronal Se reutiliza la arquitectura del caso base para aislar el efecto de la actualización suave del target.

3.5.2 Implementación del agente DQN

[ ]:
```
# Configuración de la memoria del algoritmo:
#    limit:         número máximo de transacciones
#    window_length: fotogramas consecutivos utilizados para formar un estado
memory_propuesta_2 = SequentialMemory(limit=1000000,␣
 ↪window_length=WINDOW_LENGTH)
# Procesado de imágenes: convertir imagenes a escala de grises y normalización␣
 ↪de los píxeles
processor_propuesta_2 = AtariProcessor()

# Estrategia de exploración:
#    EpsGreedyQPolicy: estrategia basada en epsilon
#    attr:             atributo epsilon
#    value_max:        valor inicial (máximo)
#    value_min:        valor mínimo (final) con el tiempo irá disminuyendo␣
 ↪hasta este valor
```

```
#    value_test:       valor eps para el modo de prueba
#    nb_steps:         número de pasos, la disminución de eps será gradual␣
↪durante este número de pasos
policy_propuesta_2 = LinearAnnealedPolicy(EpsGreedyQPolicy(), attr='eps',
                              value_max=1., value_min=.1, value_test=.05,
                              nb_steps=2000000)

# Configuración del agente DQN
#    model:                red neuronal
#    nb_actions:           número de acciones del juego
#    policy:               estrategia de exploración
#    memory:               memoria del algoritmo
#    processor:            procesado de imágenes
#    nb_steps_warmup:      número de pasos en los que el algoritmo no entrena,␣
↪solo juega para llenar la memoria con experiencias iniciales
#    gamma:                factor de actualización de las recompensas
#    target_model_update: número de pasos hasta actualizar los valores de la␣
↪red de predicción a la red de target
#    train_interval:       la red se entrena en intervalo de fotogramas para␣
↪optimización y alineamiento con los fotogramas del estado
dqn_propuesta_2 = DQNAgent(model=model, nb_actions=nb_actions,␣
↪policy=policy_propuesta_2,
               memory=memory_propuesta_2, processor=processor_propuesta_2,
               nb_steps_warmup=50000, gamma=.99,
               target_model_update=1e-3, train_interval=4)

# Compilación del agente
#    Adam:           optimizador Adam
#    learning_rate: tasa de aprendizaje
#    metrics:        métrica MAE (Error Absoluto Medio)
dqn_propuesta_2.compile(Adam(learning_rate=.00025), metrics=['mae'])
```

[ ]:
```
# Training part
weights_filename = 'dqn_{}_weights.h5f'.format(env_name)
checkpoint_weights_filename = 'dqn_' + env_name + '_weights_{step}.h5f'
log_filename = 'dqn_{}_log.json'.format(env_name)
callbacks = [ModelIntervalCheckpoint(checkpoint_weights_filename,␣
↪interval=100000)]
callbacks += [FileLogger(log_filename, interval=100)]

# Comenzar el entrenamiento
dqn_propuesta_2.fit(env, callbacks=callbacks, nb_steps=1000000,␣
↪log_interval=10000, visualize=False)
#dqn_propuesta_2.fit(env, callbacks=callbacks, nb_steps=20000,␣
↪log_interval=100, visualize=False)
```

```python
dqn_propuesta_2.save_weights(weights_filename, overwrite=True)
```

```
Training for 1000000 steps …
Interval 1 (0 steps performed)

/home/mrlyon/miniconda3/envs/miar_rl/lib/python3.11/site-
packages/keras/src/engine/training_v1.py:2359: UserWarning:
`Model.state_updates` will be removed in a future version. This property should
not be used in TensorFlow 2.0, as `updates` are applied automatically.
  updates=self.state_updates,

10000/10000 [==============================] - 37s 4ms/step - reward: 0.0140
14 episodes - episode_reward: 8.643 [5.000, 16.000] - ale.lives: 1.984

Interval 2 (10000 steps performed)
10000/10000 [==============================] - 36s 4ms/step - reward: 0.0142
15 episodes - episode_reward: 10.733 [5.000, 22.000] - ale.lives: 2.042

Interval 3 (20000 steps performed)
10000/10000 [==============================] - 40s 4ms/step - reward: 0.0137
13 episodes - episode_reward: 10.308 [3.000, 22.000] - ale.lives: 2.082

Interval 4 (30000 steps performed)
10000/10000 [==============================] - 44s 4ms/step - reward: 0.0132
17 episodes - episode_reward: 7.882 [4.000, 17.000] - ale.lives: 1.995

Interval 5 (40000 steps performed)
10000/10000 [==============================] - 42s 4ms/step - reward: 0.0133
14 episodes - episode_reward: 9.214 [3.000, 17.000] - ale.lives: 2.096

Interval 6 (50000 steps performed)
10000/10000 [==============================] - 252s 25ms/step - reward: 0.0119
16 episodes - episode_reward: 7.688 [3.000, 12.000] - loss: 0.007 - mae: 0.050 -
mean_q: 0.067 - mean_eps: 0.975 - ale.lives: 2.034

Interval 7 (60000 steps performed)
10000/10000 [==============================] - 257s 26ms/step - reward: 0.0126
15 episodes - episode_reward: 8.200 [4.000, 14.000] - loss: 0.006 - mae: 0.088 -
mean_q: 0.112 - mean_eps: 0.971 - ale.lives: 2.140

Interval 8 (70000 steps performed)
10000/10000 [==============================] - 253s 25ms/step - reward: 0.0145
15 episodes - episode_reward: 9.600 [3.000, 20.000] - loss: 0.006 - mae: 0.137 -
mean_q: 0.172 - mean_eps: 0.966 - ale.lives: 2.163

Interval 9 (80000 steps performed)
10000/10000 [==============================] - 247s 25ms/step - reward: 0.0158
12 episodes - episode_reward: 13.000 [7.000, 24.000] - loss: 0.006 - mae: 0.188
```

```
- mean_q: 0.235 - mean_eps: 0.962 - ale.lives: 2.011

Interval 10 (90000 steps performed)
10000/10000 [==============================] - 246s 25ms/step - reward: 0.0132
15 episodes - episode_reward: 9.133 [3.000, 17.000] - loss: 0.007 - mae: 0.244 -
mean_q: 0.302 - mean_eps: 0.957 - ale.lives: 2.059

Interval 11 (100000 steps performed)
10000/10000 [==============================] - 250s 25ms/step - reward: 0.0143
15 episodes - episode_reward: 9.667 [4.000, 20.000] - loss: 0.007 - mae: 0.298 -
mean_q: 0.367 - mean_eps: 0.953 - ale.lives: 2.181

Interval 12 (110000 steps performed)
10000/10000 [==============================] - 258s 26ms/step - reward: 0.0120
15 episodes - episode_reward: 7.733 [2.000, 14.000] - loss: 0.007 - mae: 0.351 -
mean_q: 0.432 - mean_eps: 0.948 - ale.lives: 2.127

Interval 13 (120000 steps performed)
10000/10000 [==============================] - 250s 25ms/step - reward: 0.0136
16 episodes - episode_reward: 8.188 [3.000, 13.000] - loss: 0.008 - mae: 0.401 -
mean_q: 0.491 - mean_eps: 0.944 - ale.lives: 2.220

Interval 14 (130000 steps performed)
10000/10000 [==============================] - 246s 25ms/step - reward: 0.0145
14 episodes - episode_reward: 10.643 [5.000, 17.000] - loss: 0.008 - mae: 0.454
- mean_q: 0.556 - mean_eps: 0.939 - ale.lives: 2.121

Interval 15 (140000 steps performed)
10000/10000 [==============================] - 245s 24ms/step - reward: 0.0128
15 episodes - episode_reward: 8.867 [4.000, 14.000] - loss: 0.008 - mae: 0.510 -
mean_q: 0.625 - mean_eps: 0.935 - ale.lives: 2.121

Interval 16 (150000 steps performed)
10000/10000 [==============================] - 248s 25ms/step - reward: 0.0139
16 episodes - episode_reward: 8.375 [3.000, 21.000] - loss: 0.009 - mae: 0.567 -
mean_q: 0.696 - mean_eps: 0.930 - ale.lives: 2.104

Interval 17 (160000 steps performed)
10000/10000 [==============================] - 250s 25ms/step - reward: 0.0138
15 episodes - episode_reward: 8.733 [5.000, 17.000] - loss: 0.009 - mae: 0.624 -
mean_q: 0.765 - mean_eps: 0.926 - ale.lives: 1.987

Interval 18 (170000 steps performed)
10000/10000 [==============================] - 251s 25ms/step - reward: 0.0171
14 episodes - episode_reward: 12.214 [6.000, 25.000] - loss: 0.010 - mae: 0.686
- mean_q: 0.841 - mean_eps: 0.921 - ale.lives: 2.012

Interval 19 (180000 steps performed)
```

```
10000/10000 [==============================] - 251s 25ms/step - reward: 0.0150
15 episodes - episode_reward: 10.733 [3.000, 26.000] - loss: 0.011 - mae: 0.758
- mean_q: 0.931 - mean_eps: 0.917 - ale.lives: 2.069

Interval 20 (190000 steps performed)
10000/10000 [==============================] - 251s 25ms/step - reward: 0.0148
11 episodes - episode_reward: 13.000 [9.000, 20.000] - loss: 0.012 - mae: 0.838
- mean_q: 1.029 - mean_eps: 0.912 - ale.lives: 2.283

Interval 21 (200000 steps performed)
10000/10000 [==============================] - 251s 25ms/step - reward: 0.0158
16 episodes - episode_reward: 10.250 [5.000, 27.000] - loss: 0.012 - mae: 0.908
- mean_q: 1.115 - mean_eps: 0.908 - ale.lives: 2.041

Interval 22 (210000 steps performed)
10000/10000 [==============================] - 253s 25ms/step - reward: 0.0149
15 episodes - episode_reward: 9.933 [3.000, 17.000] - loss: 0.013 - mae: 0.994 -
mean_q: 1.221 - mean_eps: 0.903 - ale.lives: 2.151

Interval 23 (220000 steps performed)
10000/10000 [==============================] - 257s 26ms/step - reward: 0.0140
12 episodes - episode_reward: 11.083 [4.000, 18.000] - loss: 0.013 - mae: 1.080
- mean_q: 1.323 - mean_eps: 0.899 - ale.lives: 2.046

Interval 24 (230000 steps performed)
10000/10000 [==============================] - 256s 26ms/step - reward: 0.0155
16 episodes - episode_reward: 10.000 [4.000, 22.000] - loss: 0.014 - mae: 1.156
- mean_q: 1.415 - mean_eps: 0.894 - ale.lives: 2.177

Interval 25 (240000 steps performed)
10000/10000 [==============================] - 258s 26ms/step - reward: 0.0150
16 episodes - episode_reward: 8.875 [4.000, 14.000] - loss: 0.014 - mae: 1.235 -
mean_q: 1.509 - mean_eps: 0.890 - ale.lives: 2.044

Interval 26 (250000 steps performed)
10000/10000 [==============================] - 269s 27ms/step - reward: 0.0151
13 episodes - episode_reward: 11.615 [5.000, 18.000] - loss: 0.015 - mae: 1.307
- mean_q: 1.596 - mean_eps: 0.885 - ale.lives: 2.219

Interval 27 (260000 steps performed)
10000/10000 [==============================] - 271s 27ms/step - reward: 0.0158
14 episodes - episode_reward: 11.643 [6.000, 24.000] - loss: 0.015 - mae: 1.380
- mean_q: 1.683 - mean_eps: 0.881 - ale.lives: 1.992

Interval 28 (270000 steps performed)
10000/10000 [==============================] - 274s 27ms/step - reward: 0.0172
15 episodes - episode_reward: 11.000 [2.000, 24.000] - loss: 0.016 - mae: 1.457
- mean_q: 1.776 - mean_eps: 0.876 - ale.lives: 2.172
```

```
Interval 29 (280000 steps performed)
10000/10000 [==============================] - 274s 27ms/step - reward: 0.0150
14 episodes - episode_reward: 11.143 [4.000, 19.000] - loss: 0.016 - mae: 1.516
- mean_q: 1.847 - mean_eps: 0.872 - ale.lives: 2.151

Interval 30 (290000 steps performed)
10000/10000 [==============================] - 276s 28ms/step - reward: 0.0156
15 episodes - episode_reward: 10.800 [4.000, 17.000] - loss: 0.016 - mae: 1.592
- mean_q: 1.939 - mean_eps: 0.867 - ale.lives: 1.931

Interval 31 (300000 steps performed)
10000/10000 [==============================] - 289s 29ms/step - reward: 0.0151
16 episodes - episode_reward: 9.125 [5.000, 14.000] - loss: 0.017 - mae: 1.653 -
mean_q: 2.012 - mean_eps: 0.863 - ale.lives: 2.145

Interval 32 (310000 steps performed)
10000/10000 [==============================] - 280s 28ms/step - reward: 0.0162
17 episodes - episode_reward: 9.824 [5.000, 22.000] - loss: 0.017 - mae: 1.726 -
mean_q: 2.101 - mean_eps: 0.858 - ale.lives: 2.027

Interval 33 (320000 steps performed)
10000/10000 [==============================] - 280s 28ms/step - reward: 0.0149
15 episodes - episode_reward: 9.533 [6.000, 17.000] - loss: 0.017 - mae: 1.792 -
mean_q: 2.179 - mean_eps: 0.854 - ale.lives: 2.192

Interval 34 (330000 steps performed)
10000/10000 [==============================] - 280s 28ms/step - reward: 0.0142
18 episodes - episode_reward: 7.889 [3.000, 12.000] - loss: 0.017 - mae: 1.872 -
mean_q: 2.277 - mean_eps: 0.849 - ale.lives: 2.105

Interval 35 (340000 steps performed)
10000/10000 [==============================] - 272s 27ms/step - reward: 0.0169
15 episodes - episode_reward: 11.200 [6.000, 19.000] - loss: 0.019 - mae: 1.935
- mean_q: 2.354 - mean_eps: 0.845 - ale.lives: 2.107

Interval 36 (350000 steps performed)
10000/10000 [==============================] - 272s 27ms/step - reward: 0.0176
13 episodes - episode_reward: 13.077 [5.000, 24.000] - loss: 0.019 - mae: 2.012
- mean_q: 2.447 - mean_eps: 0.840 - ale.lives: 2.024

Interval 37 (360000 steps performed)
10000/10000 [==============================] - 273s 27ms/step - reward: 0.0164
14 episodes - episode_reward: 12.643 [6.000, 19.000] - loss: 0.020 - mae: 2.082
- mean_q: 2.531 - mean_eps: 0.836 - ale.lives: 1.992

Interval 38 (370000 steps performed)
10000/10000 [==============================] - 274s 27ms/step - reward: 0.0162
```

```
15 episodes - episode_reward: 10.533 [4.000, 16.000] - loss: 0.021 - mae: 2.154
- mean_q: 2.617 - mean_eps: 0.831 - ale.lives: 2.095

Interval 39 (380000 steps performed)
10000/10000 [==============================] - 276s 28ms/step - reward: 0.0158
16 episodes - episode_reward: 10.062 [4.000, 16.000] - loss: 0.022 - mae: 2.217
- mean_q: 2.693 - mean_eps: 0.827 - ale.lives: 2.132

Interval 40 (390000 steps performed)
10000/10000 [==============================] - 277s 28ms/step - reward: 0.0154
14 episodes - episode_reward: 11.071 [4.000, 22.000] - loss: 0.022 - mae: 2.284
- mean_q: 2.774 - mean_eps: 0.822 - ale.lives: 1.989

Interval 41 (400000 steps performed)
10000/10000 [==============================] - 279s 28ms/step - reward: 0.0170
14 episodes - episode_reward: 11.357 [6.000, 21.000] - loss: 0.022 - mae: 2.344
- mean_q: 2.847 - mean_eps: 0.818 - ale.lives: 2.067

Interval 42 (410000 steps performed)
10000/10000 [==============================] - 283s 28ms/step - reward: 0.0161
17 episodes - episode_reward: 9.882 [3.000, 23.000] - loss: 0.023 - mae: 2.415 -
mean_q: 2.932 - mean_eps: 0.813 - ale.lives: 2.188

Interval 43 (420000 steps performed)
10000/10000 [==============================] - 290s 29ms/step - reward: 0.0160
16 episodes - episode_reward: 10.250 [4.000, 20.000] - loss: 0.023 - mae: 2.491
- mean_q: 3.023 - mean_eps: 0.809 - ale.lives: 2.089

Interval 44 (430000 steps performed)
10000/10000 [==============================] - 300s 30ms/step - reward: 0.0172
14 episodes - episode_reward: 12.286 [7.000, 19.000] - loss: 0.023 - mae: 2.542
- mean_q: 3.082 - mean_eps: 0.804 - ale.lives: 2.117

Interval 45 (440000 steps performed)
10000/10000 [==============================] - 288s 29ms/step - reward: 0.0171
15 episodes - episode_reward: 10.933 [5.000, 21.000] - loss: 0.024 - mae: 2.606
- mean_q: 3.160 - mean_eps: 0.800 - ale.lives: 2.243

Interval 46 (450000 steps performed)
10000/10000 [==============================] - 289s 29ms/step - reward: 0.0171
12 episodes - episode_reward: 13.667 [8.000, 24.000] - loss: 0.025 - mae: 2.665
- mean_q: 3.231 - mean_eps: 0.795 - ale.lives: 2.056

Interval 47 (460000 steps performed)
10000/10000 [==============================] - 299s 30ms/step - reward: 0.0156
15 episodes - episode_reward: 10.467 [4.000, 19.000] - loss: 0.025 - mae: 2.714
- mean_q: 3.289 - mean_eps: 0.791 - ale.lives: 2.057
```

```
Interval 48 (470000 steps performed)
10000/10000 [==============================] - 299s 30ms/step - reward: 0.0188
13 episodes - episode_reward: 14.769 [9.000, 22.000] - loss: 0.025 - mae: 2.770
- mean_q: 3.356 - mean_eps: 0.786 - ale.lives: 2.204

Interval 49 (480000 steps performed)
10000/10000 [==============================] - 303s 30ms/step - reward: 0.0171
14 episodes - episode_reward: 12.429 [7.000, 19.000] - loss: 0.026 - mae: 2.810
- mean_q: 3.402 - mean_eps: 0.782 - ale.lives: 2.009

Interval 50 (490000 steps performed)
10000/10000 [==============================] - 300s 30ms/step - reward: 0.0175
15 episodes - episode_reward: 11.933 [4.000, 22.000] - loss: 0.026 - mae: 2.866
- mean_q: 3.469 - mean_eps: 0.777 - ale.lives: 2.131

Interval 51 (500000 steps performed)
10000/10000 [==============================] - 316s 32ms/step - reward: 0.0179
12 episodes - episode_reward: 14.000 [8.000, 20.000] - loss: 0.027 - mae: 2.914
- mean_q: 3.527 - mean_eps: 0.773 - ale.lives: 2.021

Interval 52 (510000 steps performed)
10000/10000 [==============================] - 311s 31ms/step - reward: 0.0177
15 episodes - episode_reward: 12.667 [7.000, 22.000] - loss: 0.028 - mae: 2.956
- mean_q: 3.578 - mean_eps: 0.768 - ale.lives: 2.244

Interval 53 (520000 steps performed)
10000/10000 [==============================] - 315s 31ms/step - reward: 0.0177
14 episodes - episode_reward: 12.429 [6.000, 23.000] - loss: 0.029 - mae: 3.015
- mean_q: 3.648 - mean_eps: 0.764 - ale.lives: 2.090

Interval 54 (530000 steps performed)
10000/10000 [==============================] - 313s 31ms/step - reward: 0.0172
13 episodes - episode_reward: 11.923 [5.000, 22.000] - loss: 0.028 - mae: 3.046
- mean_q: 3.685 - mean_eps: 0.759 - ale.lives: 2.085

Interval 55 (540000 steps performed)
10000/10000 [==============================] - 314s 31ms/step - reward: 0.0168
13 episodes - episode_reward: 14.000 [5.000, 22.000] - loss: 0.029 - mae: 3.108
- mean_q: 3.760 - mean_eps: 0.755 - ale.lives: 2.013

Interval 56 (550000 steps performed)
10000/10000 [==============================] - 316s 32ms/step - reward: 0.0162
15 episodes - episode_reward: 10.667 [4.000, 21.000] - loss: 0.030 - mae: 3.141
- mean_q: 3.800 - mean_eps: 0.750 - ale.lives: 2.012

Interval 57 (560000 steps performed)
10000/10000 [==============================] - 317s 32ms/step - reward: 0.0188
14 episodes - episode_reward: 13.929 [7.000, 24.000] - loss: 0.030 - mae: 3.183
```

```
- mean_q: 3.849 - mean_eps: 0.746 - ale.lives: 2.081

Interval 58 (570000 steps performed)
10000/10000 [==============================] - 318s 32ms/step - reward: 0.0141
15 episodes - episode_reward: 9.000 [2.000, 18.000] - loss: 0.032 - mae: 3.225 -
mean_q: 3.900 - mean_eps: 0.741 - ale.lives: 2.108

Interval 59 (580000 steps performed)
10000/10000 [==============================] - 320s 32ms/step - reward: 0.0178
12 episodes - episode_reward: 13.833 [7.000, 20.000] - loss: 0.032 - mae: 3.276
- mean_q: 3.962 - mean_eps: 0.737 - ale.lives: 1.962

Interval 60 (590000 steps performed)
10000/10000 [==============================] - 322s 32ms/step - reward: 0.0171
16 episodes - episode_reward: 11.688 [6.000, 21.000] - loss: 0.032 - mae: 3.313
- mean_q: 4.006 - mean_eps: 0.732 - ale.lives: 2.075

Interval 61 (600000 steps performed)
10000/10000 [==============================] - 322s 32ms/step - reward: 0.0171
12 episodes - episode_reward: 13.167 [3.000, 23.000] - loss: 0.034 - mae: 3.344
- mean_q: 4.043 - mean_eps: 0.728 - ale.lives: 2.145

Interval 62 (610000 steps performed)
10000/10000 [==============================] - 324s 32ms/step - reward: 0.0163
13 episodes - episode_reward: 12.231 [5.000, 17.000] - loss: 0.033 - mae: 3.388
- mean_q: 4.096 - mean_eps: 0.723 - ale.lives: 2.079

Interval 63 (620000 steps performed)
10000/10000 [==============================] - 312s 31ms/step - reward: 0.0164
15 episodes - episode_reward: 11.800 [4.000, 24.000] - loss: 0.036 - mae: 3.432
- mean_q: 4.148 - mean_eps: 0.719 - ale.lives: 2.235

Interval 64 (630000 steps performed)
10000/10000 [==============================] - 314s 31ms/step - reward: 0.0189
15 episodes - episode_reward: 12.933 [4.000, 20.000] - loss: 0.037 - mae: 3.470
- mean_q: 4.194 - mean_eps: 0.714 - ale.lives: 2.045

Interval 65 (640000 steps performed)
10000/10000 [==============================] - 317s 32ms/step - reward: 0.0165
12 episodes - episode_reward: 13.333 [6.000, 21.000] - loss: 0.037 - mae: 3.510
- mean_q: 4.243 - mean_eps: 0.710 - ale.lives: 2.151

Interval 66 (650000 steps performed)
10000/10000 [==============================] - 320s 32ms/step - reward: 0.0162
14 episodes - episode_reward: 11.643 [3.000, 18.000] - loss: 0.037 - mae: 3.549
- mean_q: 4.291 - mean_eps: 0.705 - ale.lives: 2.196

Interval 67 (660000 steps performed)
```

```
10000/10000 [==============================] - 336s 34ms/step - reward: 0.0151
14 episodes - episode_reward: 10.786 [3.000, 19.000] - loss: 0.035 - mae: 3.593
- mean_q: 4.342 - mean_eps: 0.701 - ale.lives: 2.212

Interval 68 (670000 steps performed)
10000/10000 [==============================] - 344s 34ms/step - reward: 0.0160
14 episodes - episode_reward: 11.857 [6.000, 18.000] - loss: 0.036 - mae: 3.632
- mean_q: 4.389 - mean_eps: 0.696 - ale.lives: 2.063

Interval 69 (680000 steps performed)
10000/10000 [==============================] - 331s 33ms/step - reward: 0.0192
12 episodes - episode_reward: 15.250 [5.000, 27.000] - loss: 0.036 - mae: 3.660
- mean_q: 4.424 - mean_eps: 0.692 - ale.lives: 2.195

Interval 70 (690000 steps performed)
10000/10000 [==============================] - 338s 34ms/step - reward: 0.0171
12 episodes - episode_reward: 13.500 [8.000, 24.000] - loss: 0.036 - mae: 3.718
- mean_q: 4.494 - mean_eps: 0.687 - ale.lives: 2.228

Interval 71 (700000 steps performed)
10000/10000 [==============================] - 323s 32ms/step - reward: 0.0191
13 episodes - episode_reward: 16.000 [5.000, 24.000] - loss: 0.037 - mae: 3.741
- mean_q: 4.521 - mean_eps: 0.683 - ale.lives: 2.221

Interval 72 (710000 steps performed)
10000/10000 [==============================] - 324s 32ms/step - reward: 0.0174
12 episodes - episode_reward: 13.583 [7.000, 23.000] - loss: 0.038 - mae: 3.774
- mean_q: 4.559 - mean_eps: 0.678 - ale.lives: 2.319

Interval 73 (720000 steps performed)
10000/10000 [==============================] - 327s 33ms/step - reward: 0.0183
14 episodes - episode_reward: 13.429 [5.000, 22.000] - loss: 0.038 - mae: 3.803
- mean_q: 4.597 - mean_eps: 0.674 - ale.lives: 2.168

Interval 74 (730000 steps performed)
10000/10000 [==============================] - 328s 33ms/step - reward: 0.0182
14 episodes - episode_reward: 12.571 [3.000, 26.000] - loss: 0.038 - mae: 3.845
- mean_q: 4.646 - mean_eps: 0.669 - ale.lives: 2.046

Interval 75 (740000 steps performed)
10000/10000 [==============================] - 330s 33ms/step - reward: 0.0180
12 episodes - episode_reward: 15.583 [3.000, 22.000] - loss: 0.039 - mae: 3.871
- mean_q: 4.677 - mean_eps: 0.665 - ale.lives: 2.323

Interval 76 (750000 steps performed)
10000/10000 [==============================] - 331s 33ms/step - reward: 0.0162
11 episodes - episode_reward: 14.818 [7.000, 22.000] - loss: 0.040 - mae: 3.901
- mean_q: 4.713 - mean_eps: 0.660 - ale.lives: 2.099
```

```
Interval 77 (760000 steps performed)
10000/10000 [==============================] - 332s 33ms/step - reward: 0.0176
12 episodes - episode_reward: 14.583 [2.000, 24.000] - loss: 0.038 - mae: 3.941
- mean_q: 4.763 - mean_eps: 0.656 - ale.lives: 2.260

Interval 78 (770000 steps performed)
10000/10000 [==============================] - 334s 33ms/step - reward: 0.0184
12 episodes - episode_reward: 14.667 [5.000, 25.000] - loss: 0.039 - mae: 3.970
- mean_q: 4.798 - mean_eps: 0.651 - ale.lives: 1.955

Interval 79 (780000 steps performed)
10000/10000 [==============================] - 336s 34ms/step - reward: 0.0158
14 episodes - episode_reward: 11.857 [3.000, 28.000] - loss: 0.039 - mae: 4.007
- mean_q: 4.843 - mean_eps: 0.647 - ale.lives: 2.308

Interval 80 (790000 steps performed)
10000/10000 [==============================] - 404s 40ms/step - reward: 0.0167
13 episodes - episode_reward: 12.231 [5.000, 19.000] - loss: 0.040 - mae: 4.031
- mean_q: 4.870 - mean_eps: 0.642 - ale.lives: 2.150

Interval 81 (800000 steps performed)
10000/10000 [==============================] - 398s 40ms/step - reward: 0.0167
13 episodes - episode_reward: 12.846 [6.000, 21.000] - loss: 0.041 - mae: 4.062
- mean_q: 4.907 - mean_eps: 0.638 - ale.lives: 2.239

Interval 82 (810000 steps performed)
10000/10000 [==============================] - 359s 36ms/step - reward: 0.0188
13 episodes - episode_reward: 15.077 [5.000, 26.000] - loss: 0.040 - mae: 4.090
- mean_q: 4.941 - mean_eps: 0.633 - ale.lives: 2.041

Interval 83 (820000 steps performed)
10000/10000 [==============================] - 361s 36ms/step - reward: 0.0175
14 episodes - episode_reward: 12.500 [5.000, 24.000] - loss: 0.039 - mae: 4.126
- mean_q: 4.984 - mean_eps: 0.629 - ale.lives: 2.174

Interval 84 (830000 steps performed)
10000/10000 [==============================] - 410s 41ms/step - reward: 0.0172
13 episodes - episode_reward: 13.462 [7.000, 20.000] - loss: 0.039 - mae: 4.145
- mean_q: 5.009 - mean_eps: 0.624 - ale.lives: 2.098

Interval 85 (840000 steps performed)
10000/10000 [==============================] - 411s 41ms/step - reward: 0.0157
13 episodes - episode_reward: 11.308 [4.000, 21.000] - loss: 0.041 - mae: 4.169
- mean_q: 5.036 - mean_eps: 0.620 - ale.lives: 2.205

Interval 86 (850000 steps performed)
10000/10000 [==============================] - 428s 43ms/step - reward: 0.0205
```

```
13 episodes - episode_reward: 16.231 [5.000, 27.000] - loss: 0.040 - mae: 4.206
- mean_q: 5.082 - mean_eps: 0.615 - ale.lives: 2.115

Interval 87 (860000 steps performed)
10000/10000 [==============================] - 394s 39ms/step - reward: 0.0190
12 episodes - episode_reward: 15.417 [5.000, 23.000] - loss: 0.040 - mae: 4.223
- mean_q: 5.103 - mean_eps: 0.611 - ale.lives: 2.135

Interval 88 (870000 steps performed)
10000/10000 [==============================] - 425s 43ms/step - reward: 0.0146
14 episodes - episode_reward: 11.071 [1.000, 22.000] - loss: 0.042 - mae: 4.263
- mean_q: 5.150 - mean_eps: 0.606 - ale.lives: 2.140

Interval 89 (880000 steps performed)
10000/10000 [==============================] - 407s 41ms/step - reward: 0.0181
13 episodes - episode_reward: 13.692 [7.000, 21.000] - loss: 0.042 - mae: 4.284
- mean_q: 5.174 - mean_eps: 0.602 - ale.lives: 2.130

Interval 90 (890000 steps performed)
10000/10000 [==============================] - 434s 43ms/step - reward: 0.0182
13 episodes - episode_reward: 14.231 [4.000, 25.000] - loss: 0.042 - mae: 4.307
- mean_q: 5.205 - mean_eps: 0.597 - ale.lives: 2.154

Interval 91 (900000 steps performed)
10000/10000 [==============================] - 382s 38ms/step - reward: 0.0171
14 episodes - episode_reward: 12.000 [6.000, 22.000] - loss: 0.042 - mae: 4.346
- mean_q: 5.253 - mean_eps: 0.593 - ale.lives: 2.007

Interval 92 (910000 steps performed)
10000/10000 [==============================] - 365s 36ms/step - reward: 0.0192
13 episodes - episode_reward: 14.923 [6.000, 21.000] - loss: 0.044 - mae: 4.383
- mean_q: 5.298 - mean_eps: 0.588 - ale.lives: 2.054

Interval 93 (920000 steps performed)
10000/10000 [==============================] - 363s 36ms/step - reward: 0.0188
12 episodes - episode_reward: 15.917 [11.000, 22.000] - loss: 0.042 - mae: 4.417
- mean_q: 5.338 - mean_eps: 0.584 - ale.lives: 2.174

Interval 94 (930000 steps performed)
10000/10000 [==============================] - 430s 43ms/step - reward: 0.0177
14 episodes - episode_reward: 11.143 [4.000, 23.000] - loss: 0.043 - mae: 4.430
- mean_q: 5.355 - mean_eps: 0.579 - ale.lives: 2.001

Interval 95 (940000 steps performed)
10000/10000 [==============================] - 410s 41ms/step - reward: 0.0184
15 episodes - episode_reward: 12.867 [3.000, 23.000] - loss: 0.045 - mae: 4.475
- mean_q: 5.412 - mean_eps: 0.575 - ale.lives: 2.171
```

```
Interval 96 (950000 steps performed)
10000/10000 [==============================] - 455s 45ms/step - reward: 0.0190
12 episodes - episode_reward: 16.500 [5.000, 22.000] - loss: 0.043 - mae: 4.511
- mean_q: 5.455 - mean_eps: 0.570 - ale.lives: 2.183

Interval 97 (960000 steps performed)
10000/10000 [==============================] - 479s 48ms/step - reward: 0.0167
14 episodes - episode_reward: 11.571 [4.000, 19.000] - loss: 0.046 - mae: 4.545
- mean_q: 5.493 - mean_eps: 0.566 - ale.lives: 2.195

Interval 98 (970000 steps performed)
10000/10000 [==============================] - 480s 48ms/step - reward: 0.0173
14 episodes - episode_reward: 12.286 [5.000, 22.000] - loss: 0.046 - mae: 4.582
- mean_q: 5.538 - mean_eps: 0.561 - ale.lives: 2.123

Interval 99 (980000 steps performed)
10000/10000 [==============================] - 436s 44ms/step - reward: 0.0180
15 episodes - episode_reward: 12.467 [6.000, 25.000] - loss: 0.047 - mae: 4.603
- mean_q: 5.562 - mean_eps: 0.557 - ale.lives: 2.057

Interval 100 (990000 steps performed)
10000/10000 [==============================] - 467s 47ms/step - reward: 0.0150
done, took 30379.496 seconds
```

[ ]:
```python
# Testing part to calculate the mean reward
weights_filename = 'dqn_{}_weights.h5f'.format(env_name)
dqn_propuesta_2.load_weights(weights_filename)
dqn_propuesta_2.test(env, nb_episodes=10, visualize=False)
```

```
Testing for 10 episodes …
Episode 1: reward: 19.000, steps: 1230
Episode 2: reward: 12.000, steps: 964
Episode 3: reward: 2.000, steps: 359
Episode 4: reward: 15.000, steps: 810
Episode 5: reward: 11.000, steps: 799
Episode 6: reward: 8.000, steps: 674
Episode 7: reward: 14.000, steps: 917
Episode 8: reward: 15.000, steps: 886
Episode 9: reward: 12.000, steps: 962
Episode 10: reward: 11.000, steps: 1173
```

[ ]:
```
<keras.src.callbacks.History at 0x7a82f2d6ee90>
```

3.5.3 Generación de vídeo

[ ]:
```python
import tensorflow as tf
tf.config.list_physical_devices('GPU')
```

[ ]: [PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU')]

[ ]:
```
# Testing part
from gym import wrappers

if IN_COLAB:
    env = wrappers.Monitor(env, "/content/drive/", force=True)
else:
    env = wrappers.Monitor(env, "./videos_grabados/", force=True)
env.reset()

weights_filename = 'dqn_{}_weights.h5f'.format(env_name)
dqn_propuesta_2.load_weights(weights_filename)
dqn_propuesta_2.test(env, nb_episodes=10, visualize=False)
```

```
Testing for 10 episodes …
Episode 1: reward: 4.000, steps: 383
Episode 2: reward: 15.000, steps: 962
Episode 3: reward: 12.000, steps: 917
Episode 4: reward: 6.000, steps: 598
Episode 5: reward: 10.000, steps: 827
Episode 6: reward: 15.000, steps: 1062
Episode 7: reward: 9.000, steps: 782
Episode 8: reward: 19.000, steps: 1109
Episode 9: reward: 19.000, steps: 1119
Episode 10: reward: 10.000, steps: 948
```

[ ]: <keras.src.callbacks.History at 0x7a828dbe0450>

[ ]:
```
import io
import base64
from IPython.display import HTML

if IN_COLAB:
    video = io.open('/content/drive/openaigym.video.%s.video000008.mp4' % env.
 ↪file_infix, 'r+b').read()
else:
    video = io.open('./videos_grabados/openaigym.video.%s.video000008.mp4' %␣
 ↪env.file_infix, 'r+b').read()

encoded = base64.b64encode(video)
HTML(data='''
    <video width="360" height="auto" alt="test" controls><source src="data:
 ↪video/mp4;base64,{0}" type="video/mp4" /></video>'''
.format(encoded.decode('ascii')))
```

[ ]: <IPython.core.display.HTML object>

1.4.6 3.6 Propuesta 3: red con mayor capacidad de representación

Como propuesta final planteamos una cuestión de escala: la inclusión de una capa densa de 1024 neuronas a la salida del flatten.

El procesamiento visual (extracción de patrones básicos, enemigos, disparos, posición…) se lleva a cabo por la parte convolucional de la red, la cual no se ve modificada. Nuestro énfasis se pone en la parte densamente conectada posterior, que es la encargada de combinar las características extraídas por las convoluciones y traducirlas en valores Q más precisos para cada acción.

Por tanto, con esta mejora incremental no estamos evaluando el funcionamiento del extractor visual, sino que los cambios que observemos podrán ser principalmente atribuibles a las mejoras en la capacidad de estimar los valores de Q.

La justificación de esta decisión la encontramos en el hecho de que las dinámicas espaciotemporales de un entorno como es el juego Space Invaders (proyectiles rápidos, barreras destructibles, cambios bruscos tras perder vidas…) tienen una complejidad suficientemente alta como para que la sencillez de la red original pueda estar suponiendo una limitación en nuestro objetivo.

Implementación: se modifica la arquitectura de la red añadiendo mayor capacidad de representación respecto al modelo base.

3.6.1 Implementación de la red neuronal

[ ]:
```
# Formato de los datos de entrada
input_shape = (WINDOW_LENGTH,) + INPUT_SHAPE

# Definición del modelo
model_propuesta_3 = Sequential(name="DQN_SpaceInvaders_Mejorada")

# Datos de entrada
model_propuesta_3.add(InputLayer(input_shape=input_shape, name="input_frames"))

# Compatibilidad de la entrada
model_propuesta_3.add(Permute((2, 3, 1), name="permute_channels"))

# Capas convolucionales
model_propuesta_3.add(Conv2D(32, (8, 8), strides=(4, 4), activation="relu",␣
 ↪name="conv1"))
model_propuesta_3.add(Conv2D(64, (4, 4), strides=(2, 2), activation="relu",␣
 ↪name="conv2"))
model_propuesta_3.add(Conv2D(64, (3, 3), strides=(1, 1), activation="relu",␣
 ↪name="conv3"))

# Capa Flatten
model_propuesta_3.add(Flatten(name="flatten"))

# Capas Densas
model_propuesta_3.add(Dense(1024, activation="relu", name="new_dense_hidden"))
model_propuesta_3.add(Dense(512, activation="relu", name="dense_hidden"))
```

```
model_propuesta_3.add(Dense(nb_actions, activation="linear",␣
↪name="output_values"))

# Imprimir información del modelo
print(model_propuesta_3.summary())
```

```
Model: "DQN_SpaceInvaders_Mejorada"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 permute_channels (Permute)  (None, 84, 84, 4)         0

 conv1 (Conv2D)              (None, 20, 20, 32)        8224

 conv2 (Conv2D)              (None, 9, 9, 64)          32832

 conv3 (Conv2D)              (None, 7, 7, 64)          36928

 flatten (Flatten)           (None, 3136)              0

 new_dense_hidden (Dense)    (None, 1024)              3212288

 dense_hidden (Dense)        (None, 512)               524800

 output_values (Dense)       (None, 6)                 3078

=================================================================
Total params: 3818150 (14.57 MB)
Trainable params: 3818150 (14.57 MB)
Non-trainable params: 0 (0.00 Byte)
_________________________________________________________________
None
```

3.6.2 Implementación del agente DQN

[ ]:
```
# Configuración de la memoria del algoritmo:
#    limit:         número máximo de transacciones
#    window_length: fotogramas consecutivos utilizados para formar un estado
memory_propuesta_3 = SequentialMemory(limit=1000000,␣
↪window_length=WINDOW_LENGTH)
# Procesado de imágenes: convertir imagenes a escala de grises y normalización␣
↪de los píxeles
processor_propuesta_3 = AtariProcessor()

# Estrategia de exploración:
#    EpsGreedyQPolicy: estrategia basada en epsilon
#    attr:             atributo epsilon
```

```
#    value_max:         valor inicial (máximo)
#    value_min:         valor mínimo (final) con el tiempo irá disminuyendo␣
↪hasta este valor
#    value_test:        valor eps para el modo de prueba
#    nb_steps:          número de pasos, la disminución de eps será gradual␣
↪durante este número de pasos
policy_propuesta_3 = LinearAnnealedPolicy(EpsGreedyQPolicy(), attr='eps',
                              value_max=1., value_min=.1, value_test=.05,
                              nb_steps=2000000)

# Configuración del agente DQN
#    model:                red neuronal
#    nb_actions:           número de acciones del juego
#    policy:               estrategia de exploración
#    memory:               memoria del algoritmo
#    processor:            procesado de imágenes
#    nb_steps_warmup:      número de pasos en los que el algoritmo no entrena,␣
↪solo juega para llenar la memoria con experiencias iniciales
#    gamma:                factor de actualización de las recompensas
#    target_model_update: número de pasos hasta actualizar los valores de la␣
↪red de predicción a la red de target
#    train_interval:       la red se entrena en intervalo de fotogramas para␣
↪optimización y alineamiento con los fotogramas del estado
#    enable_double_dqn:   DDQN
dqn_propuesta_3 = DQNAgent(model=model_propuesta_3, nb_actions=nb_actions,␣
↪policy=policy_propuesta_3,
               memory=memory_propuesta_3, processor=processor_propuesta_3,
               nb_steps_warmup=50000, gamma=.99,
               target_model_update=10000, train_interval=4)

# Compilación del agente
#    Adam:          optimizador Adam
#    learning_rate: tasa de aprendizaje
#    metrics:       métrica MAE (Error Absoluto Medio)
dqn_propuesta_3.compile(Adam(learning_rate=.00025), metrics=['mae'])
```

[ ]:

```
# Training part
weights_filename = 'dqn_{}_weights.h5f'.format(env_name)
checkpoint_weights_filename = 'dqn_' + env_name + '_weights_{step}.h5f'
log_filename = 'dqn_{}_log.json'.format(env_name)
callbacks = [ModelIntervalCheckpoint(checkpoint_weights_filename,␣
↪interval=100000)]
callbacks += [FileLogger(log_filename, interval=100)]

# Comenzar el entrenamiento
```

```python
dqn_propuesta_3.fit(env, callbacks=callbacks, nb_steps=1000000,␣
 ↪log_interval=10000, visualize=False)
#dqn_propuesta_3.fit(env, callbacks=callbacks, nb_steps=20000,␣
 ↪log_interval=100, visualize=False)

dqn_propuesta_3.save_weights(weights_filename, overwrite=True)
```

```
Training for 1000000 steps …
Interval 1 (0 steps performed)
```

```
/home/mrlyon/miniconda3/envs/miar_rl/lib/python3.11/site-
packages/keras/src/engine/training_v1.py:2359: UserWarning:
`Model.state_updates` will be removed in a future version. This property should
not be used in TensorFlow 2.0, as `updates` are applied automatically.
  updates=self.state_updates,
```

```
10000/10000 [==============================] - 37s 4ms/step - reward: 0.0143
14 episodes - episode_reward: 9.857 [3.000, 18.000] - ale.lives: 2.063

Interval 2 (10000 steps performed)
10000/10000 [==============================] - 45s 4ms/step - reward: 0.0150
15 episodes - episode_reward: 10.067 [3.000, 29.000] - ale.lives: 2.131

Interval 3 (20000 steps performed)
10000/10000 [==============================] - 47s 5ms/step - reward: 0.0127
15 episodes - episode_reward: 7.933 [2.000, 22.000] - ale.lives: 2.055

Interval 4 (30000 steps performed)
10000/10000 [==============================] - 46s 5ms/step - reward: 0.0133
15 episodes - episode_reward: 9.600 [6.000, 14.000] - ale.lives: 2.123

Interval 5 (40000 steps performed)
10000/10000 [==============================] - 45s 4ms/step - reward: 0.0143
13 episodes - episode_reward: 10.231 [5.000, 17.000] - ale.lives: 2.082

Interval 6 (50000 steps performed)
10000/10000 [==============================] - 278s 28ms/step - reward: 0.0151
14 episodes - episode_reward: 11.571 [6.000, 16.000] - loss: 0.007 - mae: 0.030
- mean_q: 0.041 - mean_eps: 0.975 - ale.lives: 2.151

Interval 7 (60000 steps performed)
10000/10000 [==============================] - 292s 29ms/step - reward: 0.0125
14 episodes - episode_reward: 8.429 [2.000, 19.000] - loss: 0.007 - mae: 0.044 -
mean_q: 0.058 - mean_eps: 0.971 - ale.lives: 2.083

Interval 8 (70000 steps performed)
10000/10000 [==============================] - 285s 29ms/step - reward: 0.0135
15 episodes - episode_reward: 9.267 [5.000, 16.000] - loss: 0.007 - mae: 0.064 -
```

```
mean_q: 0.082 - mean_eps: 0.966 - ale.lives: 2.007

Interval 9 (80000 steps performed)
10000/10000 [==============================] - 286s 29ms/step - reward: 0.0130
15 episodes - episode_reward: 8.333 [3.000, 17.000] - loss: 0.007 - mae: 0.091 -
mean_q: 0.116 - mean_eps: 0.962 - ale.lives: 2.172

Interval 10 (90000 steps performed)
10000/10000 [==============================] - 280s 28ms/step - reward: 0.0144
15 episodes - episode_reward: 9.267 [4.000, 21.000] - loss: 0.006 - mae: 0.108 -
mean_q: 0.135 - mean_eps: 0.957 - ale.lives: 2.048

Interval 11 (100000 steps performed)
10000/10000 [==============================] - 270s 27ms/step - reward: 0.0154
16 episodes - episode_reward: 10.000 [5.000, 20.000] - loss: 0.007 - mae: 0.125
- mean_q: 0.156 - mean_eps: 0.953 - ale.lives: 2.225

Interval 12 (110000 steps performed)
10000/10000 [==============================] - 269s 27ms/step - reward: 0.0145
15 episodes - episode_reward: 9.733 [3.000, 20.000] - loss: 0.007 - mae: 0.146 -
mean_q: 0.181 - mean_eps: 0.948 - ale.lives: 2.139

Interval 13 (120000 steps performed)
10000/10000 [==============================] - 270s 27ms/step - reward: 0.0128
14 episodes - episode_reward: 9.429 [3.000, 15.000] - loss: 0.007 - mae: 0.162 -
mean_q: 0.199 - mean_eps: 0.944 - ale.lives: 2.242

Interval 14 (130000 steps performed)
10000/10000 [==============================] - 272s 27ms/step - reward: 0.0150
16 episodes - episode_reward: 9.500 [3.000, 20.000] - loss: 0.007 - mae: 0.187 -
mean_q: 0.231 - mean_eps: 0.939 - ale.lives: 2.076

Interval 15 (140000 steps performed)
10000/10000 [==============================] - 272s 27ms/step - reward: 0.0134
14 episodes - episode_reward: 9.286 [4.000, 20.000] - loss: 0.007 - mae: 0.210 -
mean_q: 0.257 - mean_eps: 0.935 - ale.lives: 2.083

Interval 16 (150000 steps performed)
10000/10000 [==============================] - 274s 27ms/step - reward: 0.0137
12 episodes - episode_reward: 11.750 [4.000, 20.000] - loss: 0.008 - mae: 0.240
- mean_q: 0.294 - mean_eps: 0.930 - ale.lives: 2.083

Interval 17 (160000 steps performed)
10000/10000 [==============================] - 274s 27ms/step - reward: 0.0143
16 episodes - episode_reward: 8.625 [4.000, 18.000] - loss: 0.008 - mae: 0.265 -
mean_q: 0.323 - mean_eps: 0.926 - ale.lives: 2.073

Interval 18 (170000 steps performed)
```

```
10000/10000 [==============================] - 277s 28ms/step - reward: 0.0154
15 episodes - episode_reward: 10.133 [6.000, 16.000] - loss: 0.008 - mae: 0.286
- mean_q: 0.350 - mean_eps: 0.921 - ale.lives: 2.105

Interval 19 (180000 steps performed)
10000/10000 [==============================] - 271s 27ms/step - reward: 0.0145
16 episodes - episode_reward: 8.500 [4.000, 12.000] - loss: 0.008 - mae: 0.305 -
mean_q: 0.373 - mean_eps: 0.917 - ale.lives: 2.128

Interval 20 (190000 steps performed)
10000/10000 [==============================] - 263s 26ms/step - reward: 0.0148
15 episodes - episode_reward: 10.333 [4.000, 17.000] - loss: 0.008 - mae: 0.337
- mean_q: 0.413 - mean_eps: 0.912 - ale.lives: 2.062

Interval 21 (200000 steps performed)
10000/10000 [==============================] - 280s 28ms/step - reward: 0.0150
16 episodes - episode_reward: 9.562 [5.000, 18.000] - loss: 0.009 - mae: 0.361 -
mean_q: 0.441 - mean_eps: 0.908 - ale.lives: 2.116

Interval 22 (210000 steps performed)
10000/10000 [==============================] - 270s 27ms/step - reward: 0.0143
14 episodes - episode_reward: 10.643 [5.000, 19.000] - loss: 0.009 - mae: 0.392
- mean_q: 0.478 - mean_eps: 0.903 - ale.lives: 2.169

Interval 23 (220000 steps performed)
10000/10000 [==============================] - 275s 27ms/step - reward: 0.0153
14 episodes - episode_reward: 10.857 [4.000, 19.000] - loss: 0.010 - mae: 0.417
- mean_q: 0.509 - mean_eps: 0.899 - ale.lives: 2.146

Interval 24 (230000 steps performed)
10000/10000 [==============================] - 270s 27ms/step - reward: 0.0155
13 episodes - episode_reward: 11.000 [5.000, 20.000] - loss: 0.010 - mae: 0.444
- mean_q: 0.542 - mean_eps: 0.894 - ale.lives: 2.024

Interval 25 (240000 steps performed)
10000/10000 [==============================] - 271s 27ms/step - reward: 0.0118
16 episodes - episode_reward: 8.188 [2.000, 15.000] - loss: 0.010 - mae: 0.482 -
mean_q: 0.588 - mean_eps: 0.890 - ale.lives: 2.058

Interval 26 (250000 steps performed)
10000/10000 [==============================] - 262s 26ms/step - reward: 0.0153
14 episodes - episode_reward: 10.857 [6.000, 18.000] - loss: 0.010 - mae: 0.497
- mean_q: 0.607 - mean_eps: 0.885 - ale.lives: 2.050

Interval 27 (260000 steps performed)
10000/10000 [==============================] - 266s 27ms/step - reward: 0.0158
16 episodes - episode_reward: 9.125 [4.000, 20.000] - loss: 0.010 - mae: 0.531 -
mean_q: 0.649 - mean_eps: 0.881 - ale.lives: 2.247
```

```
Interval 28 (270000 steps performed)
10000/10000 [==============================] - 273s 27ms/step - reward: 0.0162
15 episodes - episode_reward: 11.667 [4.000, 24.000] - loss: 0.010 - mae: 0.556
- mean_q: 0.679 - mean_eps: 0.876 - ale.lives: 2.097

Interval 29 (280000 steps performed)
10000/10000 [==============================] - 271s 27ms/step - reward: 0.0159
16 episodes - episode_reward: 9.938 [5.000, 18.000] - loss: 0.011 - mae: 0.573 -
mean_q: 0.701 - mean_eps: 0.872 - ale.lives: 2.150

Interval 30 (290000 steps performed)
10000/10000 [==============================] - 274s 27ms/step - reward: 0.0162
14 episodes - episode_reward: 11.000 [5.000, 21.000] - loss: 0.010 - mae: 0.586
- mean_q: 0.717 - mean_eps: 0.867 - ale.lives: 2.078

Interval 31 (300000 steps performed)
10000/10000 [==============================] - 279s 28ms/step - reward: 0.0154
15 episodes - episode_reward: 10.333 [4.000, 26.000] - loss: 0.011 - mae: 0.634
- mean_q: 0.777 - mean_eps: 0.863 - ale.lives: 2.162

Interval 32 (310000 steps performed)
10000/10000 [==============================] - 290s 29ms/step - reward: 0.0160
14 episodes - episode_reward: 11.500 [6.000, 23.000] - loss: 0.011 - mae: 0.660
- mean_q: 0.809 - mean_eps: 0.858 - ale.lives: 2.050

Interval 33 (320000 steps performed)
10000/10000 [==============================] - 297s 30ms/step - reward: 0.0164
14 episodes - episode_reward: 12.000 [5.000, 18.000] - loss: 0.012 - mae: 0.702
- mean_q: 0.859 - mean_eps: 0.854 - ale.lives: 2.179

Interval 34 (330000 steps performed)
10000/10000 [==============================] - 298s 30ms/step - reward: 0.0181
15 episodes - episode_reward: 12.200 [4.000, 27.000] - loss: 0.012 - mae: 0.721
- mean_q: 0.882 - mean_eps: 0.849 - ale.lives: 2.047

Interval 35 (340000 steps performed)
10000/10000 [==============================] - 301s 30ms/step - reward: 0.0179
14 episodes - episode_reward: 12.500 [6.000, 22.000] - loss: 0.013 - mae: 0.745
- mean_q: 0.912 - mean_eps: 0.845 - ale.lives: 2.178

Interval 36 (350000 steps performed)
10000/10000 [==============================] - 300s 30ms/step - reward: 0.0174
16 episodes - episode_reward: 10.562 [4.000, 23.000] - loss: 0.013 - mae: 0.759
- mean_q: 0.929 - mean_eps: 0.840 - ale.lives: 2.122

Interval 37 (360000 steps performed)
10000/10000 [==============================] - 303s 30ms/step - reward: 0.0169
```

```
15 episodes - episode_reward: 11.267 [5.000, 18.000] - loss: 0.012 - mae: 0.776
- mean_q: 0.949 - mean_eps: 0.836 - ale.lives: 2.064

Interval 38 (370000 steps performed)
10000/10000 [==============================] - 299s 30ms/step - reward: 0.0159
14 episodes - episode_reward: 11.429 [5.000, 17.000] - loss: 0.012 - mae: 0.808
- mean_q: 0.988 - mean_eps: 0.831 - ale.lives: 2.097

Interval 39 (380000 steps performed)
10000/10000 [==============================] - 297s 30ms/step - reward: 0.0169
17 episodes - episode_reward: 10.235 [2.000, 19.000] - loss: 0.014 - mae: 0.854
- mean_q: 1.042 - mean_eps: 0.827 - ale.lives: 2.063

Interval 40 (390000 steps performed)
10000/10000 [==============================] - 297s 30ms/step - reward: 0.0171
14 episodes - episode_reward: 11.429 [6.000, 21.000] - loss: 0.014 - mae: 0.889
- mean_q: 1.087 - mean_eps: 0.822 - ale.lives: 2.199

Interval 41 (400000 steps performed)
10000/10000 [==============================] - 311s 31ms/step - reward: 0.0172
14 episodes - episode_reward: 12.429 [3.000, 24.000] - loss: 0.014 - mae: 0.902
- mean_q: 1.100 - mean_eps: 0.818 - ale.lives: 2.083

Interval 42 (410000 steps performed)
10000/10000 [==============================] - 313s 31ms/step - reward: 0.0176
17 episodes - episode_reward: 10.294 [6.000, 22.000] - loss: 0.015 - mae: 0.934
- mean_q: 1.139 - mean_eps: 0.813 - ale.lives: 2.180

Interval 43 (420000 steps performed)
10000/10000 [==============================] - 315s 31ms/step - reward: 0.0174
14 episodes - episode_reward: 12.571 [8.000, 19.000] - loss: 0.015 - mae: 0.953
- mean_q: 1.160 - mean_eps: 0.809 - ale.lives: 1.988

Interval 44 (430000 steps performed)
10000/10000 [==============================] - 307s 31ms/step - reward: 0.0173
16 episodes - episode_reward: 11.312 [7.000, 21.000] - loss: 0.015 - mae: 0.982
- mean_q: 1.196 - mean_eps: 0.804 - ale.lives: 2.071

Interval 45 (440000 steps performed)
10000/10000 [==============================] - 311s 31ms/step - reward: 0.0182
12 episodes - episode_reward: 14.167 [6.000, 24.000] - loss: 0.014 - mae: 1.014
- mean_q: 1.234 - mean_eps: 0.800 - ale.lives: 2.001

Interval 46 (450000 steps performed)
10000/10000 [==============================] - 317s 32ms/step - reward: 0.0164
16 episodes - episode_reward: 10.938 [5.000, 21.000] - loss: 0.014 - mae: 1.034
- mean_q: 1.256 - mean_eps: 0.795 - ale.lives: 2.063
```

```
Interval 47 (460000 steps performed)
10000/10000 [==============================] - 317s 32ms/step - reward: 0.0175
13 episodes - episode_reward: 11.923 [3.000, 34.000] - loss: 0.014 - mae: 1.091
- mean_q: 1.326 - mean_eps: 0.791 - ale.lives: 2.075

Interval 48 (470000 steps performed)
10000/10000 [==============================] - 314s 31ms/step - reward: 0.0166
16 episodes - episode_reward: 11.312 [4.000, 26.000] - loss: 0.015 - mae: 1.129
- mean_q: 1.372 - mean_eps: 0.786 - ale.lives: 2.079

Interval 49 (480000 steps performed)
10000/10000 [==============================] - 330s 33ms/step - reward: 0.0175
15 episodes - episode_reward: 11.533 [4.000, 25.000] - loss: 0.015 - mae: 1.134
- mean_q: 1.376 - mean_eps: 0.782 - ale.lives: 2.140

Interval 50 (490000 steps performed)
10000/10000 [==============================] - 331s 33ms/step - reward: 0.0157
17 episodes - episode_reward: 9.588 [5.000, 17.000] - loss: 0.016 - mae: 1.151 -
mean_q: 1.397 - mean_eps: 0.777 - ale.lives: 2.034

Interval 51 (500000 steps performed)
10000/10000 [==============================] - 332s 33ms/step - reward: 0.0165
14 episodes - episode_reward: 12.000 [4.000, 29.000] - loss: 0.014 - mae: 1.178
- mean_q: 1.427 - mean_eps: 0.773 - ale.lives: 2.044

Interval 52 (510000 steps performed)
10000/10000 [==============================] - 329s 33ms/step - reward: 0.0193
13 episodes - episode_reward: 14.385 [6.000, 27.000] - loss: 0.014 - mae: 1.215
- mean_q: 1.472 - mean_eps: 0.768 - ale.lives: 2.042

Interval 53 (520000 steps performed)
10000/10000 [==============================] - 325s 33ms/step - reward: 0.0190
14 episodes - episode_reward: 13.286 [5.000, 28.000] - loss: 0.013 - mae: 1.218
- mean_q: 1.474 - mean_eps: 0.764 - ale.lives: 2.196

Interval 54 (530000 steps performed)
10000/10000 [==============================] - 334s 33ms/step - reward: 0.0171
17 episodes - episode_reward: 10.765 [5.000, 22.000] - loss: 0.013 - mae: 1.245
- mean_q: 1.508 - mean_eps: 0.759 - ale.lives: 2.133

Interval 55 (540000 steps performed)
10000/10000 [==============================] - 336s 34ms/step - reward: 0.0191
14 episodes - episode_reward: 12.786 [4.000, 24.000] - loss: 0.014 - mae: 1.277
- mean_q: 1.545 - mean_eps: 0.755 - ale.lives: 2.341

Interval 56 (550000 steps performed)
10000/10000 [==============================] - 335s 33ms/step - reward: 0.0166
15 episodes - episode_reward: 11.267 [3.000, 21.000] - loss: 0.014 - mae: 1.309
```

```
- mean_q: 1.585 - mean_eps: 0.750 - ale.lives: 2.027

Interval 57 (560000 steps performed)
10000/10000 [==============================] - 328s 33ms/step - reward: 0.0169
15 episodes - episode_reward: 10.600 [4.000, 20.000] - loss: 0.013 - mae: 1.332
- mean_q: 1.614 - mean_eps: 0.746 - ale.lives: 2.080

Interval 58 (570000 steps performed)
10000/10000 [==============================] - 334s 33ms/step - reward: 0.0191
15 episodes - episode_reward: 13.667 [4.000, 24.000] - loss: 0.013 - mae: 1.355
- mean_q: 1.640 - mean_eps: 0.741 - ale.lives: 2.200

Interval 59 (580000 steps performed)
10000/10000 [==============================] - 342s 34ms/step - reward: 0.0199
14 episodes - episode_reward: 13.786 [6.000, 24.000] - loss: 0.013 - mae: 1.393
- mean_q: 1.685 - mean_eps: 0.737 - ale.lives: 2.130

Interval 60 (590000 steps performed)
10000/10000 [==============================] - 348s 35ms/step - reward: 0.0188
14 episodes - episode_reward: 13.286 [5.000, 27.000] - loss: 0.014 - mae: 1.418
- mean_q: 1.716 - mean_eps: 0.732 - ale.lives: 2.046

Interval 61 (600000 steps performed)
10000/10000 [==============================] - 347s 35ms/step - reward: 0.0173
16 episodes - episode_reward: 11.125 [4.000, 17.000] - loss: 0.016 - mae: 1.441
- mean_q: 1.744 - mean_eps: 0.728 - ale.lives: 2.071

Interval 62 (610000 steps performed)
10000/10000 [==============================] - 353s 35ms/step - reward: 0.0177
16 episodes - episode_reward: 11.188 [5.000, 23.000] - loss: 0.014 - mae: 1.472
- mean_q: 1.780 - mean_eps: 0.723 - ale.lives: 2.158

Interval 63 (620000 steps performed)
10000/10000 [==============================] - 335s 34ms/step - reward: 0.0188
14 episodes - episode_reward: 13.857 [4.000, 24.000] - loss: 0.014 - mae: 1.502
- mean_q: 1.819 - mean_eps: 0.719 - ale.lives: 2.074

Interval 64 (630000 steps performed)
10000/10000 [==============================] - 347s 35ms/step - reward: 0.0181
14 episodes - episode_reward: 12.929 [5.000, 26.000] - loss: 0.015 - mae: 1.521
- mean_q: 1.841 - mean_eps: 0.714 - ale.lives: 2.127

Interval 65 (640000 steps performed)
10000/10000 [==============================] - 376s 38ms/step - reward: 0.0182
14 episodes - episode_reward: 12.429 [5.000, 24.000] - loss: 0.015 - mae: 1.551
- mean_q: 1.879 - mean_eps: 0.710 - ale.lives: 2.125

Interval 66 (650000 steps performed)
```

```
10000/10000 [==============================] - 369s 37ms/step - reward: 0.0187
14 episodes - episode_reward: 13.357 [7.000, 25.000] - loss: 0.015 - mae: 1.557
- mean_q: 1.883 - mean_eps: 0.705 - ale.lives: 2.092

Interval 67 (660000 steps performed)
10000/10000 [==============================] - 380s 38ms/step - reward: 0.0183
16 episodes - episode_reward: 11.750 [3.000, 21.000] - loss: 0.016 - mae: 1.575
- mean_q: 1.907 - mean_eps: 0.701 - ale.lives: 2.055

Interval 68 (670000 steps performed)
10000/10000 [==============================] - 364s 36ms/step - reward: 0.0189
15 episodes - episode_reward: 12.267 [7.000, 31.000] - loss: 0.015 - mae: 1.588
- mean_q: 1.918 - mean_eps: 0.696 - ale.lives: 2.129

Interval 69 (680000 steps performed)
10000/10000 [==============================] - 361s 36ms/step - reward: 0.0165
15 episodes - episode_reward: 11.133 [5.000, 25.000] - loss: 0.014 - mae: 1.617
- mean_q: 1.953 - mean_eps: 0.692 - ale.lives: 2.162

Interval 70 (690000 steps performed)
10000/10000 [==============================] - 363s 36ms/step - reward: 0.0179
14 episodes - episode_reward: 11.929 [5.000, 20.000] - loss: 0.020 - mae: 1.634
- mean_q: 1.978 - mean_eps: 0.687 - ale.lives: 2.225

Interval 71 (700000 steps performed)
10000/10000 [==============================] - 365s 37ms/step - reward: 0.0187
13 episodes - episode_reward: 15.154 [5.000, 28.000] - loss: 0.014 - mae: 1.652
- mean_q: 1.996 - mean_eps: 0.683 - ale.lives: 2.047

Interval 72 (710000 steps performed)
10000/10000 [==============================] - 368s 37ms/step - reward: 0.0197
15 episodes - episode_reward: 13.400 [7.000, 27.000] - loss: 0.018 - mae: 1.666
- mean_q: 2.014 - mean_eps: 0.678 - ale.lives: 2.037

Interval 73 (720000 steps performed)
10000/10000 [==============================] - 372s 37ms/step - reward: 0.0187
15 episodes - episode_reward: 12.533 [6.000, 26.000] - loss: 0.021 - mae: 1.683
- mean_q: 2.036 - mean_eps: 0.674 - ale.lives: 2.262

Interval 74 (730000 steps performed)
10000/10000 [==============================] - 373s 37ms/step - reward: 0.0185
15 episodes - episode_reward: 12.000 [4.000, 23.000] - loss: 0.017 - mae: 1.695
- mean_q: 2.048 - mean_eps: 0.669 - ale.lives: 2.043

Interval 75 (740000 steps performed)
10000/10000 [==============================] - 378s 38ms/step - reward: 0.0160
14 episodes - episode_reward: 11.643 [3.000, 23.000] - loss: 0.021 - mae: 1.707
- mean_q: 2.064 - mean_eps: 0.665 - ale.lives: 2.107
```

```
Interval 76 (750000 steps performed)
10000/10000 [==============================] - 391s 39ms/step - reward: 0.0164
15 episodes - episode_reward: 10.467 [4.000, 21.000] - loss: 0.022 - mae: 1.721
- mean_q: 2.083 - mean_eps: 0.660 - ale.lives: 2.068

Interval 77 (760000 steps performed)
10000/10000 [==============================] - 381s 38ms/step - reward: 0.0176
15 episodes - episode_reward: 12.333 [4.000, 23.000] - loss: 0.031 - mae: 1.744
- mean_q: 2.113 - mean_eps: 0.656 - ale.lives: 2.136

Interval 78 (770000 steps performed)
10000/10000 [==============================] - 382s 38ms/step - reward: 0.0183
14 episodes - episode_reward: 13.286 [5.000, 22.000] - loss: 0.111 - mae: 1.780
- mean_q: 2.165 - mean_eps: 0.651 - ale.lives: 2.172

Interval 79 (780000 steps performed)
10000/10000 [==============================] - 390s 39ms/step - reward: 0.0183
13 episodes - episode_reward: 13.077 [7.000, 21.000] - loss: 0.035 - mae: 1.816
- mean_q: 2.200 - mean_eps: 0.647 - ale.lives: 2.093

Interval 80 (790000 steps performed)
10000/10000 [==============================] - 391s 39ms/step - reward: 0.0190
15 episodes - episode_reward: 13.267 [8.000, 20.000] - loss: 0.022 - mae: 1.829
- mean_q: 2.210 - mean_eps: 0.642 - ale.lives: 2.101

Interval 81 (800000 steps performed)
10000/10000 [==============================] - 391s 39ms/step - reward: 0.0178
15 episodes - episode_reward: 12.133 [5.000, 19.000] - loss: 0.026 - mae: 1.867
- mean_q: 2.259 - mean_eps: 0.638 - ale.lives: 2.115

Interval 82 (810000 steps performed)
10000/10000 [==============================] - 397s 40ms/step - reward: 0.0194
13 episodes - episode_reward: 14.385 [7.000, 30.000] - loss: 0.018 - mae: 1.887
- mean_q: 2.280 - mean_eps: 0.633 - ale.lives: 2.127

Interval 83 (820000 steps performed)
10000/10000 [==============================] - 420s 42ms/step - reward: 0.0197
14 episodes - episode_reward: 12.786 [7.000, 25.000] - loss: 0.033 - mae: 1.937
- mean_q: 2.341 - mean_eps: 0.629 - ale.lives: 2.221

Interval 84 (830000 steps performed)
10000/10000 [==============================] - 418s 42ms/step - reward: 0.0191
13 episodes - episode_reward: 15.077 [5.000, 32.000] - loss: 0.021 - mae: 1.997
- mean_q: 2.413 - mean_eps: 0.624 - ale.lives: 2.098

Interval 85 (840000 steps performed)
10000/10000 [==============================] - 424s 42ms/step - reward: 0.0195
```

```
15 episodes - episode_reward: 13.733 [5.000, 22.000] - loss: 0.037 - mae: 2.030
- mean_q: 2.457 - mean_eps: 0.620 - ale.lives: 2.074

Interval 86 (850000 steps performed)
10000/10000 [==============================] - 410s 41ms/step - reward: 0.0223
13 episodes - episode_reward: 17.462 [10.000, 31.000] - loss: 0.020 - mae: 2.049
- mean_q: 2.477 - mean_eps: 0.615 - ale.lives: 1.985

Interval 87 (860000 steps performed)
10000/10000 [==============================] - 411s 41ms/step - reward: 0.0200
14 episodes - episode_reward: 14.500 [7.000, 25.000] - loss: 0.023 - mae: 2.090
- mean_q: 2.524 - mean_eps: 0.611 - ale.lives: 2.197

Interval 88 (870000 steps performed)
10000/10000 [==============================] - 414s 41ms/step - reward: 0.0218
13 episodes - episode_reward: 16.154 [6.000, 25.000] - loss: 0.021 - mae: 2.117
- mean_q: 2.557 - mean_eps: 0.606 - ale.lives: 2.009

Interval 89 (880000 steps performed)
10000/10000 [==============================] - 411s 41ms/step - reward: 0.0198
14 episodes - episode_reward: 14.857 [7.000, 24.000] - loss: 0.019 - mae: 2.126
- mean_q: 2.567 - mean_eps: 0.602 - ale.lives: 2.084

Interval 90 (890000 steps performed)
10000/10000 [==============================] - 407s 41ms/step - reward: 0.0213
12 episodes - episode_reward: 17.250 [8.000, 34.000] - loss: 0.020 - mae: 2.156
- mean_q: 2.605 - mean_eps: 0.597 - ale.lives: 2.166

Interval 91 (900000 steps performed)
10000/10000 [==============================] - 411s 41ms/step - reward: 0.0213
12 episodes - episode_reward: 17.500 [8.000, 29.000] - loss: 0.019 - mae: 2.198
- mean_q: 2.655 - mean_eps: 0.593 - ale.lives: 2.199

Interval 92 (910000 steps performed)
10000/10000 [==============================] - 402s 40ms/step - reward: 0.0203
14 episodes - episode_reward: 14.714 [7.000, 21.000] - loss: 0.019 - mae: 2.193
- mean_q: 2.648 - mean_eps: 0.588 - ale.lives: 2.120

Interval 93 (920000 steps performed)
10000/10000 [==============================] - 405s 40ms/step - reward: 0.0200
13 episodes - episode_reward: 14.692 [7.000, 22.000] - loss: 0.018 - mae: 2.217
- mean_q: 2.678 - mean_eps: 0.584 - ale.lives: 2.065

Interval 94 (930000 steps performed)
10000/10000 [==============================] - 416s 42ms/step - reward: 0.0203
14 episodes - episode_reward: 15.214 [8.000, 23.000] - loss: 0.020 - mae: 2.228
- mean_q: 2.692 - mean_eps: 0.579 - ale.lives: 1.981
```

```
Interval 95 (940000 steps performed)
10000/10000 [==============================] - 428s 43ms/step - reward: 0.0211
13 episodes - episode_reward: 16.154 [8.000, 23.000] - loss: 0.018 - mae: 2.258
- mean_q: 2.726 - mean_eps: 0.575 - ale.lives: 2.204

Interval 96 (950000 steps performed)
10000/10000 [==============================] - 413s 41ms/step - reward: 0.0216
12 episodes - episode_reward: 17.000 [6.000, 35.000] - loss: 0.018 - mae: 2.267
- mean_q: 2.735 - mean_eps: 0.570 - ale.lives: 2.076

Interval 97 (960000 steps performed)
10000/10000 [==============================] - 414s 41ms/step - reward: 0.0192
15 episodes - episode_reward: 13.133 [6.000, 20.000] - loss: 0.019 - mae: 2.283
- mean_q: 2.756 - mean_eps: 0.566 - ale.lives: 2.103

Interval 98 (970000 steps performed)
10000/10000 [==============================] - 442s 44ms/step - reward: 0.0201
13 episodes - episode_reward: 15.692 [7.000, 24.000] - loss: 0.019 - mae: 2.285
- mean_q: 2.757 - mean_eps: 0.561 - ale.lives: 2.042

Interval 99 (980000 steps performed)
10000/10000 [==============================] - 456s 46ms/step - reward: 0.0226
14 episodes - episode_reward: 16.071 [8.000, 27.000] - loss: 0.018 - mae: 2.301
- mean_q: 2.775 - mean_eps: 0.557 - ale.lives: 2.256

Interval 100 (990000 steps performed)
10000/10000 [==============================] - 464s 46ms/step - reward: 0.0200
done, took 32275.314 seconds
```

[ ]:
```python
# Testing part to calculate the mean reward
weights_filename = 'dqn_{}_weights.h5f'.format(env_name)
dqn_propuesta_3.load_weights(weights_filename)
dqn_propuesta_3.test(env, nb_episodes=10, visualize=False)
```

```
Testing for 10 episodes …
Episode 1: reward: 22.000, steps: 974
Episode 2: reward: 18.000, steps: 857
Episode 3: reward: 22.000, steps: 837
Episode 4: reward: 18.000, steps: 846
Episode 5: reward: 9.000, steps: 522
Episode 6: reward: 25.000, steps: 977
Episode 7: reward: 24.000, steps: 850
Episode 8: reward: 23.000, steps: 913
Episode 9: reward: 25.000, steps: 979
Episode 10: reward: 27.000, steps: 933
```

[ ]:
```
<keras.src.callbacks.History at 0x71a5d76c9ed0>
```

3.6.3 Generación de vídeo

[ ]:
```
import tensorflow as tf
tf.config.list_physical_devices('GPU')
```

[ ]: [PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU')]

[ ]:
```
# Testing part
from gym import wrappers

if IN_COLAB:
    env = wrappers.Monitor(env, "/content/drive/", force=True)
else:
    env = wrappers.Monitor(env, "./videos_grabados/", force=True)
env.reset()

weights_filename = 'dqn_{}_weights.h5f'.format(env_name)
dqn_propuesta_3.load_weights(weights_filename)
dqn_propuesta_3.test(env, nb_episodes=10, visualize=False)
```

```
Testing for 10 episodes …
Episode 1: reward: 19.000, steps: 671
Episode 2: reward: 15.000, steps: 579
Episode 3: reward: 18.000, steps: 904
Episode 4: reward: 12.000, steps: 666
Episode 5: reward: 21.000, steps: 935
Episode 6: reward: 22.000, steps: 787
Episode 7: reward: 22.000, steps: 994
Episode 8: reward: 30.000, steps: 1364
Episode 9: reward: 11.000, steps: 453
Episode 10: reward: 21.000, steps: 806
```

[ ]: <keras.src.callbacks.History at 0x71a5d763b690>

[ ]:
```
import io
import base64
from IPython.display import HTML

if IN_COLAB:
    video = io.open('/content/drive/openaigym.video.%s.video000008.mp4' % env.
 ↪file_infix, 'r+b').read()
else:
    video = io.open('./videos_grabados/openaigym.video.%s.video000008.mp4' %␣
 ↪env.file_infix, 'r+b').read()

encoded = base64.b64encode(video)
HTML(data='''
```

```
    <video width="360" height="auto" alt="test" controls><source src="data:
↪video/mp4;base64,{0}" type="video/mp4" /></video>'''
.format(encoded.decode('ascii')))
```

[ ]: <IPython.core.display.HTML object>

1.5 PARTE 4. Evaluación comparativa y análisis de resultados

En esta parte se comparan de forma homogénea el DQN base y las tres propuestas de mejora. La comparación se realiza sobre recompensas en modo test, no sobre la recompensa acumulada durante entrenamiento.

1.5.1 4.1 Reconstrucción de agentes para evaluación homogénea

Se vuelven a definir las arquitecturas y agentes necesarios para evaluar los cuatro modelos bajo el mismo procedimiento. Esto evita depender de objetos intermedios generados durante el entrenamiento y permite cargar los pesos de cada propuesta de forma ordenada.

La duración de la evaluación será la establecida en los criterios (200 episodios).

[ ]:
```
# Red neuronal inicial (DQN Base, Propuesta 1 y Propuesta 2)
def red_neuronal_inicial():
    input_shape = (WINDOW_LENGTH,) + INPUT_SHAPE
    model = Sequential(name="DQN_SpaceInvaders")
    model.add(InputLayer(input_shape=input_shape, name="input_frames"))
    model.add(Permute((2, 3, 1), name="permute_channels"))
    model.add(Conv2D(32, (8, 8), strides=(4, 4), activation="relu",␣
↪name="conv1"))
    model.add(Conv2D(64, (4, 4), strides=(2, 2), activation="relu",␣
↪name="conv2"))
    model.add(Conv2D(64, (3, 3), strides=(1, 1), activation="relu",␣
↪name="conv3"))
    model.add(Flatten(name="flatten"))
    model.add(Dense(512, activation="relu", name="dense_hidden"))
    model.add(Dense(nb_actions, activation="linear", name="output_values"))
    return model


# Configuración DQN Base
memory = SequentialMemory(limit=1000000, window_length=WINDOW_LENGTH)
processor = AtariProcessor()
policy = LinearAnnealedPolicy(EpsGreedyQPolicy(), attr='eps',
                              value_max=1., value_min=.1, value_test=.05,
                              nb_steps=2000000)
dqn = DQNAgent(model=red_neuronal_inicial(), nb_actions=nb_actions,␣
↪policy=policy,
```

```
               memory=memory, processor=processor,
               nb_steps_warmup=50000, gamma=.99,
               target_model_update=10000,
               train_interval=4)
dqn.compile(Adam(learning_rate=.00025), metrics=['mae'])


# Configuración DQN Propuesta 1
memory_propuesta_1 = SequentialMemory(limit=1000000,␣
 ↪window_length=WINDOW_LENGTH)
processor_propuesta_1 = AtariProcessor()
policy_propuesta_1 = LinearAnnealedPolicy(EpsGreedyQPolicy(), attr='eps',
                              value_max=1., value_min=.1, value_test=.05,
                              nb_steps=2000000)
dqn_propuesta_1 = DQNAgent(model=red_neuronal_inicial(), nb_actions=nb_actions,␣
 ↪policy=policy_propuesta_1,
               memory=memory_propuesta_1, processor=processor_propuesta_1,
               nb_steps_warmup=50000, gamma=.99,
               target_model_update=10000,
               train_interval=4, enable_double_dqn=True)
dqn_propuesta_1.compile(Adam(learning_rate=.00025), metrics=['mae'])


# Configuración DQN Propuesta 2
memory_propuesta_2 = SequentialMemory(limit=1000000,␣
 ↪window_length=WINDOW_LENGTH)
processor_propuesta_2 = AtariProcessor()
policy_propuesta_2 = LinearAnnealedPolicy(EpsGreedyQPolicy(), attr='eps',
                              value_max=1., value_min=.1, value_test=.05,
                              nb_steps=2000000)
dqn_propuesta_2 = DQNAgent(model=red_neuronal_inicial(), nb_actions=nb_actions,␣
 ↪policy=policy_propuesta_2,
               memory=memory_propuesta_2, processor=processor_propuesta_2,
               nb_steps_warmup=50000, gamma=.99,
               target_model_update=1e-3, train_interval=4)
dqn_propuesta_2.compile(Adam(learning_rate=.00025), metrics=['mae'])


# Red neuronal segunda (DQN Propuesta 3)
def red_neuronal_propuesta3():
    input_shape = (WINDOW_LENGTH,) + INPUT_SHAPE
    model_propuesta_3 = Sequential(name="DQN_SpaceInvaders_Mejorada")
    model_propuesta_3.add(InputLayer(input_shape=input_shape,␣
 ↪name="input_frames3"))
    model_propuesta_3.add(Permute((2, 3, 1), name="permute_channels3"))
    model_propuesta_3.add(Conv2D(32, (8, 8), strides=(4, 4), activation="relu",␣
 ↪name="conv13"))
```

```
    model_propuesta_3.add(Conv2D(64, (4, 4), strides=(2, 2), activation="relu",␣
 ↪name="conv23"))
    model_propuesta_3.add(Conv2D(64, (3, 3), strides=(1, 1), activation="relu",␣
 ↪name="conv33"))
    model_propuesta_3.add(Flatten(name="flatten3"))
    model_propuesta_3.add(Dense(1024, activation="relu",␣
 ↪name="new_dense_hidden3"))
    model_propuesta_3.add(Dense(512, activation="relu", name="dense_hidden3"))
    model_propuesta_3.add(Dense(nb_actions, activation="linear",␣
 ↪name="output_values3"))
    return model_propuesta_3


# Configuración DQN Propuesta 3
memory_propuesta_3 = SequentialMemory(limit=1000000,␣
 ↪window_length=WINDOW_LENGTH)
processor_propuesta_3 = AtariProcessor()
policy_propuesta_3 = LinearAnnealedPolicy(EpsGreedyQPolicy(), attr='eps',
                              value_max=1., value_min=.1, value_test=.05,
                              nb_steps=2000000)
dqn_propuesta_3 = DQNAgent(model=red_neuronal_propuesta3(),␣
 ↪nb_actions=nb_actions, policy=policy_propuesta_3,
               memory=memory_propuesta_3, processor=processor_propuesta_3,
               nb_steps_warmup=50000, gamma=.99,
               target_model_update=10000, train_interval=4)
dqn_propuesta_3.compile(Adam(learning_rate=.00025), metrics=['mae'])
```

1.5.2 4.2 Test de los agentes

Se evalúa cada agente durante 200 episodios de test. Las recompensas de cada episodio se almacenan en el diccionario `resultados`, que será la base de la tabla resumen y de las gráficas comparativas.

[ ]:
```
import os
import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.callbacks import Callback

# Definir el Callback recolector
class GuardarResultadosCallback(Callback):
    def __init__(self):
        super().__init__()
        self.rewards = []
    def on_episode_end(self, episode, logs={}):
        self.rewards.append(logs.get('episode_reward'))

propuestas = [
#   (agente_dqn,      carpeta_de_los_pesos,                    etiqueta),
```

```python
    (dqn,              "dqn_SpaceInvaders-v0_Train_Base",       "DQN Base"),
    (dqn_propuesta_1, "dqn_SpaceInvaders-v0_Train_Propuesta1", "DQN Propuesta␣
 ↪1"),
    (dqn_propuesta_2, "dqn_SpaceInvaders-v0_Train_Propuesta2", "DQN Propuesta␣
 ↪2"),
    (dqn_propuesta_3, "dqn_SpaceInvaders-v0_Train_Propuesta3", "DQN Propuesta␣
 ↪3")
]

resultados = {}

# Bucle para testeo automatico de los resultados
for agente_dqn, carpeta_pesos, etiqueta in propuestas:
    print(f"\n--- Evaluando: {etiqueta} ---")

    # Carpeta de los pesos de cada agente
    nombre_archivo = 'dqn_{}_weights.h5f'.format(env_name)
    weights_filename = os.path.join(carpeta_pesos, nombre_archivo)

    # Cargar los pesos de cada agente
    agente_dqn.load_weights(weights_filename)
    callback_recolector = GuardarResultadosCallback()

    # Evaluar cada agente
    #agente_dqn.test(env, nb_episodes=10, visualize=False,␣
 ↪callbacks=[callback_recolector])
    agente_dqn.test(env, nb_episodes=200, visualize=False,␣
 ↪callbacks=[callback_recolector])

    # Guardar en el array los resultados por cada agente
    resultados[etiqueta] = np.array(callback_recolector.rewards)
```

```
--- Evaluando: DQN Base ---
Testing for 200 episodes …
Episode 1: reward: 15.000, steps: 954
Episode 2: reward: 9.000, steps: 638
Episode 3: reward: 10.000, steps: 645
Episode 4: reward: 3.000, steps: 410
Episode 5: reward: 15.000, steps: 720
Episode 6: reward: 15.000, steps: 777
Episode 7: reward: 22.000, steps: 1388
Episode 8: reward: 6.000, steps: 377
Episode 9: reward: 16.000, steps: 1007
Episode 10: reward: 10.000, steps: 626
Episode 11: reward: 11.000, steps: 868
Episode 12: reward: 6.000, steps: 506
```

```
Episode 13: reward: 18.000, steps: 973
Episode 14: reward: 15.000, steps: 786
Episode 15: reward: 7.000, steps: 505
Episode 16: reward: 18.000, steps: 1153
Episode 17: reward: 22.000, steps: 1126
Episode 18: reward: 6.000, steps: 486
Episode 19: reward: 10.000, steps: 888
Episode 20: reward: 15.000, steps: 907
Episode 21: reward: 21.000, steps: 1273
Episode 22: reward: 14.000, steps: 675
Episode 23: reward: 22.000, steps: 1178
Episode 24: reward: 22.000, steps: 931
Episode 25: reward: 19.000, steps: 1398
Episode 26: reward: 7.000, steps: 564
Episode 27: reward: 17.000, steps: 865
Episode 28: reward: 7.000, steps: 642
Episode 29: reward: 11.000, steps: 621
Episode 30: reward: 7.000, steps: 464
Episode 31: reward: 9.000, steps: 595
Episode 32: reward: 6.000, steps: 358
Episode 33: reward: 7.000, steps: 634
Episode 34: reward: 16.000, steps: 963
Episode 35: reward: 21.000, steps: 1204
Episode 36: reward: 6.000, steps: 479
Episode 37: reward: 24.000, steps: 1132
Episode 38: reward: 17.000, steps: 734
Episode 39: reward: 11.000, steps: 785
Episode 40: reward: 15.000, steps: 1017
Episode 41: reward: 12.000, steps: 655
Episode 42: reward: 8.000, steps: 633
Episode 43: reward: 10.000, steps: 646
Episode 44: reward: 10.000, steps: 656
Episode 45: reward: 12.000, steps: 805
Episode 46: reward: 25.000, steps: 1291
Episode 47: reward: 18.000, steps: 988
Episode 48: reward: 11.000, steps: 836
Episode 49: reward: 7.000, steps: 619
Episode 50: reward: 13.000, steps: 808
Episode 51: reward: 14.000, steps: 959
Episode 52: reward: 14.000, steps: 826
Episode 53: reward: 4.000, steps: 515
Episode 54: reward: 8.000, steps: 668
Episode 55: reward: 27.000, steps: 1215
Episode 56: reward: 13.000, steps: 1007
Episode 57: reward: 17.000, steps: 1101
Episode 58: reward: 17.000, steps: 905
Episode 59: reward: 23.000, steps: 1127
Episode 60: reward: 13.000, steps: 957
```

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Episode 61: reward: 28.000, steps: 1282
Episode 62: reward: 9.000, steps: 716
Episode 63: reward: 7.000, steps: 491
Episode 64: reward: 6.000, steps: 477
Episode 65: reward: 11.000, steps: 685
Episode 66: reward: 20.000, steps: 1055
Episode 67: reward: 4.000, steps: 489
Episode 68: reward: 15.000, steps: 1076
Episode 69: reward: 16.000, steps: 1110
Episode 70: reward: 17.000, steps: 1072
Episode 71: reward: 13.000, steps: 891
Episode 72: reward: 28.000, steps: 1520
Episode 73: reward: 14.000, steps: 918
Episode 74: reward: 7.000, steps: 607
Episode 75: reward: 9.000, steps: 574
Episode 76: reward: 16.000, steps: 949
Episode 77: reward: 7.000, steps: 676
Episode 78: reward: 12.000, steps: 677
Episode 79: reward: 28.000, steps: 1392
Episode 80: reward: 13.000, steps: 815
Episode 81: reward: 14.000, steps: 888
Episode 82: reward: 9.000, steps: 795
Episode 83: reward: 9.000, steps: 617
Episode 84: reward: 10.000, steps: 789
Episode 85: reward: 22.000, steps: 1036
Episode 86: reward: 23.000, steps: 1144
Episode 87: reward: 5.000, steps: 632
Episode 88: reward: 16.000, steps: 1130
Episode 89: reward: 17.000, steps: 1004
Episode 90: reward: 10.000, steps: 646
Episode 91: reward: 20.000, steps: 1128
Episode 92: reward: 8.000, steps: 716
Episode 93: reward: 22.000, steps: 1239
Episode 94: reward: 2.000, steps: 381
Episode 95: reward: 26.000, steps: 1528
Episode 96: reward: 17.000, steps: 1004
Episode 97: reward: 14.000, steps: 953
Episode 98: reward: 20.000, steps: 1019
Episode 99: reward: 12.000, steps: 1079
Episode 100: reward: 23.000, steps: 1074
Episode 101: reward: 19.000, steps: 924
Episode 102: reward: 14.000, steps: 923
Episode 103: reward: 5.000, steps: 479
Episode 104: reward: 9.000, steps: 766
Episode 105: reward: 16.000, steps: 1045
Episode 106: reward: 15.000, steps: 897
Episode 107: reward: 15.000, steps: 953
Episode 108: reward: 11.000, steps: 806
```

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Episode 109: reward: 16.000, steps: 1222
Episode 110: reward: 13.000, steps: 664
Episode 111: reward: 18.000, steps: 1179
Episode 112: reward: 18.000, steps: 896
Episode 113: reward: 7.000, steps: 632
Episode 114: reward: 8.000, steps: 651
Episode 115: reward: 4.000, steps: 536
Episode 116: reward: 11.000, steps: 708
Episode 117: reward: 23.000, steps: 1263
Episode 118: reward: 5.000, steps: 487
Episode 119: reward: 10.000, steps: 717
Episode 120: reward: 4.000, steps: 471
Episode 121: reward: 15.000, steps: 906
Episode 122: reward: 24.000, steps: 1083
Episode 123: reward: 4.000, steps: 493
Episode 124: reward: 12.000, steps: 1257
Episode 125: reward: 20.000, steps: 995
Episode 126: reward: 16.000, steps: 866
Episode 127: reward: 10.000, steps: 806
Episode 128: reward: 17.000, steps: 1007
Episode 129: reward: 18.000, steps: 1199
Episode 130: reward: 17.000, steps: 1090
Episode 131: reward: 7.000, steps: 652
Episode 132: reward: 25.000, steps: 1193
Episode 133: reward: 9.000, steps: 668
Episode 134: reward: 7.000, steps: 670
Episode 135: reward: 16.000, steps: 1178
Episode 136: reward: 13.000, steps: 793
Episode 137: reward: 7.000, steps: 482
Episode 138: reward: 12.000, steps: 952
Episode 139: reward: 23.000, steps: 1029
Episode 140: reward: 18.000, steps: 1025
Episode 141: reward: 15.000, steps: 909
Episode 142: reward: 18.000, steps: 998
Episode 143: reward: 24.000, steps: 1314
Episode 144: reward: 24.000, steps: 1165
Episode 145: reward: 13.000, steps: 712
Episode 146: reward: 11.000, steps: 888
Episode 147: reward: 18.000, steps: 905
Episode 148: reward: 17.000, steps: 1106
Episode 149: reward: 21.000, steps: 1038
Episode 150: reward: 22.000, steps: 1031
Episode 151: reward: 15.000, steps: 1018
Episode 152: reward: 12.000, steps: 683
Episode 153: reward: 9.000, steps: 701
Episode 154: reward: 13.000, steps: 668
Episode 155: reward: 13.000, steps: 750
Episode 156: reward: 15.000, steps: 934
```

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Episode 157: reward: 21.000, steps: 1014
Episode 158: reward: 12.000, steps: 713
Episode 159: reward: 17.000, steps: 995
Episode 160: reward: 26.000, steps: 1074
Episode 161: reward: 11.000, steps: 639
Episode 162: reward: 8.000, steps: 664
Episode 163: reward: 24.000, steps: 1248
Episode 164: reward: 7.000, steps: 660
Episode 165: reward: 20.000, steps: 1165
Episode 166: reward: 16.000, steps: 1015
Episode 167: reward: 11.000, steps: 526
Episode 168: reward: 15.000, steps: 875
Episode 169: reward: 6.000, steps: 517
Episode 170: reward: 17.000, steps: 1197
Episode 171: reward: 14.000, steps: 950
Episode 172: reward: 9.000, steps: 966
Episode 173: reward: 19.000, steps: 1188
Episode 174: reward: 9.000, steps: 656
Episode 175: reward: 8.000, steps: 838
Episode 176: reward: 17.000, steps: 1036
Episode 177: reward: 17.000, steps: 835
Episode 178: reward: 7.000, steps: 495
Episode 179: reward: 9.000, steps: 692
Episode 180: reward: 5.000, steps: 774
Episode 181: reward: 8.000, steps: 914
Episode 182: reward: 13.000, steps: 722
Episode 183: reward: 17.000, steps: 877
Episode 184: reward: 23.000, steps: 1125
Episode 185: reward: 6.000, steps: 621
Episode 186: reward: 19.000, steps: 1236
Episode 187: reward: 26.000, steps: 1431
Episode 188: reward: 10.000, steps: 810
Episode 189: reward: 19.000, steps: 977
Episode 190: reward: 12.000, steps: 760
Episode 191: reward: 7.000, steps: 559
Episode 192: reward: 17.000, steps: 1010
Episode 193: reward: 20.000, steps: 1052
Episode 194: reward: 22.000, steps: 1083
Episode 195: reward: 12.000, steps: 937
Episode 196: reward: 23.000, steps: 1217
Episode 197: reward: 5.000, steps: 653
Episode 198: reward: 16.000, steps: 1019
Episode 199: reward: 12.000, steps: 942
Episode 200: reward: 12.000, steps: 937

--- Evaluando: DQN Propuesta 1 ---
Testing for 200 episodes …
Episode 1: reward: 24.000, steps: 1258
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```
Episode 2: reward: 15.000, steps: 657
Episode 3: reward: 13.000, steps: 610
Episode 4: reward: 7.000, steps: 358
Episode 5: reward: 11.000, steps: 548
Episode 6: reward: 20.000, steps: 938
Episode 7: reward: 20.000, steps: 1400
Episode 8: reward: 19.000, steps: 1049
Episode 9: reward: 22.000, steps: 942
Episode 10: reward: 18.000, steps: 732
Episode 11: reward: 11.000, steps: 504
Episode 12: reward: 11.000, steps: 566
Episode 13: reward: 13.000, steps: 886
Episode 14: reward: 11.000, steps: 818
Episode 15: reward: 11.000, steps: 546
Episode 16: reward: 13.000, steps: 983
Episode 17: reward: 10.000, steps: 532
Episode 18: reward: 22.000, steps: 1094
Episode 19: reward: 13.000, steps: 844
Episode 20: reward: 15.000, steps: 894
Episode 21: reward: 12.000, steps: 874
Episode 22: reward: 13.000, steps: 701
Episode 23: reward: 16.000, steps: 1077
Episode 24: reward: 17.000, steps: 920
Episode 25: reward: 14.000, steps: 974
Episode 26: reward: 18.000, steps: 702
Episode 27: reward: 15.000, steps: 1054
Episode 28: reward: 9.000, steps: 655
Episode 29: reward: 12.000, steps: 904
Episode 30: reward: 17.000, steps: 927
Episode 31: reward: 10.000, steps: 577
Episode 32: reward: 21.000, steps: 1323
Episode 33: reward: 12.000, steps: 531
Episode 34: reward: 21.000, steps: 1160
Episode 35: reward: 11.000, steps: 481
Episode 36: reward: 17.000, steps: 958
Episode 37: reward: 12.000, steps: 655
Episode 38: reward: 15.000, steps: 990
Episode 39: reward: 7.000, steps: 554
Episode 40: reward: 9.000, steps: 510
Episode 41: reward: 18.000, steps: 1028
Episode 42: reward: 18.000, steps: 804
Episode 43: reward: 10.000, steps: 528
Episode 44: reward: 17.000, steps: 799
Episode 45: reward: 32.000, steps: 1613
Episode 46: reward: 19.000, steps: 781
Episode 47: reward: 16.000, steps: 906
Episode 48: reward: 11.000, steps: 650
Episode 49: reward: 10.000, steps: 659
```

```
Episode 50: reward: 11.000, steps: 540
Episode 51: reward: 25.000, steps: 957
Episode 52: reward: 11.000, steps: 500
Episode 53: reward: 17.000, steps: 1119
Episode 54: reward: 13.000, steps: 853
Episode 55: reward: 8.000, steps: 490
Episode 56: reward: 9.000, steps: 683
Episode 57: reward: 29.000, steps: 1448
Episode 58: reward: 19.000, steps: 707
Episode 59: reward: 21.000, steps: 1302
Episode 60: reward: 17.000, steps: 721
Episode 61: reward: 12.000, steps: 867
Episode 62: reward: 12.000, steps: 616
Episode 63: reward: 11.000, steps: 494
Episode 64: reward: 22.000, steps: 898
Episode 65: reward: 12.000, steps: 761
Episode 66: reward: 15.000, steps: 1068
Episode 67: reward: 20.000, steps: 1102
Episode 68: reward: 14.000, steps: 654
Episode 69: reward: 18.000, steps: 1235
Episode 70: reward: 12.000, steps: 968
Episode 71: reward: 25.000, steps: 1429
Episode 72: reward: 17.000, steps: 878
Episode 73: reward: 19.000, steps: 851
Episode 74: reward: 15.000, steps: 884
Episode 75: reward: 12.000, steps: 850
Episode 76: reward: 7.000, steps: 406
Episode 77: reward: 16.000, steps: 740
Episode 78: reward: 12.000, steps: 995
Episode 79: reward: 12.000, steps: 987
Episode 80: reward: 11.000, steps: 619
Episode 81: reward: 15.000, steps: 943
Episode 82: reward: 13.000, steps: 951
Episode 83: reward: 11.000, steps: 550
Episode 84: reward: 12.000, steps: 672
Episode 85: reward: 26.000, steps: 1338
Episode 86: reward: 8.000, steps: 495
Episode 87: reward: 11.000, steps: 555
Episode 88: reward: 8.000, steps: 440
Episode 89: reward: 5.000, steps: 365
Episode 90: reward: 6.000, steps: 363
Episode 91: reward: 14.000, steps: 888
Episode 92: reward: 20.000, steps: 999
Episode 93: reward: 16.000, steps: 1170
Episode 94: reward: 14.000, steps: 894
Episode 95: reward: 27.000, steps: 1190
Episode 96: reward: 12.000, steps: 843
Episode 97: reward: 12.000, steps: 849
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Episode 98: reward: 9.000, steps: 620
Episode 99: reward: 16.000, steps: 1068
Episode 100: reward: 16.000, steps: 761
Episode 101: reward: 20.000, steps: 870
Episode 102: reward: 9.000, steps: 563
Episode 103: reward: 10.000, steps: 560
Episode 104: reward: 21.000, steps: 929
Episode 105: reward: 16.000, steps: 846
Episode 106: reward: 13.000, steps: 950
Episode 107: reward: 14.000, steps: 627
Episode 108: reward: 20.000, steps: 1057
Episode 109: reward: 14.000, steps: 875
Episode 110: reward: 16.000, steps: 711
Episode 111: reward: 17.000, steps: 958
Episode 112: reward: 28.000, steps: 1196
Episode 113: reward: 18.000, steps: 795
Episode 114: reward: 20.000, steps: 842
Episode 115: reward: 17.000, steps: 957
Episode 116: reward: 13.000, steps: 957
Episode 117: reward: 7.000, steps: 378
Episode 118: reward: 22.000, steps: 1357
Episode 119: reward: 16.000, steps: 664
Episode 120: reward: 5.000, steps: 480
Episode 121: reward: 15.000, steps: 629
Episode 122: reward: 15.000, steps: 931
Episode 123: reward: 15.000, steps: 896
Episode 124: reward: 6.000, steps: 379
Episode 125: reward: 17.000, steps: 1018
Episode 126: reward: 19.000, steps: 928
Episode 127: reward: 11.000, steps: 564
Episode 128: reward: 12.000, steps: 583
Episode 129: reward: 21.000, steps: 1390
Episode 130: reward: 26.000, steps: 1251
Episode 131: reward: 15.000, steps: 634
Episode 132: reward: 8.000, steps: 483
Episode 133: reward: 12.000, steps: 569
Episode 134: reward: 17.000, steps: 1108
Episode 135: reward: 14.000, steps: 873
Episode 136: reward: 9.000, steps: 531
Episode 137: reward: 22.000, steps: 1078
Episode 138: reward: 19.000, steps: 782
Episode 139: reward: 17.000, steps: 744
Episode 140: reward: 7.000, steps: 372
Episode 141: reward: 13.000, steps: 1003
Episode 142: reward: 10.000, steps: 547
Episode 143: reward: 16.000, steps: 1064
Episode 144: reward: 10.000, steps: 496
Episode 145: reward: 12.000, steps: 877
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Episode 146: reward: 16.000, steps: 769
Episode 147: reward: 15.000, steps: 943
Episode 148: reward: 23.000, steps: 1392
Episode 149: reward: 16.000, steps: 774
Episode 150: reward: 28.000, steps: 1091
Episode 151: reward: 16.000, steps: 617
Episode 152: reward: 12.000, steps: 651
Episode 153: reward: 17.000, steps: 1067
Episode 154: reward: 19.000, steps: 867
Episode 155: reward: 22.000, steps: 1312
Episode 156: reward: 18.000, steps: 849
Episode 157: reward: 27.000, steps: 1493
Episode 158: reward: 26.000, steps: 1339
Episode 159: reward: 13.000, steps: 832
Episode 160: reward: 22.000, steps: 1104
Episode 161: reward: 25.000, steps: 1293
Episode 162: reward: 15.000, steps: 907
Episode 163: reward: 16.000, steps: 1032
Episode 164: reward: 16.000, steps: 961
Episode 165: reward: 20.000, steps: 1260
Episode 166: reward: 13.000, steps: 856
Episode 167: reward: 11.000, steps: 499
Episode 168: reward: 16.000, steps: 727
Episode 169: reward: 12.000, steps: 970
Episode 170: reward: 6.000, steps: 376
Episode 171: reward: 18.000, steps: 1128
Episode 172: reward: 24.000, steps: 1439
Episode 173: reward: 12.000, steps: 877
Episode 174: reward: 28.000, steps: 1461
Episode 175: reward: 17.000, steps: 922
Episode 176: reward: 10.000, steps: 453
Episode 177: reward: 12.000, steps: 644
Episode 178: reward: 15.000, steps: 924
Episode 179: reward: 13.000, steps: 517
Episode 180: reward: 14.000, steps: 908
Episode 181: reward: 5.000, steps: 500
Episode 182: reward: 15.000, steps: 852
Episode 183: reward: 12.000, steps: 585
Episode 184: reward: 18.000, steps: 1113
Episode 185: reward: 17.000, steps: 916
Episode 186: reward: 17.000, steps: 1066
Episode 187: reward: 9.000, steps: 437
Episode 188: reward: 14.000, steps: 919
Episode 189: reward: 12.000, steps: 837
Episode 190: reward: 29.000, steps: 1225
Episode 191: reward: 17.000, steps: 954
Episode 192: reward: 16.000, steps: 812
Episode 193: reward: 12.000, steps: 555
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Episode 194: reward: 22.000, steps: 1510
Episode 195: reward: 16.000, steps: 1082
Episode 196: reward: 15.000, steps: 765
Episode 197: reward: 17.000, steps: 708
Episode 198: reward: 11.000, steps: 537
Episode 199: reward: 12.000, steps: 513
Episode 200: reward: 16.000, steps: 758

--- Evaluando: DQN Propuesta 2 ---
Testing for 200 episodes …
Episode 1: reward: 12.000, steps: 720
Episode 2: reward: 7.000, steps: 493
Episode 3: reward: 12.000, steps: 713
Episode 4: reward: 19.000, steps: 1179
Episode 5: reward: 21.000, steps: 1317
Episode 6: reward: 9.000, steps: 987
Episode 7: reward: 2.000, steps: 488
Episode 8: reward: 16.000, steps: 918
Episode 9: reward: 17.000, steps: 1045
Episode 10: reward: 11.000, steps: 943
Episode 11: reward: 4.000, steps: 686
Episode 12: reward: 8.000, steps: 604
Episode 13: reward: 12.000, steps: 644
Episode 14: reward: 8.000, steps: 693
Episode 15: reward: 11.000, steps: 706
Episode 16: reward: 9.000, steps: 679
Episode 17: reward: 13.000, steps: 1264
Episode 18: reward: 20.000, steps: 947
Episode 19: reward: 14.000, steps: 921
Episode 20: reward: 9.000, steps: 799
Episode 21: reward: 9.000, steps: 708
Episode 22: reward: 18.000, steps: 1075
Episode 23: reward: 14.000, steps: 646
Episode 24: reward: 6.000, steps: 489
Episode 25: reward: 4.000, steps: 526
Episode 26: reward: 6.000, steps: 631
Episode 27: reward: 15.000, steps: 1051
Episode 28: reward: 9.000, steps: 723
Episode 29: reward: 7.000, steps: 621
Episode 30: reward: 15.000, steps: 983
Episode 31: reward: 14.000, steps: 813
Episode 32: reward: 30.000, steps: 1526
Episode 33: reward: 11.000, steps: 843
Episode 34: reward: 13.000, steps: 880
Episode 35: reward: 12.000, steps: 671
Episode 36: reward: 12.000, steps: 862
Episode 37: reward: 12.000, steps: 1017
Episode 38: reward: 19.000, steps: 1247
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Episode 39: reward: 15.000, steps: 934
Episode 40: reward: 2.000, steps: 616
Episode 41: reward: 7.000, steps: 921
Episode 42: reward: 13.000, steps: 655
Episode 43: reward: 12.000, steps: 665
Episode 44: reward: 13.000, steps: 916
Episode 45: reward: 4.000, steps: 494
Episode 46: reward: 8.000, steps: 889
Episode 47: reward: 7.000, steps: 606
Episode 48: reward: 8.000, steps: 843
Episode 49: reward: 17.000, steps: 1130
Episode 50: reward: 16.000, steps: 922
Episode 51: reward: 6.000, steps: 618
Episode 52: reward: 11.000, steps: 682
Episode 53: reward: 7.000, steps: 642
Episode 54: reward: 15.000, steps: 879
Episode 55: reward: 11.000, steps: 611
Episode 56: reward: 8.000, steps: 513
Episode 57: reward: 17.000, steps: 983
Episode 58: reward: 7.000, steps: 727
Episode 59: reward: 18.000, steps: 1083
Episode 60: reward: 10.000, steps: 708
Episode 61: reward: 27.000, steps: 1349
Episode 62: reward: 7.000, steps: 637
Episode 63: reward: 20.000, steps: 1100
Episode 64: reward: 3.000, steps: 631
Episode 65: reward: 8.000, steps: 527
Episode 66: reward: 1.000, steps: 388
Episode 67: reward: 4.000, steps: 675
Episode 68: reward: 25.000, steps: 1202
Episode 69: reward: 18.000, steps: 1172
Episode 70: reward: 24.000, steps: 1252
Episode 71: reward: 11.000, steps: 641
Episode 72: reward: 8.000, steps: 523
Episode 73: reward: 9.000, steps: 807
Episode 74: reward: 8.000, steps: 630
Episode 75: reward: 10.000, steps: 946
Episode 76: reward: 6.000, steps: 869
Episode 77: reward: 12.000, steps: 952
Episode 78: reward: 3.000, steps: 506
Episode 79: reward: 12.000, steps: 761
Episode 80: reward: 15.000, steps: 968
Episode 81: reward: 23.000, steps: 1273
Episode 82: reward: 12.000, steps: 1104
Episode 83: reward: 9.000, steps: 998
Episode 84: reward: 13.000, steps: 740
Episode 85: reward: 16.000, steps: 1215
Episode 86: reward: 11.000, steps: 811
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Episode 87: reward: 11.000, steps: 877
Episode 88: reward: 10.000, steps: 701
Episode 89: reward: 14.000, steps: 1056
Episode 90: reward: 17.000, steps: 1123
Episode 91: reward: 18.000, steps: 1123
Episode 92: reward: 7.000, steps: 501
Episode 93: reward: 9.000, steps: 910
Episode 94: reward: 13.000, steps: 1081
Episode 95: reward: 7.000, steps: 767
Episode 96: reward: 4.000, steps: 796
Episode 97: reward: 7.000, steps: 508
Episode 98: reward: 10.000, steps: 919
Episode 99: reward: 3.000, steps: 367
Episode 100: reward: 14.000, steps: 1280
Episode 101: reward: 11.000, steps: 786
Episode 102: reward: 14.000, steps: 1087
Episode 103: reward: 18.000, steps: 1170
Episode 104: reward: 13.000, steps: 702
Episode 105: reward: 17.000, steps: 1361
Episode 106: reward: 15.000, steps: 787
Episode 107: reward: 4.000, steps: 526
Episode 108: reward: 13.000, steps: 776
Episode 109: reward: 15.000, steps: 1267
Episode 110: reward: 14.000, steps: 1029
Episode 111: reward: 15.000, steps: 1006
Episode 112: reward: 13.000, steps: 843
Episode 113: reward: 11.000, steps: 615
Episode 114: reward: 4.000, steps: 580
Episode 115: reward: 9.000, steps: 949
Episode 116: reward: 12.000, steps: 921
Episode 117: reward: 5.000, steps: 626
Episode 118: reward: 20.000, steps: 1046
Episode 119: reward: 8.000, steps: 510
Episode 120: reward: 12.000, steps: 1179
Episode 121: reward: 19.000, steps: 1040
Episode 122: reward: 7.000, steps: 603
Episode 123: reward: 4.000, steps: 530
Episode 124: reward: 17.000, steps: 1275
Episode 125: reward: 8.000, steps: 717
Episode 126: reward: 15.000, steps: 813
Episode 127: reward: 8.000, steps: 651
Episode 128: reward: 7.000, steps: 496
Episode 129: reward: 23.000, steps: 1322
Episode 130: reward: 17.000, steps: 1134
Episode 131: reward: 20.000, steps: 1256
Episode 132: reward: 16.000, steps: 1097
Episode 133: reward: 15.000, steps: 926
Episode 134: reward: 10.000, steps: 915
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Episode 135: reward: 15.000, steps: 991
Episode 136: reward: 8.000, steps: 672
Episode 137: reward: 18.000, steps: 1242
Episode 138: reward: 3.000, steps: 626
Episode 139: reward: 13.000, steps: 1070
Episode 140: reward: 13.000, steps: 889
Episode 141: reward: 11.000, steps: 933
Episode 142: reward: 24.000, steps: 985
Episode 143: reward: 19.000, steps: 1075
Episode 144: reward: 21.000, steps: 978
Episode 145: reward: 2.000, steps: 485
Episode 146: reward: 16.000, steps: 979
Episode 147: reward: 9.000, steps: 503
Episode 148: reward: 11.000, steps: 1000
Episode 149: reward: 13.000, steps: 764
Episode 150: reward: 15.000, steps: 922
Episode 151: reward: 23.000, steps: 1021
Episode 152: reward: 2.000, steps: 685
Episode 153: reward: 12.000, steps: 898
Episode 154: reward: 12.000, steps: 1164
Episode 155: reward: 25.000, steps: 1107
Episode 156: reward: 7.000, steps: 728
Episode 157: reward: 19.000, steps: 1112
Episode 158: reward: 10.000, steps: 722
Episode 159: reward: 7.000, steps: 831
Episode 160: reward: 10.000, steps: 950
Episode 161: reward: 10.000, steps: 686
Episode 162: reward: 10.000, steps: 839
Episode 163: reward: 13.000, steps: 967
Episode 164: reward: 12.000, steps: 1156
Episode 165: reward: 14.000, steps: 1218
Episode 166: reward: 16.000, steps: 856
Episode 167: reward: 9.000, steps: 722
Episode 168: reward: 15.000, steps: 1098
Episode 169: reward: 15.000, steps: 935
Episode 170: reward: 22.000, steps: 1178
Episode 171: reward: 14.000, steps: 934
Episode 172: reward: 7.000, steps: 854
Episode 173: reward: 16.000, steps: 989
Episode 174: reward: 4.000, steps: 515
Episode 175: reward: 10.000, steps: 949
Episode 176: reward: 13.000, steps: 834
Episode 177: reward: 16.000, steps: 995
Episode 178: reward: 13.000, steps: 1082
Episode 179: reward: 6.000, steps: 745
Episode 180: reward: 15.000, steps: 1452
Episode 181: reward: 23.000, steps: 1016
Episode 182: reward: 10.000, steps: 645
```

```
Episode 183: reward: 5.000, steps: 485
Episode 184: reward: 15.000, steps: 1029
Episode 185: reward: 10.000, steps: 885
Episode 186: reward: 13.000, steps: 946
Episode 187: reward: 13.000, steps: 922
Episode 188: reward: 5.000, steps: 560
Episode 189: reward: 11.000, steps: 664
Episode 190: reward: 9.000, steps: 621
Episode 191: reward: 12.000, steps: 956
Episode 192: reward: 7.000, steps: 681
Episode 193: reward: 10.000, steps: 788
Episode 194: reward: 10.000, steps: 626
Episode 195: reward: 16.000, steps: 922
Episode 196: reward: 8.000, steps: 908
Episode 197: reward: 11.000, steps: 811
Episode 198: reward: 13.000, steps: 1181
Episode 199: reward: 10.000, steps: 558
Episode 200: reward: 15.000, steps: 837

--- Evaluando: DQN Propuesta 3 ---
Testing for 200 episodes …
Episode 1: reward: 17.000, steps: 844
Episode 2: reward: 28.000, steps: 1323
Episode 3: reward: 25.000, steps: 1062
Episode 4: reward: 15.000, steps: 669
Episode 5: reward: 13.000, steps: 562
Episode 6: reward: 27.000, steps: 1046
Episode 7: reward: 32.000, steps: 1448
Episode 8: reward: 15.000, steps: 673
Episode 9: reward: 20.000, steps: 729
Episode 10: reward: 23.000, steps: 948
Episode 11: reward: 27.000, steps: 1537
Episode 12: reward: 20.000, steps: 881
Episode 13: reward: 29.000, steps: 1298
Episode 14: reward: 24.000, steps: 888
Episode 15: reward: 23.000, steps: 963
Episode 16: reward: 22.000, steps: 938
Episode 17: reward: 21.000, steps: 909
Episode 18: reward: 19.000, steps: 814
Episode 19: reward: 25.000, steps: 1111
Episode 20: reward: 19.000, steps: 864
Episode 21: reward: 25.000, steps: 936
Episode 22: reward: 18.000, steps: 803
Episode 23: reward: 24.000, steps: 968
Episode 24: reward: 17.000, steps: 676
Episode 25: reward: 20.000, steps: 934
Episode 26: reward: 15.000, steps: 727
Episode 27: reward: 21.000, steps: 836
```

```
Episode 28: reward: 25.000, steps: 995
Episode 29: reward: 13.000, steps: 667
Episode 30: reward: 23.000, steps: 864
Episode 31: reward: 16.000, steps: 663
Episode 32: reward: 22.000, steps: 869
Episode 33: reward: 28.000, steps: 1077
Episode 34: reward: 16.000, steps: 700
Episode 35: reward: 24.000, steps: 989
Episode 36: reward: 24.000, steps: 827
Episode 37: reward: 31.000, steps: 1455
Episode 38: reward: 21.000, steps: 890
Episode 39: reward: 17.000, steps: 737
Episode 40: reward: 22.000, steps: 781
Episode 41: reward: 33.000, steps: 1254
Episode 42: reward: 24.000, steps: 931
Episode 43: reward: 16.000, steps: 701
Episode 44: reward: 24.000, steps: 871
Episode 45: reward: 21.000, steps: 848
Episode 46: reward: 20.000, steps: 765
Episode 47: reward: 29.000, steps: 1399
Episode 48: reward: 18.000, steps: 647
Episode 49: reward: 22.000, steps: 911
Episode 50: reward: 22.000, steps: 1003
Episode 51: reward: 27.000, steps: 877
Episode 52: reward: 21.000, steps: 932
Episode 53: reward: 19.000, steps: 814
Episode 54: reward: 16.000, steps: 621
Episode 55: reward: 19.000, steps: 774
Episode 56: reward: 26.000, steps: 1068
Episode 57: reward: 22.000, steps: 874
Episode 58: reward: 20.000, steps: 805
Episode 59: reward: 15.000, steps: 717
Episode 60: reward: 20.000, steps: 853
Episode 61: reward: 13.000, steps: 626
Episode 62: reward: 19.000, steps: 879
Episode 63: reward: 21.000, steps: 869
Episode 64: reward: 28.000, steps: 1308
Episode 65: reward: 11.000, steps: 470
Episode 66: reward: 19.000, steps: 828
Episode 67: reward: 19.000, steps: 806
Episode 68: reward: 19.000, steps: 717
Episode 69: reward: 19.000, steps: 913
Episode 70: reward: 23.000, steps: 915
Episode 71: reward: 20.000, steps: 863
Episode 72: reward: 26.000, steps: 951
Episode 73: reward: 22.000, steps: 1091
Episode 74: reward: 21.000, steps: 893
Episode 75: reward: 25.000, steps: 982
```

```
Episode 76: reward: 23.000, steps: 820
Episode 77: reward: 22.000, steps: 723
Episode 78: reward: 15.000, steps: 782
Episode 79: reward: 20.000, steps: 798
Episode 80: reward: 26.000, steps: 1037
Episode 81: reward: 23.000, steps: 1028
Episode 82: reward: 12.000, steps: 629
Episode 83: reward: 22.000, steps: 847
Episode 84: reward: 23.000, steps: 962
Episode 85: reward: 25.000, steps: 955
Episode 86: reward: 17.000, steps: 865
Episode 87: reward: 22.000, steps: 817
Episode 88: reward: 16.000, steps: 661
Episode 89: reward: 29.000, steps: 1310
Episode 90: reward: 22.000, steps: 929
Episode 91: reward: 16.000, steps: 619
Episode 92: reward: 24.000, steps: 872
Episode 93: reward: 13.000, steps: 556
Episode 94: reward: 13.000, steps: 612
Episode 95: reward: 17.000, steps: 793
Episode 96: reward: 19.000, steps: 982
Episode 97: reward: 16.000, steps: 719
Episode 98: reward: 24.000, steps: 891
Episode 99: reward: 15.000, steps: 636
Episode 100: reward: 18.000, steps: 700
Episode 101: reward: 25.000, steps: 982
Episode 102: reward: 17.000, steps: 689
Episode 103: reward: 18.000, steps: 922
Episode 104: reward: 23.000, steps: 1035
Episode 105: reward: 21.000, steps: 841
Episode 106: reward: 19.000, steps: 907
Episode 107: reward: 13.000, steps: 576
Episode 108: reward: 24.000, steps: 904
Episode 109: reward: 17.000, steps: 957
Episode 110: reward: 14.000, steps: 574
Episode 111: reward: 19.000, steps: 794
Episode 112: reward: 14.000, steps: 573
Episode 113: reward: 16.000, steps: 628
Episode 114: reward: 24.000, steps: 876
Episode 115: reward: 14.000, steps: 801
Episode 116: reward: 26.000, steps: 1037
Episode 117: reward: 28.000, steps: 860
Episode 118: reward: 16.000, steps: 706
Episode 119: reward: 20.000, steps: 754
Episode 120: reward: 25.000, steps: 1276
Episode 121: reward: 11.000, steps: 657
Episode 122: reward: 17.000, steps: 787
Episode 123: reward: 17.000, steps: 637
```

```
Episode 124: reward: 32.000, steps: 1333
Episode 125: reward: 25.000, steps: 932
Episode 126: reward: 28.000, steps: 1128
Episode 127: reward: 30.000, steps: 1250
Episode 128: reward: 16.000, steps: 706
Episode 129: reward: 24.000, steps: 1000
Episode 130: reward: 17.000, steps: 763
Episode 131: reward: 20.000, steps: 881
Episode 132: reward: 21.000, steps: 955
Episode 133: reward: 22.000, steps: 954
Episode 134: reward: 16.000, steps: 665
Episode 135: reward: 22.000, steps: 805
Episode 136: reward: 22.000, steps: 1024
Episode 137: reward: 13.000, steps: 495
Episode 138: reward: 24.000, steps: 912
Episode 139: reward: 23.000, steps: 975
Episode 140: reward: 25.000, steps: 937
Episode 141: reward: 17.000, steps: 677
Episode 142: reward: 26.000, steps: 922
Episode 143: reward: 27.000, steps: 1125
Episode 144: reward: 25.000, steps: 902
Episode 145: reward: 15.000, steps: 585
Episode 146: reward: 22.000, steps: 901
Episode 147: reward: 23.000, steps: 880
Episode 148: reward: 21.000, steps: 990
Episode 149: reward: 24.000, steps: 882
Episode 150: reward: 22.000, steps: 880
Episode 151: reward: 31.000, steps: 1091
Episode 152: reward: 25.000, steps: 1035
Episode 153: reward: 21.000, steps: 883
Episode 154: reward: 29.000, steps: 1070
Episode 155: reward: 16.000, steps: 775
Episode 156: reward: 20.000, steps: 893
Episode 157: reward: 15.000, steps: 706
Episode 158: reward: 32.000, steps: 1296
Episode 159: reward: 26.000, steps: 983
Episode 160: reward: 18.000, steps: 876
Episode 161: reward: 29.000, steps: 1159
Episode 162: reward: 15.000, steps: 675
Episode 163: reward: 19.000, steps: 784
Episode 164: reward: 25.000, steps: 1089
Episode 165: reward: 17.000, steps: 715
Episode 166: reward: 12.000, steps: 574
Episode 167: reward: 24.000, steps: 979
Episode 168: reward: 27.000, steps: 1004
Episode 169: reward: 22.000, steps: 912
Episode 170: reward: 17.000, steps: 828
Episode 171: reward: 29.000, steps: 1218
```

```
Episode 172: reward: 22.000, steps: 969
Episode 173: reward: 18.000, steps: 659
Episode 174: reward: 15.000, steps: 658
Episode 175: reward: 22.000, steps: 901
Episode 176: reward: 8.000, steps: 531
Episode 177: reward: 26.000, steps: 1050
Episode 178: reward: 22.000, steps: 921
Episode 179: reward: 19.000, steps: 754
Episode 180: reward: 22.000, steps: 920
Episode 181: reward: 18.000, steps: 927
Episode 182: reward: 26.000, steps: 879
Episode 183: reward: 20.000, steps: 865
Episode 184: reward: 23.000, steps: 883
Episode 185: reward: 22.000, steps: 859
Episode 186: reward: 15.000, steps: 681
Episode 187: reward: 13.000, steps: 581
Episode 188: reward: 20.000, steps: 965
Episode 189: reward: 20.000, steps: 997
Episode 190: reward: 16.000, steps: 873
Episode 191: reward: 18.000, steps: 886
Episode 192: reward: 14.000, steps: 581
Episode 193: reward: 19.000, steps: 881
Episode 194: reward: 18.000, steps: 784
Episode 195: reward: 24.000, steps: 874
Episode 196: reward: 19.000, steps: 868
Episode 197: reward: 19.000, steps: 921
Episode 198: reward: 19.000, steps: 717
Episode 199: reward: 19.000, steps: 632
Episode 200: reward: 16.000, steps: 759
```

1.5.3 4.3 Guardado y carga de resultados

En esta celda, guardamos el diccionario `resultados` en un archivo `.npy`, útil para reutilizar las recompensas sin tener que volver a cargar pesos ni repetir la evaluación.

[ ]:
```python
import os
import numpy as np

carpeta_resultados = 'dqn_SpaceInvaders-v0_Resultados'
os.makedirs(carpeta_resultados, exist_ok=True)

resultados_filename = os.path.join(carpeta_resultados,␣
↪'dqn_SpaceInvaders-v0_Resultados.npy')
np.save(resultados_filename, resultados)

print(f"Resultados guardados en: {resultados_filename}")
for nombre, recompensas in resultados.items():
```

```
    print(f"{nombre}: {len(recompensas)} episodios, recompensa media = {np.
↪mean(recompensas):.2f}")
```

```
Resultados guardados en:
dqn_SpaceInvaders-v0_Resultados\dqn_SpaceInvaders-v0_Resultados.npy
DQN Base: 200 episodios, recompensa media = 13.99
DQN Propuesta 1: 200 episodios, recompensa media = 15.26
DQN Propuesta 2: 200 episodios, recompensa media = 11.97
DQN Propuesta 3: 200 episodios, recompensa media = 20.79
```

En esta celda, levamos a cabo la carga del archivo `.npy` con las recompensas de test, para poder llevar a cabo las gráficas y comparación de resultados en caso de que queramos cargarlos directamente.

[ ]:
```
import os
import numpy as np

resultados_filename = os.path.join('dqn_SpaceInvaders-v0_Resultados',␣
↪'dqn_SpaceInvaders-v0_Resultados.npy')
resultados = np.load(resultados_filename, allow_pickle=True).item()
```

1.5.4 4.4 Tabla resumen de estadísticas

En la siguiente tabla se resumen las estadísticas principales de la evaluación en **200 episodios de test** para cada agente. Se incluye la recompensa mínima, máxima y media, la moda, la máxima racha de episodios consecutivos con recompensa mayor o igual que 20 y el porcentaje de episodios que alcanzan dicho umbral.

Se incluyen los resultados obtenidos durante esta ejecución concreta del entrenamiento de los modelos para poder ser usados en los análisiis gráficos y comparativos en caso de no disponer de los archivos donde están contenidos los pesos.

[ ]:
```
# resultados = {
#     'DQN Base': np.array([
#         15, 9, 10, 3, 15, 15, 22, 6, 16, 10, 11, 6, 18, 15, 7, 18, 22, 6, 10,␣
↪15,
#         21, 14, 22, 22, 19, 7, 17, 7, 11, 7, 9, 6, 7, 16, 21, 6, 24, 17, 11,␣
↪15,
#         12, 8, 10, 10, 12, 25, 18, 11, 7, 13, 14, 14, 4, 8, 27, 13, 17, 17,␣
↪23, 13,
#         28, 9, 7, 6, 11, 20, 4, 15, 16, 17, 13, 28, 14, 7, 9, 16, 7, 12, 28,␣
↪13,
#         14, 9, 9, 10, 22, 23, 5, 16, 17, 10, 20, 8, 22, 2, 26, 17, 14, 20,␣
↪12, 23,
#         19, 14, 5, 9, 16, 15, 15, 11, 16, 13, 18, 18, 7, 8, 4, 11, 23, 5, 10,␣
↪4,
#         15, 24, 4, 12, 20, 16, 10, 17, 18, 17, 7, 25, 9, 7, 16, 13, 7, 12,␣
↪23, 18,
```

```
#         15, 18, 24, 24, 13, 11, 18, 17, 21, 22, 15, 12, 9, 13, 13, 15, 21, 12, 17, 26,
#         11, 8, 24, 7, 20, 16, 11, 15, 6, 17, 14, 9, 19, 9, 8, 17, 17, 7, 9, 5,
#         8, 13, 17, 23, 6, 19, 26, 10, 19, 12, 7, 17, 20, 22, 12, 23, 5, 16, 12, 12
#     ], dtype=float),
#     'DQN Propuesta 1': np.array([
#         24, 15, 13, 7, 11, 20, 20, 19, 22, 18, 11, 11, 13, 11, 11, 13, 10, 22, 13, 15,
#         12, 13, 16, 17, 14, 18, 15, 9, 12, 17, 10, 21, 12, 21, 11, 17, 12, 15, 7, 9,
#         18, 18, 10, 17, 32, 19, 16, 11, 10, 11, 25, 11, 17, 13, 8, 9, 29, 19, 21, 17,
#         12, 12, 11, 22, 12, 15, 20, 14, 18, 12, 25, 17, 19, 15, 12, 7, 16, 12, 12, 11,
#         15, 13, 11, 12, 26, 8, 11, 8, 5, 6, 14, 20, 16, 14, 27, 12, 12, 9, 16, 16,
#         20, 9, 10, 21, 16, 13, 14, 20, 14, 16, 17, 28, 18, 20, 17, 13, 7, 22, 16, 5,
#         15, 15, 15, 6, 17, 19, 11, 12, 21, 26, 15, 8, 12, 17, 14, 9, 22, 19, 17, 7,
#         13, 10, 16, 10, 12, 16, 15, 23, 16, 28, 16, 12, 17, 19, 22, 18, 27, 26, 13, 22,
#         25, 15, 16, 16, 20, 13, 11, 16, 12, 6, 18, 24, 12, 28, 17, 10, 12, 15, 13, 14,
#         5, 15, 12, 18, 17, 17, 9, 14, 12, 29, 17, 16, 12, 22, 16, 15, 17, 11, 12, 16
#     ], dtype=float),
#     'DQN Propuesta 2': np.array([
#         12, 7, 12, 19, 21, 9, 2, 16, 17, 11, 4, 8, 12, 8, 11, 9, 13, 20, 14, 9,
#         9, 18, 14, 6, 4, 6, 15, 9, 7, 15, 14, 30, 11, 13, 12, 12, 12, 19, 15, 2,
#         7, 13, 12, 13, 4, 8, 7, 8, 17, 16, 6, 11, 7, 15, 11, 8, 17, 7, 18, 10,
#         27, 7, 20, 3, 8, 1, 4, 25, 18, 24, 11, 8, 9, 8, 10, 6, 12, 3, 12, 15,
#         23, 12, 9, 13, 16, 11, 11, 10, 14, 17, 18, 7, 9, 13, 7, 4, 7, 10, 3, 14,
#         11, 14, 18, 13, 17, 15, 4, 13, 15, 14, 15, 13, 11, 4, 9, 12, 5, 20, 8, 12,
#         19, 7, 4, 17, 8, 15, 8, 7, 23, 17, 20, 16, 15, 10, 15, 8, 18, 3, 13, 13,
#         11, 24, 19, 21, 2, 16, 9, 11, 13, 15, 23, 2, 12, 12, 25, 7, 19, 10, 7, 10,
#         10, 10, 13, 12, 14, 16, 9, 15, 15, 22, 14, 7, 16, 4, 10, 13, 16, 13, 6, 15,
```

```
#         23, 10, 5, 15, 10, 13, 13, 5, 11, 9, 12, 7, 10, 10, 16, 8, 11, 13,␣
↪10, 15
#     ], dtype=float),
#     'DQN Propuesta 3': np.array([
#         17, 28, 25, 15, 13, 27, 32, 15, 20, 23, 27, 20, 29, 24, 23, 22, 21,␣
↪19, 25, 19,
#         25, 18, 24, 17, 20, 15, 21, 25, 13, 23, 16, 22, 28, 16, 24, 24, 31,␣
↪21, 17, 22,
#         33, 24, 16, 24, 21, 20, 29, 18, 22, 22, 27, 21, 19, 16, 19, 26, 22,␣
↪20, 15, 20,
#         13, 19, 21, 28, 11, 19, 19, 19, 19, 23, 20, 26, 22, 21, 25, 23, 22,␣
↪15, 20, 26,
#         23, 12, 22, 23, 25, 17, 22, 16, 29, 22, 16, 24, 13, 13, 17, 19, 16,␣
↪24, 15, 18,
#         25, 17, 18, 23, 21, 19, 13, 24, 17, 14, 19, 14, 16, 24, 14, 26, 28,␣
↪16, 20, 25,
#         11, 17, 17, 32, 25, 28, 30, 16, 24, 17, 20, 21, 22, 16, 22, 22, 13,␣
↪24, 23, 25,
#         17, 26, 27, 25, 15, 22, 23, 21, 24, 22, 31, 25, 21, 29, 16, 20, 15,␣
↪32, 26, 18,
#         29, 15, 19, 25, 17, 12, 24, 27, 22, 17, 29, 22, 18, 15, 22, 8, 26,␣
↪22, 19, 22,
#         18, 26, 20, 23, 22, 15, 13, 20, 20, 16, 18, 14, 19, 18, 24, 19, 19,␣
↪19, 19, 16
#     ], dtype=float),
# }
```

[ ]:

```
import numpy as np
import pandas as pd
from IPython.display import display

UMBRAL_OBJETIVO = 20

def max_episodios_seguidos(recompensas_episodios, umbral=UMBRAL_OBJETIVO):
    """Calcula la racha máxima de episodios consecutivos con recompensa >=␣
↪umbral."""
    seguidos_maximo = 0
    seguidos_actual = 0

    for recompensa_episodio in recompensas_episodios:
        if recompensa_episodio >= umbral:
            seguidos_actual += 1
            seguidos_maximo = max(seguidos_maximo, seguidos_actual)
        else:
            seguidos_actual = 0
```

```python
    return seguidos_maximo

def porcentaje_episodios_umbral(recompensas_episodios, umbral=UMBRAL_OBJETIVO):
    """Calcula el porcentaje de episodios con recompensa >= umbral."""
    recompensas_episodios = np.asarray(recompensas_episodios, dtype=float)
    return np.mean(recompensas_episodios >= umbral) * 100

matriz_estadisticas = []

for agente, recompensas in resultados.items():
    recompensas = np.asarray(recompensas, dtype=float)
    moda = pd.Series(recompensas).mode().iloc[0]

    matriz_estadisticas.append({
        "Propuesta": agente,
        "Episodios test": len(recompensas),
        "Recompensa mínima": np.min(recompensas),
        "Recompensa máxima": np.max(recompensas),
        "Recompensa media": round(np.mean(recompensas), 2),
        "Moda": moda,
        "Racha máxima >= 20": max_episodios_seguidos(recompensas),
        "Episodios >= 20 (%)": round(porcentaje_episodios_umbral(recompensas),
 ↪2),
    })

tabla_resumen_estadisticas = pd.DataFrame(matriz_estadisticas)
display(tabla_resumen_estadisticas)
```

```
         Propuesta  Episodios test  Recompensa mínima  Recompensa máxima  \
0         DQN Base             200                2.0               28.0
1  DQN Propuesta 1             200                5.0               32.0
2  DQN Propuesta 2             200                1.0               30.0
3  DQN Propuesta 3             200                8.0               33.0

   Recompensa media  Moda  Racha máxima >= 20  Episodios >= 20 (%)
0             13.99   7.0                   2                 19.5
1             15.26  12.0                   2                 19.0
2             11.97  13.0                   1                  8.5
3             20.79  22.0                   9                 58.5
```

1.5.5 4.5 Representación visual de los resultados

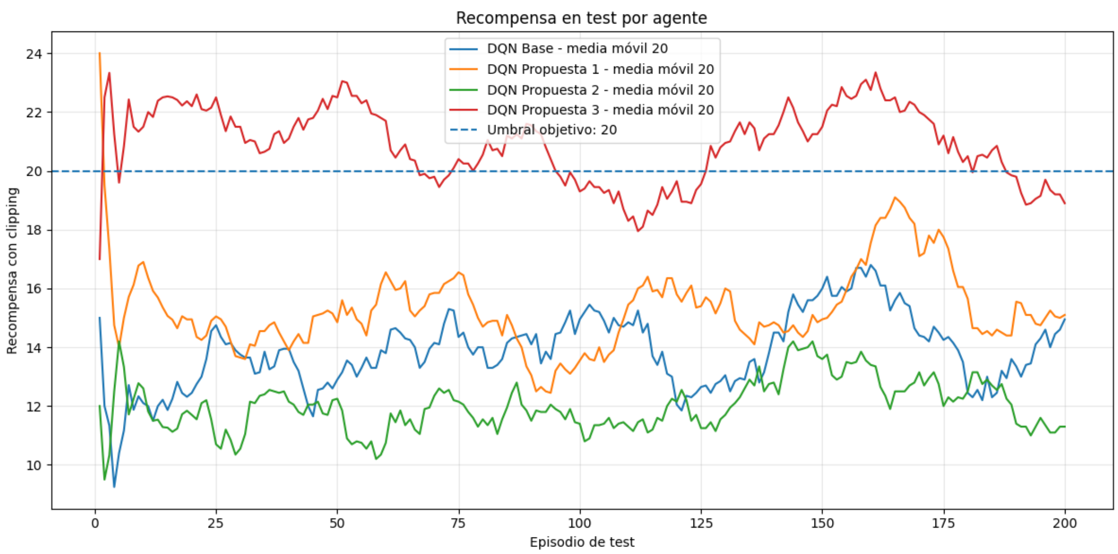
A continuación se presentan las gráficas comparativas obtenidas a partir de las recompensas registradas durante la evaluación en modo test de los distintos agentes. Estas visualizaciones permiten analizar no solo la recompensa media de cada propuesta, sino también la estabilidad del comportamiento aprendido, la distribución de resultados, el porcentaje de episodios que supera el umbral de 20 puntos y la capacidad de mantener dicho rendimiento de forma consecutiva.

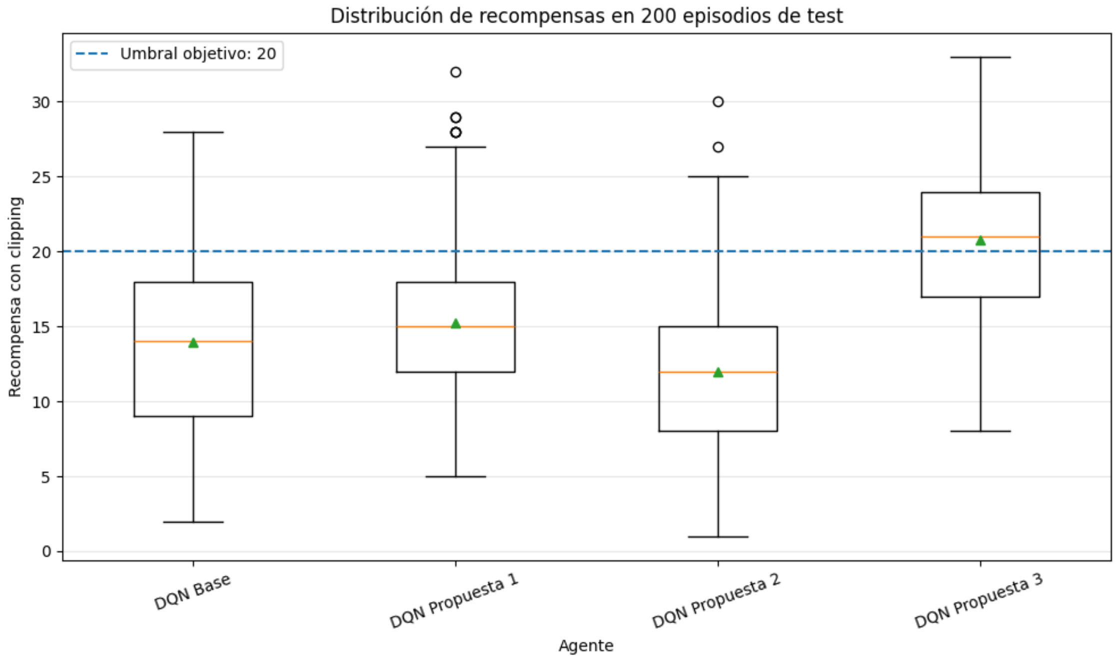
[ ]:
```python
import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

UMBRAL_OBJETIVO = 20
VENTANA_MEDIA_MOVIL = 20
CARPETA_FIGURAS = "figuras_resultados"
os.makedirs(CARPETA_FIGURAS, exist_ok=True)

def media_movil(x, ventana=VENTANA_MEDIA_MOVIL):
    x = np.asarray(x, dtype=float)
    if len(x) < ventana:
        return x
    return pd.Series(x).rolling(window=ventana, min_periods=1).mean().to_numpy()

def max_episodios_seguidos(recompensas, umbral=UMBRAL_OBJETIVO):
    max_racha = 0
    racha_actual = 0

    for recompensa in recompensas:
        if recompensa >= umbral:
            racha_actual += 1
            max_racha = max(max_racha, racha_actual)
        else:
            racha_actual = 0

    return max_racha

# ---------------------------------------------------------------------------
# Gráfica 1: evolución de la recompensa por episodio y media móvil
# ---------------------------------------------------------------------------
plt.figure(figsize=(12, 6))

for nombre, recompensas in resultados.items():
    episodios = np.arange(1, len(recompensas) + 1)
    plt.plot(episodios, media_movil(recompensas), label=f"{nombre} - media␣
 ↪móvil {VENTANA_MEDIA_MOVIL}")

plt.axhline(UMBRAL_OBJETIVO, linestyle="--", label=f"Umbral objetivo:␣
 ↪{UMBRAL_OBJETIVO}")
plt.title("Recompensa en test por agente")
plt.xlabel("Episodio de test")
plt.ylabel("Recompensa con clipping")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
```

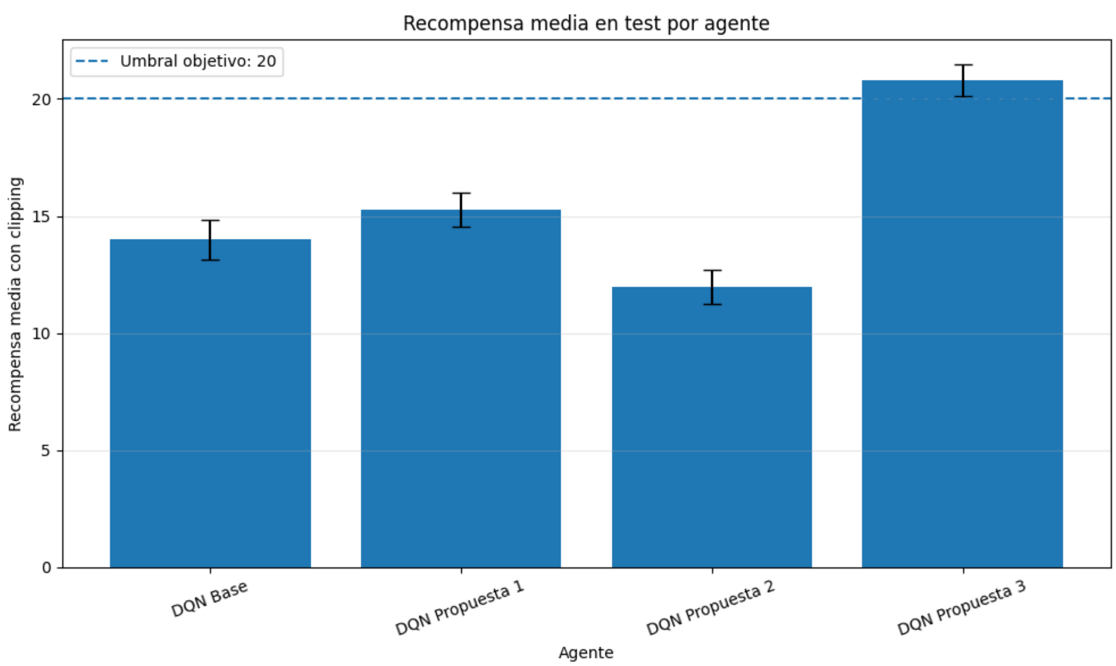
```
plt.savefig(os.path.join(CARPETA_FIGURAS, "01_media_movil_recompensa_test.
↪png"), dpi=150)
plt.show()

# ---------------------------------------------------------------------------
# Gráfica 2: distribución de recompensas por agente
# ---------------------------------------------------------------------------
plt.figure(figsize=(10, 6))
datos_boxplot = [np.asarray(recompensas, dtype=float) for recompensas in␣
↪resultados.values()]
etiquetas = list(resultados.keys())

try:
    plt.boxplot(datos_boxplot, tick_labels=etiquetas, showmeans=True)
except TypeError:
    # Compatibilidad con versiones antiguas de Matplotlib
    plt.boxplot(datos_boxplot, labels=etiquetas, showmeans=True)
plt.axhline(UMBRAL_OBJETIVO, linestyle="--", label=f"Umbral objetivo:␣
↪{UMBRAL_OBJETIVO}")
plt.title("Distribución de recompensas en 200 episodios de test")
plt.xlabel("Agente")
plt.ylabel("Recompensa con clipping")
plt.xticks(rotation=20)
plt.legend()
plt.grid(True, axis="y", alpha=0.3)
plt.tight_layout()
plt.savefig(os.path.join(CARPETA_FIGURAS, "02_distribucion_recompensas_test.
↪png"), dpi=150)
plt.show()

# ---------------------------------------------------------------------------
# Gráfica 3: recompensa media con intervalo de confianza aproximado al 95 %
# ---------------------------------------------------------------------------
nombres = []
medias = []
errores_95 = []

for nombre, recompensas in resultados.items():
    recompensas = np.asarray(recompensas, dtype=float)
    nombres.append(nombre)
    medias.append(np.mean(recompensas))
    errores_95.append(1.96 * np.std(recompensas, ddof=1) / np.
↪sqrt(len(recompensas)))

plt.figure(figsize=(10, 6))
plt.bar(nombres, medias, yerr=errores_95, capsize=6)
```

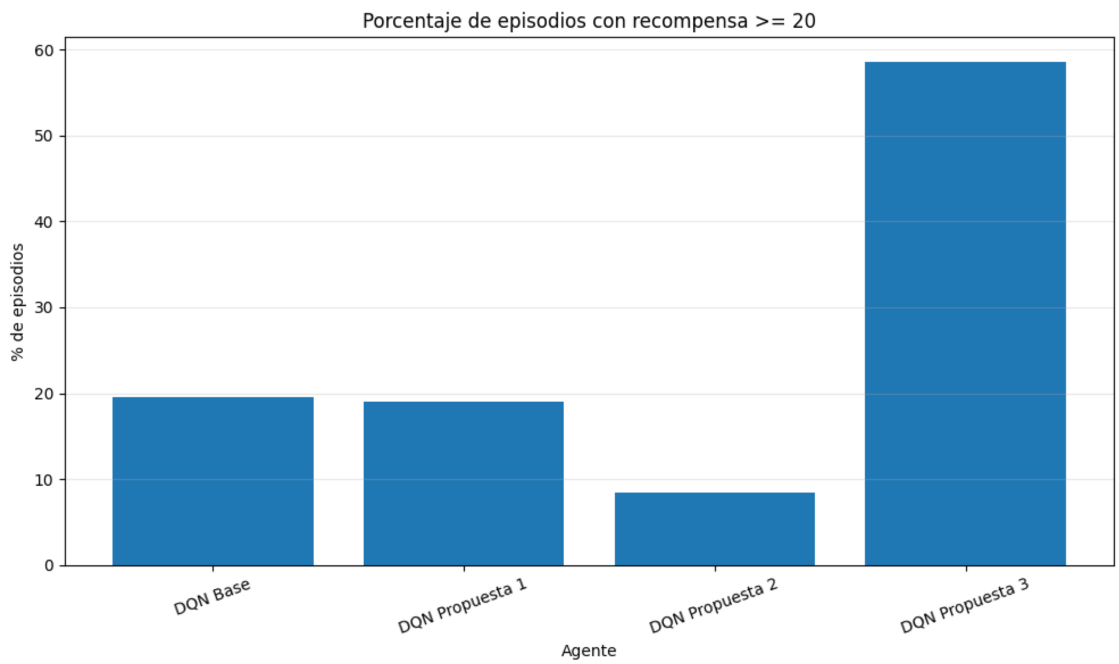
```python
plt.axhline(UMBRAL_OBJETIVO, linestyle="--", label=f"Umbral objetivo:␣
↪{UMBRAL_OBJETIVO}")
plt.title("Recompensa media en test por agente")
plt.xlabel("Agente")
plt.ylabel("Recompensa media con clipping")
plt.xticks(rotation=20)
plt.legend()
plt.grid(True, axis="y", alpha=0.3)
plt.tight_layout()
plt.savefig(os.path.join(CARPETA_FIGURAS, "03_recompensa_media_ic95.png"),␣
↪dpi=150)
plt.show()

# -----------------------------------------------------------------------------
# Gráfica 4: porcentaje de episodios que alcanzan el umbral
# -----------------------------------------------------------------------------
porcentajes = []

for nombre, recompensas in resultados.items():
    recompensas = np.asarray(recompensas, dtype=float)
    porcentajes.append(np.mean(recompensas >= UMBRAL_OBJETIVO) * 100)

plt.figure(figsize=(10, 6))
plt.bar(nombres, porcentajes)
plt.title(f"Porcentaje de episodios con recompensa >= {UMBRAL_OBJETIVO}")
plt.xlabel("Agente")
plt.ylabel("% de episodios")
plt.xticks(rotation=20)
plt.grid(True, axis="y", alpha=0.3)
plt.tight_layout()
plt.savefig(os.path.join(CARPETA_FIGURAS, "04_porcentaje_episodios_umbral.
↪png"), dpi=150)
plt.show()

# -----------------------------------------------------------------------------
# Gráfica 5: racha máxima de episodios consecutivos por encima del umbral
# -----------------------------------------------------------------------------
rachas = [
    max_episodios_seguidos(np.asarray(recompensas, dtype=float),␣
↪UMBRAL_OBJETIVO)
    for recompensas in resultados.values()
]

plt.figure(figsize=(10, 6))
plt.bar(nombres, rachas)
plt.axhline(100, linestyle="--", label="Objetivo: 100 episodios consecutivos")
```

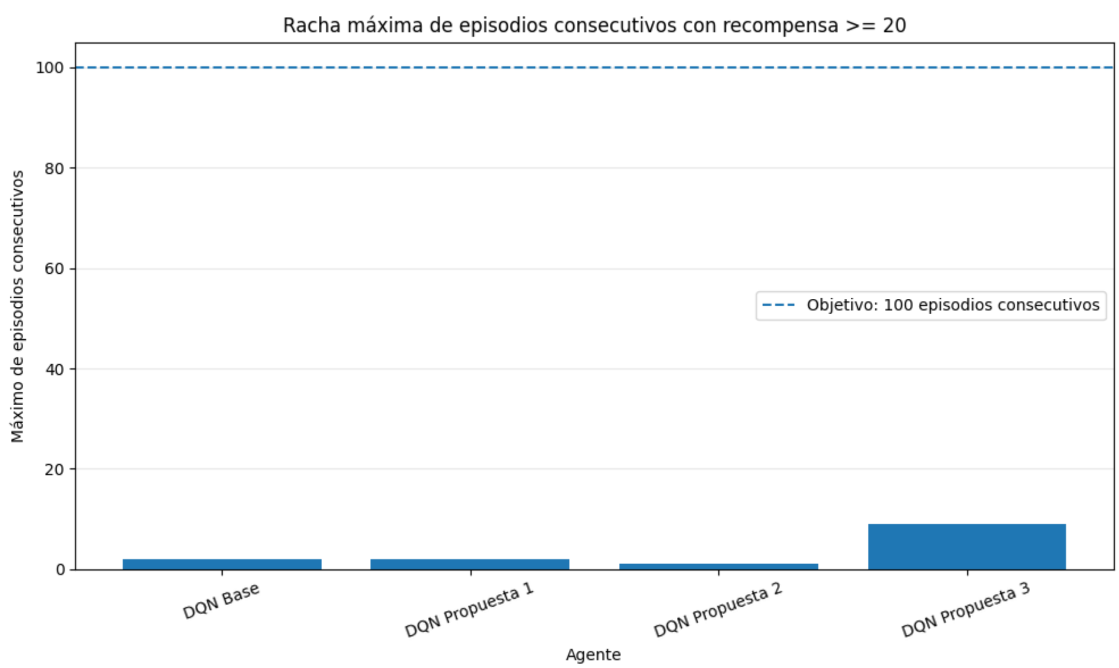
```python
plt.title(f"Racha máxima de episodios consecutivos con recompensa >=␣
    ↪{UMBRAL_OBJETIVO}")
plt.xlabel("Agente")
plt.ylabel("Máximo de episodios consecutivos")
plt.xticks(rotation=20)
plt.legend()
plt.grid(True, axis="y", alpha=0.3)
plt.tight_layout()
plt.savefig(os.path.join(CARPETA_FIGURAS, "05_rachas_maximas_umbral.png"),␣
    ↪dpi=150)
plt.show()


print(f"Figuras guardadas en la carpeta: {CARPETA_FIGURAS}")
```




Distribución de recompensas en 200 episodios de test
Umbral objetivo: 20
Recompensa con clipping
0
5
10
15
20
25
30
DQN Base
DQN Propuesta 1
DQN Propuesta 2
DQN Propuesta 3
Agente


Recompensa media en test por agente
Umbral objetivo: 20
Recompensa media con clipping
0
5
10
15
20
DQN Base
DQN Propuesta 1
DQN Propuesta 2
DQN Propuesta 3
Agente






```
Figuras guardadas en la carpeta: figuras_resultados
```

1.6 PARTE 5. Discusión de los resultados

1.6.1 5.1 Consideraciones sobre las limitaciones de tiempo y computación

Antes de pasar a discutir los resultados obtenidos y el éxito o no de las mejoras implementadas, nos gustaría usar esta sección inicial de la que posiblemente sea la parte más importante del trabajo para resaltar algunas consideraciones importantes.

Debido a limitaciones temporales y de computación, no hemos conseguido alcanzar los resultados que nos habría gustado, entre ellos la consecución de forma estable de puntuaciones iguales o superiores a 20 durante 100 episodios consecutivos. Sin embargo, en la sección final del trabajo, discutiremos qué cambios sobre nuestro trabajo podrían llevarnos a alcanzar este objetivo.

Como consecuencia de estas limitaciones, nos hemos visto obligados a limitar el entrenamiento de nuestras propuestas de mejora. Mientras que el caso base fue entrenado con un `nb_steps=2000000`, para hacer viable el entrenamiento de las propuestas de mejora en tiempo hemos decidido modificarlo a `nb_steps=1000000`.

Esta consideración deberá ser tenida en cuenta a la hora de comparar los resultados del baseline con las propuestas de mejora.

[ ]:
```
display(tabla_resumen_estadisticas)
```

```
        Propuesta  Episodios test  Recompensa mínima  Recompensa máxima  \
0        DQN Base             200                2.0               28.0
1  DQN Propuesta 1            200                5.0               32.0
2  DQN Propuesta 2            200                1.0               30.0
3  DQN Propuesta 3            200                8.0               33.0

   Recompensa media  Moda  Racha máxima >= 20  Episodios >= 20 (%)
0             13.99   7.0                   2                 19.5
1             15.26  12.0                   2                 19.0
2             11.97  13.0                   1                  8.5
3             20.79  22.0                   9                 58.5
```

1.6.2 5.2 Comparación de resultados

En esta sección se comparan los cuatro agentes entrenados: el DQN base y las tres propuestas de mejora. La evaluación se realiza en modo test durante 200 episodios por agente, utilizando las recompensas almacenadas en el diccionario `resultados`.

El objetivo externo del proyecto era alcanzar al menos 20 puntos durante 100 episodios consecutivos. Este criterio no se ha alcanzado: el mejor agente, la Propuesta 3, obtiene un máximo de 9 episodios consecutivos con recompensa mayor o igual que 20. Aun así, la Propuesta 3 sí muestra una mejora clara frente al resto de alternativas, con una recompensa media de 20.79 puntos, un 58.5 % de episodios con recompensa mayor o igual que 20 y una recompensa mínima superior a la de los demás modelos.

Una de las mejoras que nos parece más significativa es, tal como se aprecia en la gráfica 4, que esta mejora consigue triplicar el porcentaje de episodios en que se consigue alcanzar el umbral de 20 puntos respecto al baseline, a pesar de contar con la mitad de pasos de entrenamiento.

Un resultado interesante, tal como se aprecia en la gráfica de distribución de recompensas, es que, a pesar de que la propuesta 3 arroja resultados que mejoran de forma consistente a sus predecesores, las puntuaciones máximas alcanzadas no distan tanto. Por tanto, podríamos decir que, aunque todas tienen una capacidad similar de alcanzar puntuaciones máximas, la propuesta 3 consigue sobre todo consistencia en un rango superior de puntuaciones.

Evaluando por separado nuestras propuestas podríamos concluir: - **DDQN**: resulta ser una medida efectiva dado que conseguimos un aumento de aproximadamente el **9,1%** a pesar de emplear la mitad de tiempo en el entrenamiento del modelo. Este resultado parece suficientemente sólido como para considerar la medida exitosa y contemplarla para posibles desarrollos futuros.

- **Actualización suave de la red target**: en este caso, dado que la recompensa media no consigue superar la del baseline, no podemos afirmar con certeza que la mejora haya sido exitosa, dado que partimos de horizontes temporales diferentes. La actualización suave tiene sentido desde el punto de vista teórico, pero el resultado obtenido sugiere que el valor elegido para `target_model_update`, el número de pasos de entrenamiento o su interacción con el resto de hiperparámetros no fueron adecuados. Si bien no ha quedado demostrado, consideramos que no es descartable que a igualdad de cómputo pudiese suponer aportar una mejora, por lo que aún podría ser de utilidad en un desarrollo futuro.
- **Escalado de la red**: la adición de una capa densamente conectada extra ha demostrado ser el cambio de mayor efecto, ya que consigue mejorar a sus predecesoras en todos los indicadores calculados en la tabla comparativa anterior. De estos resultados podríamos inferir que las capacidades de nuestra red no estaban siendo suficientes para aproximar con un éxito suficiente el valor de la función Q, lo cual nos puede indicar una posible dirección de mejora para el futuro.

1.6.3 5.3 Propuestas de mejora y desarrollos futuros

Como se comentó anteriormente, el coste computacional del entrenamiento ha supuesto una limitación para el éxito de nuestro trabajo. Los entornos Atari requieren muchos pasos de interacción para que DQN aprenda políticas estables, especialmente cuando se parte de píxeles y se usa replay memory.

No obstante, aunque no hemos alcanzado el objetivo fijado, consideramos que no se trata de un objetivo inabordable y, de cara a posibles desarrollos futuros, estas son algunas de las líneas de mejora o propuestas que tendríamos en cuenta.

1. En primer lugar y, sin duda, una **cuestión de escala**. Si dispusiésemos de recursos computacionales y tiempo adicionales, el resultado del experimento 3 creemos que apunta en la dirección de que una mayor complejidad en la red permitiría obtener un mejor resultado. Esta podría venir por una doble vertiente: en la parte de extracción visual (parte convolucional) o en la combinación de las features obtenidas (parte densamente conectada).
2. Al hilo de la mejora anterior, creemos que un aumento en el **número de pasos** durante los cuales se entrena el modelo también contribuiría a mejorar los resultados.
3. Aún sin necesidad de introducir innovaciones, una **combinación de las 3 mejoras propuestas** en una única solución creemos que contribuiría a una mejor solución: mejores estimaciones de la función Q proporcionadas por DDQN, aumento de estabilidad por la suavización de las actualizaciones de los pesos y las mejoras derivadas del escalado del modelo.
4. Como novedades no implementadas en este trabajo, una primera alternativa sería usar una

arquitectura **Dueling DQN**, separando la estimación del valor del estado y la ventaja de cada acción.
5. Incorporar **Prioritized Experience Replay** para que el agente aprenda con mayor frecuencia de transiciones informativas o con mayor error TD.
6. Evaluar distintas planificaciones de epsilon, variando el **balance entre exploración y explotación** que se busca en el transcurso del entrenamiento.

Con estas mejoras, especialmente combinando una red con mayor capacidad, Double DQN, Dueling DQN, replay priorizado y más tiempo de entrenamiento, sería razonable esperar una política más estable y más cercana al objetivo de 100 episodios consecutivos con recompensa mayor o igual que 20.

1.7 Anexos

1.7.1 Anexo A. Prompts usados durante la resolución del proyecto

Creación de la red neuronal Responde como un experto en Redes Neuronales. Codifica en lenguaje Python la red neuronal diseñada para el artículo original de Volodymyr Mnih "Massively Parallel Methods for Deep Reinforcement Learning", indicando claramente las características de la red neuronal. Utiliza la librería Sequential y la función `add` para añadir las capas y mejorar la claridad de lectura.

1.7.2 Anexo B. Exportación opcional a PDF

Las siguientes celdas son auxiliares y solo son necesarias si se desea exportar el notebook a PDF desde Colab. No forman parte del entrenamiento ni de la evaluación del agente.

[1]:
```
!apt-get update
```

```
Get:1 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/ InRelease
[3,632 B]
Get:2 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64
InRelease [1,581 B]
Get:3 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:4 https://r2u.stat.illinois.edu/ubuntu jammy InRelease [6,555 B]
Hit:5 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:7 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy InRelease
[18.1 kB]
Get:8 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu jammy
InRelease [24.3 kB]
Get:9 https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu jammy InRelease
[24.6 kB]
Get:10 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/ Packages [100
kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:12
https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64
```

```
Packages [2,820 kB]
Get:13 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64
Packages [77.8 kB]
Get:14 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages
[4,089 kB]
Get:15 https://r2u.stat.illinois.edu/ubuntu jammy/main amd64 Packages [3,109 kB]
Get:16 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages
[1,312 kB]
Get:17 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64
Packages [7,332 kB]
Get:18 https://r2u.stat.illinois.edu/ubuntu jammy/main all Packages [10.5 MB]
Get:19 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy/main amd64
Packages [38.6 kB]
Get:20 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu jammy/main
amd64 Packages [45.0 kB]
Get:21 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages
[7,619 kB]
Get:22 https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu jammy/main amd64
Packages [75.3 kB]
Get:23 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages
[1,615 kB]
Get:24 http://archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages
[86.4 kB]
Get:25 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [4,426
kB]
Get:26 http://archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages
[35.6 kB]
Get:27 http://archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages
[82.8 kB]
Fetched 43.9 MB in 11s (4,128 kB/s)
Reading package lists… Done
W: Skipping acquire of configured file 'main/source/Sources' as repository
'https://r2u.stat.illinois.edu/ubuntu jammy InRelease' does not seem to provide
it (sources.list entry misspelt?)
```

[2]:
```
!apt-get install -y texlive-xetex texlive-fonts-recommended
 ↪texlive-plain-generic pandoc
```

```
Reading package lists… Done
Building dependency tree… Done
Reading state information… Done
The following additional packages will be installed:
  dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
  fonts-texgyre fonts-urw-base35 libapache-pom-java
  libcmark-gfm-extensions0.29.0.gfm.3 libcmark-gfm0.29.0.gfm.3
  libcommons-logging-java libcommons-parent-java libfontbox-java libgs9
  libgs9-common libidn12 libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java
  libptexenc1 libruby3.0 libsynctex2 libteckit0 libtexlua53 libtexluajit2
```

```
  libwoff1 libzzip-0-13 lmodern pandoc-data poppler-data preview-latex-style
  rake ruby ruby-net-telnet ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0
  rubygems-integration t1utils teckit tex-common tex-gyre texlive-base
  texlive-binaries texlive-latex-base texlive-latex-extra
  texlive-latex-recommended texlive-pictures tipa xfonts-encodings
  xfonts-utils
Suggested packages:
  fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
  libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
  texlive-luatex pandoc-citeproc context wkhtmltopdf librsvg2-bin groff ghc
  nodejs php python libjs-mathjax libjs-katex citation-style-language-styles
  poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
  fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
  fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
  | postscript-viewer perl-tk xpdf | pdf-viewer xzdec
  texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
  icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
  texlive-latex-extra-doc texlive-latex-recommended-doc texlive-pstricks
  dot2tex prerex texlive-pictures-doc vprerex default-jre-headless tipa-doc
The following NEW packages will be installed:
  dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
  fonts-texgyre fonts-urw-base35 libapache-pom-java
  libcmark-gfm-extensions0.29.0.gfm.3 libcmark-gfm0.29.0.gfm.3
  libcommons-logging-java libcommons-parent-java libfontbox-java libgs9
  libgs9-common libidn12 libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java
  libptexenc1 libruby3.0 libsynctex2 libteckit0 libtexlua53 libtexluajit2
  libwoff1 libzzip-0-13 lmodern pandoc pandoc-data poppler-data
  preview-latex-style rake ruby ruby-net-telnet ruby-rubygems ruby-webrick
  ruby-xmlrpc ruby3.0 rubygems-integration t1utils teckit tex-common tex-gyre
  texlive-base texlive-binaries texlive-fonts-recommended texlive-latex-base
  texlive-latex-extra texlive-latex-recommended texlive-pictures
  texlive-plain-generic texlive-xetex tipa xfonts-encodings xfonts-utils
0 upgraded, 57 newly installed, 0 to remove and 347 not upgraded.
Need to get 202 MB of archives.
After this operation, 728 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-droid-fallback all
1:6.0.1r16-1.1build1 [1,805 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-lato all 2.0-2.1
[2,696 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy/main amd64 poppler-data all
0.4.11-1 [2,171 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-common all 6.17
[33.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-urw-base35 all
20200910-1 [6,367 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9-common
all 9.55.0~dfsg1-0ubuntu5.13 [753 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libidn12 amd64
```

```
1.38-4ubuntu1.1 [60.5 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy/main amd64 libijs-0.35 amd64
0.35-15build2 [16.5 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy/main amd64 libjbig2dec0 amd64
0.19-3build2 [64.7 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9 amd64
9.55.0~dfsg1-0ubuntu5.13 [5,032 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libkpathsea6
amd64 2021.20210626.59705-1ubuntu0.3 [60.6 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy/main amd64 libwoff1 amd64
1.0.2-1build4 [45.2 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy/universe amd64 dvisvgm amd64
2.13.1-1 [1,221 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy/universe amd64 fonts-lmodern all
2.004.5-6.1 [4,532 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-noto-mono all
20201225-1build1 [397 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy/universe amd64 fonts-texgyre all
20180621-3.1 [10.2 MB]
Get:17 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libapache-pom-java
all 18-1 [4,720 B]
Get:18 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libcmark-
gfm0.29.0.gfm.3 amd64 0.29.0.gfm.3-3 [115 kB]
Get:19 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libcmark-gfm-
extensions0.29.0.gfm.3 amd64 0.29.0.gfm.3-3 [25.1 kB]
Get:20 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libcommons-parent-
java all 43-1 [10.8 kB]
Get:21 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libcommons-logging-
java all 1.2-2 [60.3 kB]
Get:22 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libptexenc1
amd64 2021.20210626.59705-1ubuntu0.3 [39.1 kB]
Get:23 http://archive.ubuntu.com/ubuntu jammy/main amd64 rubygems-integration
all 1.18 [5,336 B]
Get:24 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby3.0 amd64
3.0.2-7ubuntu2.13 [50.1 kB]
Get:25 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby-rubygems
all 3.3.5-2ubuntu1.2 [228 kB]
Get:26 http://archive.ubuntu.com/ubuntu jammy/main amd64 ruby amd64 1:3.0~exp1
[5,100 B]
Get:27 http://archive.ubuntu.com/ubuntu jammy/main amd64 rake all 13.0.6-2 [61.7
kB]
Get:28 http://archive.ubuntu.com/ubuntu jammy/main amd64 ruby-net-telnet all
0.1.1-2 [12.6 kB]
Get:29 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby-webrick
all 1.7.0-3ubuntu0.2 [52.5 kB]
Get:30 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby-xmlrpc all
0.3.2-1ubuntu0.1 [24.9 kB]
Get:31 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libruby3.0
```

```
amd64 3.0.2-7ubuntu2.13 [5,116 kB]
Get:32 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libsynctex2
amd64 2021.20210626.59705-1ubuntu0.3 [55.8 kB]
Get:33 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libteckit0 amd64
2.5.11+ds1-1 [421 kB]
Get:34 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtexlua53
amd64 2021.20210626.59705-1ubuntu0.3 [120 kB]
Get:35 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtexluajit2
amd64 2021.20210626.59705-1ubuntu0.3 [267 kB]
Get:36 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libzzip-0-13 amd64
0.13.72+dfsg.1-1.1 [27.0 kB]
Get:37 http://archive.ubuntu.com/ubuntu jammy/main amd64 xfonts-encodings all
1:1.0.5-0ubuntu2 [578 kB]
Get:38 http://archive.ubuntu.com/ubuntu jammy/main amd64 xfonts-utils amd64
1:7.7+6build2 [94.6 kB]
Get:39 http://archive.ubuntu.com/ubuntu jammy/universe amd64 lmodern all
2.004.5-6.1 [9,471 kB]
Get:40 http://archive.ubuntu.com/ubuntu jammy/universe amd64 pandoc-data all
2.9.2.1-3ubuntu2 [81.8 kB]
Get:41 http://archive.ubuntu.com/ubuntu jammy/universe amd64 pandoc amd64
2.9.2.1-3ubuntu2 [20.3 MB]
Get:42 http://archive.ubuntu.com/ubuntu jammy/universe amd64 preview-latex-style
all 12.2-1ubuntu1 [185 kB]
Get:43 http://archive.ubuntu.com/ubuntu jammy/main amd64 t1utils amd64
1.41-4build2 [61.3 kB]
Get:44 http://archive.ubuntu.com/ubuntu jammy/universe amd64 teckit amd64
2.5.11+ds1-1 [699 kB]
Get:45 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-gyre all
20180621-3.1 [6,209 kB]
Get:46 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 texlive-
binaries amd64 2021.20210626.59705-1ubuntu0.3 [9,861 kB]
Get:47 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-base all
2021.20220204-1 [21.0 MB]
Get:48 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-fonts-
recommended all 2021.20220204-1 [4,972 kB]
Get:49 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-base
all 2021.20220204-1 [1,128 kB]
Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libfontbox-java all
1:1.8.16-2 [207 kB]
Get:51 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpdfbox-java all
1:1.8.16-2 [5,199 kB]
Get:52 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-
recommended all 2021.20220204-1 [14.4 MB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-pictures
all 2021.20220204-1 [8,720 kB]
Get:54 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-extra
all 2021.20220204-1 [13.9 MB]
Get:55 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-
```

```
generic all 2021.20220204-1 [27.5 MB]
Get:56 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:57 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 202 MB in 10s (21.1 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages …
Selecting previously unselected package fonts-droid-fallback.
(Reading database … 126281 files and directories currently installed.)
Preparing to unpack …/00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
…
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) …
Selecting previously unselected package fonts-lato.
Preparing to unpack …/01-fonts-lato_2.0-2.1_all.deb …
Unpacking fonts-lato (2.0-2.1) …
Selecting previously unselected package poppler-data.
Preparing to unpack …/02-poppler-data_0.4.11-1_all.deb …
Unpacking poppler-data (0.4.11-1) …
Selecting previously unselected package tex-common.
Preparing to unpack …/03-tex-common_6.17_all.deb …
Unpacking tex-common (6.17) …
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack …/04-fonts-urw-base35_20200910-1_all.deb …
Unpacking fonts-urw-base35 (20200910-1) …
Selecting previously unselected package libgs9-common.
Preparing to unpack …/05-libgs9-common_9.55.0~dfsg1-0ubuntu5.13_all.deb …
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.13) …
Selecting previously unselected package libidn12:amd64.
Preparing to unpack …/06-libidn12_1.38-4ubuntu1.1_amd64.deb …
Unpacking libidn12:amd64 (1.38-4ubuntu1.1) …
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack …/07-libijs-0.35_0.35-15build2_amd64.deb …
Unpacking libijs-0.35:amd64 (0.35-15build2) …
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack …/08-libjbig2dec0_0.19-3build2_amd64.deb …
Unpacking libjbig2dec0:amd64 (0.19-3build2) …
Selecting previously unselected package libgs9:amd64.
Preparing to unpack …/09-libgs9_9.55.0~dfsg1-0ubuntu5.13_amd64.deb …
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) …
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack …/10-libkpathsea6_2021.20210626.59705-1ubuntu0.3_amd64.deb
…
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.3) …
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack …/11-libwoff1_1.0.2-1build4_amd64.deb …
Unpacking libwoff1:amd64 (1.0.2-1build4) …
Selecting previously unselected package dvisvgm.
```

```
Preparing to unpack …/12-dvisvgm_2.13.1-1_amd64.deb …
Unpacking dvisvgm (2.13.1-1) …
Selecting previously unselected package fonts-lmodern.
Preparing to unpack …/13-fonts-lmodern_2.004.5-6.1_all.deb …
Unpacking fonts-lmodern (2.004.5-6.1) …
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack …/14-fonts-noto-mono_20201225-1build1_all.deb …
Unpacking fonts-noto-mono (20201225-1build1) …
Selecting previously unselected package fonts-texgyre.
Preparing to unpack …/15-fonts-texgyre_20180621-3.1_all.deb …
Unpacking fonts-texgyre (20180621-3.1) …
Selecting previously unselected package libapache-pom-java.
Preparing to unpack …/16-libapache-pom-java_18-1_all.deb …
Unpacking libapache-pom-java (18-1) …
Selecting previously unselected package libcmark-gfm0.29.0.gfm.3:amd64.
Preparing to unpack …/17-libcmark-gfm0.29.0.gfm.3_0.29.0.gfm.3-3_amd64.deb …
Unpacking libcmark-gfm0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) …
Selecting previously unselected package libcmark-gfm-
extensions0.29.0.gfm.3:amd64.
Preparing to unpack …/18-libcmark-gfm-
extensions0.29.0.gfm.3_0.29.0.gfm.3-3_amd64.deb …
Unpacking libcmark-gfm-extensions0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) …
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack …/19-libcommons-parent-java_43-1_all.deb …
Unpacking libcommons-parent-java (43-1) …
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack …/20-libcommons-logging-java_1.2-2_all.deb …
Unpacking libcommons-logging-java (1.2-2) …
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack …/21-libptexenc1_2021.20210626.59705-1ubuntu0.3_amd64.deb
…
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.3) …
Selecting previously unselected package rubygems-integration.
Preparing to unpack …/22-rubygems-integration_1.18_all.deb …
Unpacking rubygems-integration (1.18) …
Selecting previously unselected package ruby3.0.
Preparing to unpack …/23-ruby3.0_3.0.2-7ubuntu2.13_amd64.deb …
Unpacking ruby3.0 (3.0.2-7ubuntu2.13) …
Selecting previously unselected package ruby-rubygems.
Preparing to unpack …/24-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb …
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) …
Selecting previously unselected package ruby.
Preparing to unpack …/25-ruby_1%3a3.0~exp1_amd64.deb …
Unpacking ruby (1:3.0~exp1) …
Selecting previously unselected package rake.
Preparing to unpack …/26-rake_13.0.6-2_all.deb …
Unpacking rake (13.0.6-2) …
Selecting previously unselected package ruby-net-telnet.
```

```
Preparing to unpack …/27-ruby-net-telnet_0.1.1-2_all.deb …
Unpacking ruby-net-telnet (0.1.1-2) …
Selecting previously unselected package ruby-webrick.
Preparing to unpack …/28-ruby-webrick_1.7.0-3ubuntu0.2_all.deb …
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) …
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack …/29-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb …
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) …
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack …/30-libruby3.0_3.0.2-7ubuntu2.13_amd64.deb …
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.13) …
Selecting previously unselected package libsynctex2:amd64.
Preparing to unpack …/31-libsynctex2_2021.20210626.59705-1ubuntu0.3_amd64.deb
…
Unpacking libsynctex2:amd64 (2021.20210626.59705-1ubuntu0.3) …
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack …/32-libteckit0_2.5.11+ds1-1_amd64.deb …
Unpacking libteckit0:amd64 (2.5.11+ds1-1) …
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack …/33-libtexlua53_2021.20210626.59705-1ubuntu0.3_amd64.deb
…
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.3) …
Selecting previously unselected package libtexluajit2:amd64.
Preparing to unpack
…/34-libtexluajit2_2021.20210626.59705-1ubuntu0.3_amd64.deb …
Unpacking libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.3) …
Selecting previously unselected package libzzip-0-13:amd64.
Preparing to unpack …/35-libzzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb …
Unpacking libzzip-0-13:amd64 (0.13.72+dfsg.1-1.1) …
Selecting previously unselected package xfonts-encodings.
Preparing to unpack …/36-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb …
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) …
Selecting previously unselected package xfonts-utils.
Preparing to unpack …/37-xfonts-utils_1%3a7.7+6build2_amd64.deb …
Unpacking xfonts-utils (1:7.7+6build2) …
Selecting previously unselected package lmodern.
Preparing to unpack …/38-lmodern_2.004.5-6.1_all.deb …
Unpacking lmodern (2.004.5-6.1) …
Selecting previously unselected package pandoc-data.
Preparing to unpack …/39-pandoc-data_2.9.2.1-3ubuntu2_all.deb …
Unpacking pandoc-data (2.9.2.1-3ubuntu2) …
Selecting previously unselected package pandoc.
Preparing to unpack …/40-pandoc_2.9.2.1-3ubuntu2_amd64.deb …
Unpacking pandoc (2.9.2.1-3ubuntu2) …
Selecting previously unselected package preview-latex-style.
Preparing to unpack …/41-preview-latex-style_12.2-1ubuntu1_all.deb …
Unpacking preview-latex-style (12.2-1ubuntu1) …
Selecting previously unselected package t1utils.
```

```
Preparing to unpack …/42-t1utils_1.41-4build2_amd64.deb …
Unpacking t1utils (1.41-4build2) …
Selecting previously unselected package teckit.
Preparing to unpack …/43-teckit_2.5.11+ds1-1_amd64.deb …
Unpacking teckit (2.5.11+ds1-1) …
Selecting previously unselected package tex-gyre.
Preparing to unpack …/44-tex-gyre_20180621-3.1_all.deb …
Unpacking tex-gyre (20180621-3.1) …
Selecting previously unselected package texlive-binaries.
Preparing to unpack …/45-texlive-
binaries_2021.20210626.59705-1ubuntu0.3_amd64.deb …
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.3) …
Selecting previously unselected package texlive-base.
Preparing to unpack …/46-texlive-base_2021.20220204-1_all.deb …
Unpacking texlive-base (2021.20220204-1) …
Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack …/47-texlive-fonts-recommended_2021.20220204-1_all.deb …
Unpacking texlive-fonts-recommended (2021.20220204-1) …
Selecting previously unselected package texlive-latex-base.
Preparing to unpack …/48-texlive-latex-base_2021.20220204-1_all.deb …
Unpacking texlive-latex-base (2021.20220204-1) …
Selecting previously unselected package libfontbox-java.
Preparing to unpack …/49-libfontbox-java_1%3a1.8.16-2_all.deb …
Unpacking libfontbox-java (1:1.8.16-2) …
Selecting previously unselected package libpdfbox-java.
Preparing to unpack …/50-libpdfbox-java_1%3a1.8.16-2_all.deb …
Unpacking libpdfbox-java (1:1.8.16-2) …
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack …/51-texlive-latex-recommended_2021.20220204-1_all.deb …
Unpacking texlive-latex-recommended (2021.20220204-1) …
Selecting previously unselected package texlive-pictures.
Preparing to unpack …/52-texlive-pictures_2021.20220204-1_all.deb …
Unpacking texlive-pictures (2021.20220204-1) …
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack …/53-texlive-latex-extra_2021.20220204-1_all.deb …
Unpacking texlive-latex-extra (2021.20220204-1) …
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack …/54-texlive-plain-generic_2021.20220204-1_all.deb …
Unpacking texlive-plain-generic (2021.20220204-1) …
Selecting previously unselected package tipa.
Preparing to unpack …/55-tipa_2%3a1.3-21_all.deb …
Unpacking tipa (2:1.3-21) …
Selecting previously unselected package texlive-xetex.
Preparing to unpack …/56-texlive-xetex_2021.20220204-1_all.deb …
Unpacking texlive-xetex (2021.20220204-1) …
Setting up fonts-lato (2.0-2.1) …
Setting up fonts-noto-mono (20201225-1build1) …
Setting up libwoff1:amd64 (1.0.2-1build4) …
```

```
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.3) …
Setting up libijs-0.35:amd64 (0.35-15build2) …
Setting up libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.3) …
Setting up libfontbox-java (1:1.8.16-2) …
Setting up rubygems-integration (1.18) …
Setting up libzzip-0-13:amd64 (0.13.72+dfsg.1-1.1) …
Setting up fonts-urw-base35 (20200910-1) …
Setting up poppler-data (0.4.11-1) …
Setting up tex-common (6.17) …
update-language: texlive-base not installed and configured, doing nothing!
Setting up libjbig2dec0:amd64 (0.19-3build2) …
Setting up libteckit0:amd64 (2.5.11+ds1-1) …
Setting up libapache-pom-java (18-1) …
Setting up ruby-net-telnet (0.1.1-2) …
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) …
Setting up t1utils (1.41-4build2) …
Setting up libidn12:amd64 (1.38-4ubuntu1.1) …
Setting up fonts-texgyre (20180621-3.1) …
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.3) …
Setting up ruby-webrick (1.7.0-3ubuntu0.2) …
Setting up libcmark-gfm0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) …
Setting up fonts-lmodern (2.004.5-6.1) …
Setting up libcmark-gfm-extensions0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) …
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) …
Setting up pandoc-data (2.9.2.1-3ubuntu2) …
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) …
Setting up libsynctex2:amd64 (2021.20210626.59705-1ubuntu0.3) …
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.13) …
Setting up teckit (2.5.11+ds1-1) …
Setting up libpdfbox-java (1:1.8.16-2) …
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) …
Setting up preview-latex-style (12.2-1ubuntu1) …
Setting up libcommons-parent-java (43-1) …
Setting up dvisvgm (2.13.1-1) …
Setting up libcommons-logging-java (1.2-2) …
Setting up xfonts-utils (1:7.7+6build2) …
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.3) …
Setting up pandoc (2.9.2.1-3ubuntu2) …
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.3) …
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) …
Setting up texlive-base (2021.20220204-1) …
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
```

```
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST…
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN…
mktexlsr: Updating /var/lib/texmf/ls-R…
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) …
Setting up texlive-plain-generic (2021.20220204-1) …
Setting up texlive-latex-base (2021.20220204-1) …
Setting up texlive-latex-recommended (2021.20220204-1) …
Setting up texlive-pictures (2021.20220204-1) …
Setting up texlive-fonts-recommended (2021.20220204-1) …
Setting up tipa (2:1.3-21) …
Setting up texlive-latex-extra (2021.20220204-1) …
Setting up texlive-xetex (2021.20220204-1) …
Setting up rake (13.0.6-2) …
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.13) …
Setting up ruby3.0 (3.0.2-7ubuntu2.13) …
Setting up ruby (1:3.0~exp1) …
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) …
Processing triggers for man-db (2.10.2-1) …
Processing triggers for mailcap (3.70+nmu1ubuntu1) …
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) …
Processing triggers for libc-bin (2.35-0ubuntu3.8) …
/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic
```

```
link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

Processing triggers for tex-common (6.17) …
Running updmap-sys. This may take some time… done.
Running mktexlsr /var/lib/texmf … done.
Building format(s) --all.
        This may take some time… done.
```

[ ]:
```
name_IPYNB_file = 'Proyecto_practico.ipynb'
#BASE_FOLDER = drive_root
BASE_FOLDER = '/content/gdrive/My Drive/Colab Notebooks/VIU IA/08_AR_MIAR/
 ↪Proyecto'
get_ipython().system(
        "apt update >> /dev/null && apt install texlive-xetex␣
 ↪texlive-fonts-recommended texlive-generic-recommended >> /dev/null"
    )
get_ipython().system(
            f"jupyter nbconvert --output-dir='{BASE_FOLDER}' '{BASE_FOLDER}/
 ↪{name_IPYNB_file}' --to pdf"
        )
```

```
WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

W: Skipping acquire of configured file 'main/source/Sources' as repository
'https://r2u.stat.illinois.edu/ubuntu jammy InRelease' does not seem to provide
it (sources.list entry misspelt?)

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

E: Unable to locate package texlive-generic-recommended
[NbConvertApp] Converting notebook /content/gdrive/My Drive/Colab Notebooks/VIU
IA/08_AR_MIAR/Proyecto/Proyecto_practico.ipynb to pdf
[NbConvertApp] Support files will be in Proyecto_practico_files/
[NbConvertApp] Making directory ./Proyecto_practico_files
```

```
[NbConvertApp] Writing 374582 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
```